

# **Logic Manager Planning**

**LM02-501**



## System Site Planning

# *Logic Manager Planning*

LM02-501  
Release 500  
9/95

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## About This Publication

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This manual provides information necessary to properly plan the installation of a Logic Manager subsystem at a TDC 3000<sup>X</sup> system site. The Logic Manager subsystem encompasses the Logic Manager (LM) and the Network Interface Module (NIM), which is resident on the Universal Control Network (UCN), a network associated with the TDC 3000<sup>X</sup> Local Control Network (LCN). The amount of information that this manual provides personally depends on your experience and the process that the Logic Manager will control and monitor.

The experienced planner, a person involved in the installation of TDC Basic or LCN equipment, will find that some information is familiar. However, regardless of your past experience, you must read Section 3 in this manual to enhance your knowledge of process control connections available, and also examine the *System Site Planning*, *Universal Control Network Planning*, and *Universal Control Network Installation* manuals to prepare yourself for the connection of the Logic Manager to the Universal Control Network (UCN).

In some cases, control room expansion will be part of installing the Logic Manager. If this is the case, use the *LCN Planning and LCN System Installation* manuals to plan for expansion of the network.

This publication supports software Release 500.



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# Acronyms

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|           |                                                   |
|-----------|---------------------------------------------------|
| APM.....  | Advanced Process Manager                          |
| BCD.....  | Binary Coded Decimal                              |
| CAS.....  | Canadian Standards Association                    |
| CIM.....  | Computer Interface Module                         |
| DCM.....  | Data Collection Module                            |
| EDM.....  | Enhanced Diagnostic Module                        |
| EMI.....  | Electromagnetic Interference                      |
| FM.....   | Factory Mutual                                    |
| I/O.....  | Input/Output                                      |
| IEEE..... | Institute of Electrical and Electronics Engineers |
| IOCM..... | Input/Output Control Module                       |
| LCN.....  | Local Control Network                             |
| LED.....  | Light Emitting Diode                              |
| LMM.....  | Logic Manager Module                              |
| LM.....   | Logic Manager                                     |
| NIM.....  | Network Interface Module                          |
| PBSM..... | Parallel Bus Selector Module                      |
| PIOM..... | Parallel I/O Module                               |
| PLDM..... | Parallel Link Driver Module                       |
| PM.....   | Process Managers                                  |
| PM.....   | Processor Module                                  |
| PSM.....  | Power Supply Module                               |
| RAM.....  | Random Access Memory                              |
| RCM.....  | Redundant Control Modules                         |
| RTU.....  | Remote Terminal Unit                              |
| SCM.....  | System Control Module                             |
| SIOM..... | Serial I/O Module                                 |
| SLM.....  | Serial Link Module                                |
| SLS.....  | Serial Link Selector                              |
| TTL.....  | Transistor to Transistor Logic                    |
| UCN.....  | Universal Control Network                         |
| UL.....   | Underwriters Laboratories                         |

## References

---

| <b>Publication<br/>Title</b>                                  | <b>Publication<br/>Number</b> | <b>Binder<br/>Title</b>                | <b>Binder<br/>Number</b> |
|---------------------------------------------------------------|-------------------------------|----------------------------------------|--------------------------|
| Logic Manager Specification and Technical Data                | LM03-500                      | System Summary - 2                     | TDC 3010-2               |
| Logic Manager Installation                                    | LM20-500                      | Implementation/Logic Manager           | TDC 3070                 |
| Logic Manager Service                                         | LM13-500                      | LM Service/Logic Manager               | TDC 3073                 |
| Universal Control Network Specification and Technical Data    | UN03-500                      | System Summary - 2                     | TDC 3010-2               |
| Universal Control Network Planning                            | UN02-501                      | System Site Planning -1                | TDC 3020-1               |
| Universal Control Network Installation                        | UN20-500                      | Installation/Universal Control Network | TDC 3041                 |
| Universal Control Network Guidelines                          | UN12-510                      | Installation/Universal Control Network | TDC 3041                 |
| LCN Planning                                                  | SW02-501                      | System Site Planning -1                | TDC 3020-1               |
| LCN System Installation                                       | SW20-500                      | LCN Installation                       | TDC 3025                 |
| LCN System Checkout                                           | SW20-510                      | LCN Installation                       | TDC 3025                 |
| LCN Guidelines - Implementation, Troubleshooting, and Service | LC09-510                      | LCN Installation                       | TDC 3025                 |
| TDC 3000 <sup>X</sup> System Site Planning                    | SW02-550                      | System Site Planning -1                | TDC 3020-1               |

# Section 1 – Introduction

## 1.1 Overview

---

**Section Contents**      The topics covered in this section are:

|     | <b>Topic</b>  | <b>See Page</b> |
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**Purpose**      This manual is intended for planning the installation of a Logic Manager (LM) subsystem at a TDC 3000<sup>X</sup> site. The Logic Manager subsystem is a device on the Universal Control Network (UCN) that includes the Network Interface Module (NIM). Process Managers (PMs) and Advanced Process Managers (APMs) can also be resident on the network.

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**Planning contents**      Planning includes the consideration of the Logic Manager cabinet layout, process wiring techniques, Logic Manager processor redundancy, serial or parallel I/O, module selection, and unique hardware features.

---

**Information not covered**      Neither installation, power-on checkout or service of the Logic Manager, nor planning for the Local Control Network (LCN) and Universal Control Network (UCN) is addressed in this manual. See the related reference documentation for information concerning these topics.

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## Section 2 – Logic Manager Description

### 2.1 Overview

#### Section Contents

The topics covered in this section are:

|         | Topic                                      | See Page |
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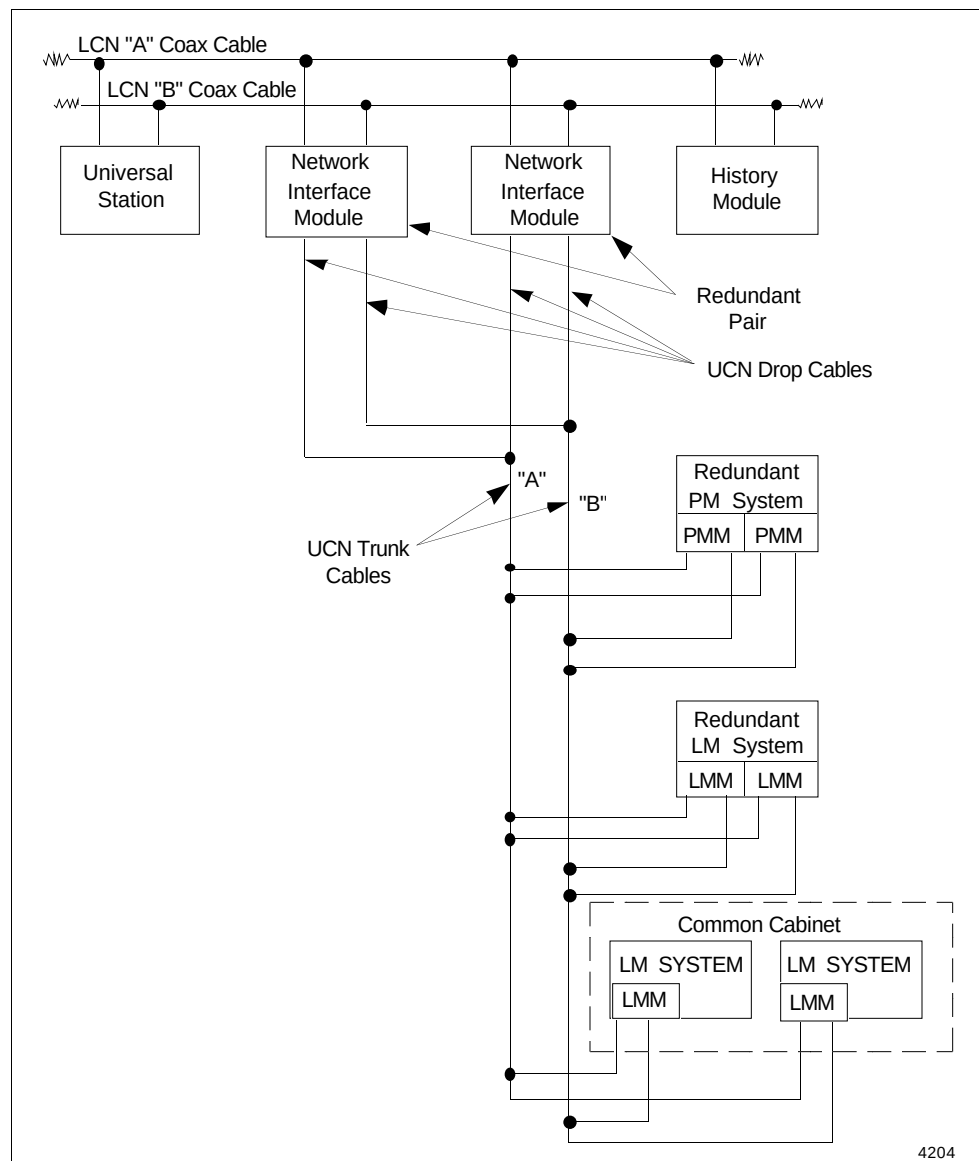
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## 2.1 Overview, Continued

### General overview

The Logic Manager subsystem (LM) performs digital and analog logic control functions as an integral part of the Honeywell TDC 3000<sup>X</sup> system. The Logic Manager provides process automation for control, communication, monitoring of specific processes, and a peer-to-peer link with the Process Manager subsystem (PM), Advanced Process Manager subsystem (APM), or other Logic Managers on the Universal Control Network (UCN). Figure 2-1 is an overview of a network with installed Logic Manager and Process Manager subsystems. The information necessary to install a Logic Manager subsystem on a Universal Control Network can be found in the *Universal Control Network Planning* and *Universal Control Network Installation* manuals.

Figure 2-1 Typical Universal Control Network Configuration



Continued on next page



## 2.1 Overview, Continued

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                  | The basic Logic Manager subsystem consists of a high-speed programmable controller processor, a Power Supply Module, optional modules, a serial or parallel I/O communication link to field devices, and a Logic Manager Module (LMM) that provides the interface between the controller processor and the UCN.                                                                                                                                  |
| <b>Uses ladder logic</b>            | The Logic Manager is programmed with ladder logic using the model 623-60 MS-DOS Loader. (Refer to Honeywell User Manual, number 623-8982). The Logic Manager Module in the Logic Manager Processor card file is configured through the Universal Station (US) on the LCN.                                                                                                                                                                        |
| <b>Scans I/O Interface</b>          | The Logic Manager subsystem scans its parallel and/or serial I/O interface, executing the control program for I/O output devices, while providing input status collected from the I/O input devices. The Logic Manager subsystem processes this information and converts it to UCN data format recognized by the TDC 3000 <sup>X</sup> LCN interface modules.                                                                                    |
| <b>Logic Manager configuration</b>  | The recommended Logic Manager subsystem consists of a single or redundant Logic Manager Processor card file(s), and either, but not both, serial or parallel I/O card file(s). That is, serial and parallel I/O should not be mixed in the same “control group” of any Logic Manager subsystem. However, although this serial/parallel mix is not recommended, it is technically permissible and can be accomplished with a “special subsystem.” |
| <b>Parallel I/O not recommended</b> | Failure in a parallel I/O subsystem is potentially more hazardous than in a serial I/O subsystem, because there is a greater risk of causing a total subsystem failure in the parallel I/O architecture. For this reason, a redundant Logic Manager with parallel I/O is not recommended. Only serial I/O is supported by a redundant Logic Manager subsystem. A redundant Logic Manager with parallel I/O is considered a “special subsystem.”  |
| <b>Logic Manager redundancy</b>     | Redundancy in the Logic Manager subsystem involves a single I/O subsystem controlled by dual (redundant) processors. The redundancy is managed by two Redundant Control Modules (RCM), one located in each Logic Manager Processor card file. Any optional modules that reside in one Logic Manager Processor card file must be duplicated in the other Processor card file.                                                                     |

## 2.2 Logic Manager Processor

---

### Overview

The Logic Manager Processor resides in a Processor card file and consists of multi-modules (cards) that provide an interface to, and control of, remote serial or local parallel I/O.

Its instruction set includes relay logic, timer, counter, and sequencer instructions in addition to four-function math, data handling capabilities, and comparisons.

The memory capability is expandable to 32 K from the standard 24 K words of memory.

With an I/O capacity of up to 2048 points and an instruction set that includes matrices, subroutines, signed mathematical operations, and indirect addressing, the Logic Manager Processor is designed to service high-speed and complex control applications.

---

*Continued on next page*

## 2.2 Logic Manager Processor, Continued

### LM Processor specifications

The specifications for the Logic Manager Processor are listed in Table 2-1.

Table 2-1 Logic Manager Processor Specifications

| Parameter                                          | Specification                                                                                                                                                           |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AC Voltage                                         | 115/230 Vac $\pm 15\%$ (selectable) 250 Vac maximum                                                                                                                     |
| Frequency                                          | 47 to 63 Hz                                                                                                                                                             |
| DC Voltage                                         | 20 to 28 Vdc                                                                                                                                                            |
| Power Fail Level                                   | 85 Vac for 115 Vac<br>190 Vac for 230 Vac<br>19 Vdc for 24 Vdc                                                                                                          |
| Operating Temperature                              | 0° to 60°C ambient                                                                                                                                                      |
| Storage Temperature                                | -40° to +70°C                                                                                                                                                           |
| Relative Humidity                                  | 5% to 95% (noncondensing)                                                                                                                                               |
| Card File Dimensions                               | 27.2 cm (10.7 in.) height<br>48.3 cm (19 in.) wide<br>19.1 cm (7.5 in.) deep                                                                                            |
| Real I/O Capacity                                  | 2048                                                                                                                                                                    |
| INTERNAL I/O CAPACITY<br>(In Addition to Real I/O) | 2048                                                                                                                                                                    |
| Data Register                                      | 4096                                                                                                                                                                    |
| Register Size                                      | 17 bits                                                                                                                                                                 |
| Memory Size (Words)                                | 24 K standard, 8 K additional                                                                                                                                           |
| Memory Usage                                       | 1 word per ladder diagram element                                                                                                                                       |
| Memory Type                                        | CMOS RAM                                                                                                                                                                |
| Battery Type                                       | Lithium – Size D in the Power Supply Module and<br>size 2/3 A in the Logic Manager Module card                                                                          |
| Battery Backup                                     | 1.5 years                                                                                                                                                               |
| Scan Rate (relay only)                             | 2.5 ms/k                                                                                                                                                                |
| Industrial Ratings                                 | NEMA-ICS2-203<br>IEC 2k V/5k V Isolation Test 255-4<br>ANSI C37.902a-1974<br>IEEE 518-1977 (Surge Withstand Test)<br>UL 508<br>CSA C22.2-142-M1983<br>FM Class No. 3820 |

## 2.3 Logic Manager Input/Output Subsystem

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Logic Manager configurations</b> | <p>The Logic Manager Input/Output (I/O) subsystem offers several options for interface of the Logic Manager's processor to the sensors and actuators at the plant. The sections that follow discuss the parallel and serial I/O subsystems that can provide two basic I/O configurations, local/parallel and remote/serial, for the Logic Manager's processor. The Logic Manager I/O subsystem discussed can also be an integral part of an additional Logic Manager configuration feature, the Logic Manager's redundancy control.</p>                                                                                                                                                                               |
| <b>I/O subsystem components</b>     | <p>The parallel and serial I/O subsystems include a wide variety of discrete, analog, and special function I/O modules, as well as related I/O hardware, such as card files, power supplies, and I/O simulator assemblies.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Configuration requirements</b>   | <p>Configurations and module types within the I/O subsystem are determined by the specific application requirements and Logic Manager's processing capability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>I/O card file overview</b>       | <p>The Logic Manager I/O subsystem makes configuring a subsystem for the Logic Manager both easy and cost effective. The I/O subsystem offers a full range of 4-, 6-, 8-, 16-, or 32-point discrete, analog, and special function modules. The full-size, 19-inch length I/O card file accommodates up to 12 I/O modules, the I/O card file Power Supply Module (PSM), and a serial or parallel interface module.</p> <p>A nonstandard half-size, 12-inch length card file accommodates up to six I/O modules, the I/O card file Power Supply Module, and a serial or parallel interface module. Tables in subsection 2.10 of this manual provide a complete list of the currently available process I/O modules.</p> |
| <b>I/O subsystem's purpose</b>      | <p>The Input/Output (I/O) subsystem, an integral part of the Logic Manager subsystem, provides the important physical link between the field devices and the control processor. This interface enables subsystem inputs to be read and outputs to be controlled by the Logic Manager's processor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Logic Manager control</b>        | <p>The I/O subsystem consists of many different input and output modules with varying numbers of points and voltage types designed for compatibility with the many different kinds of field devices and power sources to which process I/O modules are connected. The Logic Manager processor gathers and measures physical machine or process variables such as proximity (position of an object), pressure, temperature, fluid level, and voltage/current through the I/O interface. Based on these statuses, or values, the Logic Manager's processor then issues commands to control the devices being monitored, such as alarms, control relays, lights, motors, pumps, or valves.</p>                           |

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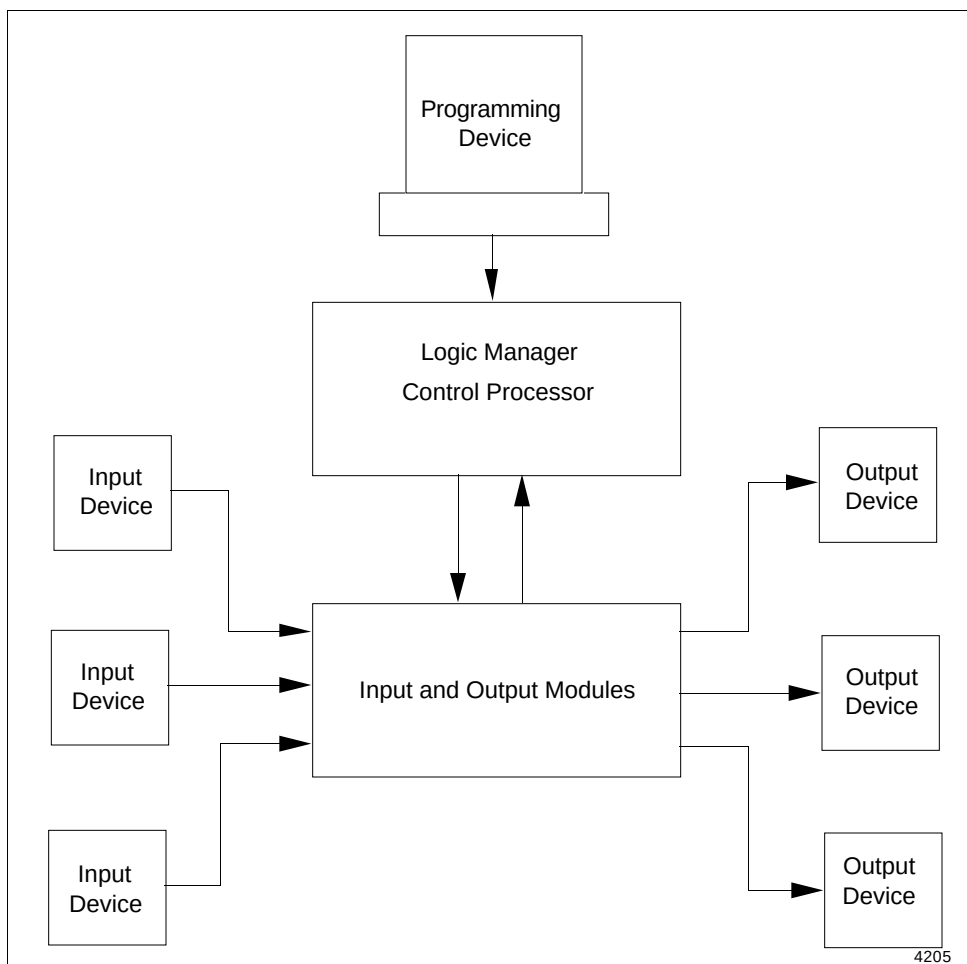
## 2.3 Logic Manager Input/Output Subsystem, Continued

### Discrete I/O control

The discrete Input/Output interface is the most basic I/O interface and involves field devices, which are connected to I/O modules, that operate with distinct, single bit discrete signals, such as on/off, open/closed, or extended/retracted. The states of these input signals are read by the I/O modules and communicated to the Logic Manager processor in a logic level format. Based on this input signal data, the processor then controls the subsystem outputs as dictated by the user program being executed by the processor.

Refer to Figure 2-2 for an illustration of the I/O interface.

Figure 2-2 Logic Manager Input/Output Subsystem



*Continued on next page*

## 2.3 Logic Manager Input/Output Subsystem, Continued

---

### Analog and special I/O control

Other more complex I/O interfaces include analog interfaces, where the modules translate analog-to-digital or digital-to-analog signals, and multi-bit digital interfaces, such as multiple bits in parallel or serial formats; for example, BCD inputs/outputs and pulse inputs/outputs. In most cases, these more complex I/O interfaces rely on PUSH/PULL communication between the I/O modules and the Logic Manager processor.

Some I/O subsystems can also contain I/O modules that perform specialized functions such as ASCII communication, subsystem diagnostics, or high-speed control that is independent of the Logic Manager's processor.

---

### Parallel and serial I/O subsystems

The Logic Manager can have several I/O subsystem configurations. A parallel I/O configuration is used for local installation of the control processor and I/O subsystem. A serial configuration permits the I/O subsystem to be located thousands of meters away from the Logic Manager processor. This configuration also minimizes field wiring.

Logic Managers can be configured for a "special I/O subsystem" with a combination of parallel and serial I/O for maximum flexibility in a subsystem layout, but this configuration is not recommended.

---

### I/O terminal blocks

The I/O modules, with varying numbers of points that include the standard 8-, 16-, and 32-point sizes, are designed for easy installation. The 8- and 16-point modules have terminal block sets that evenly split the I/O points between the top and bottom blocks of the module. These terminal block sets reduce the size of each module's field wiring bundle in half. The terminal blocks lock open without disturbing the wiring for easy module removal and replacement. The 32-point modules are provided with connectors that are easily removed.

---

## 2.4 Serial I/O Subsystem

### Overview

The remote serial I/O subsystem permits the control system to operate over long distances. Serial I/O card files can be mounted close to the machine or process being controlled to eliminate long control wire runs. The serial I/O subsystem can be set up in various multidrop configurations. See Figures 2-3 and 2-4 for two examples.

Figure 2-3 Serial I/O Communications Network (Example 1)

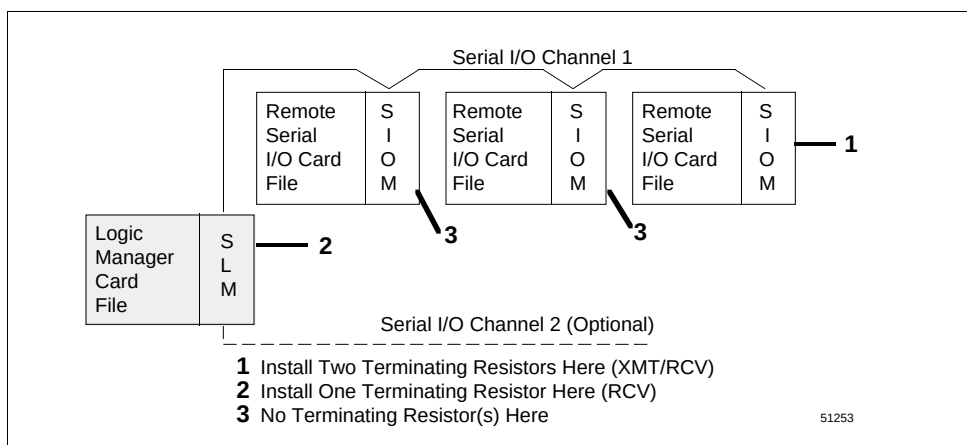
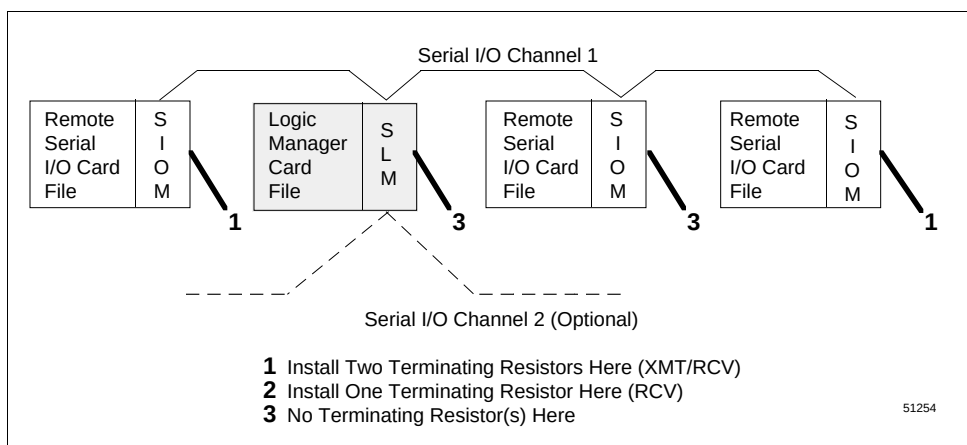


Figure 2-4 Serial I/O Communications Network (Example 2)



*Continued on next page*

## 2.4 Serial I/O Subsystem, Continued

---

### Serial I/O link control modules

The important interface control modules in a serial configuration are the model 620-9939 Serial Link Module (SLM), and the model 621-9938R or 621-9940 Serial I/O Module (SIOM). One or two SLMs, which function as the masters for the serial link, can be placed in slots L and/or M of the Logic Manager Processor card file. The SIOM resides in the rightmost slot of each Serial I/O card file.

---

### I/O data link communication

Each SLM sequentially communicates with each SIOM in its communications link. PUSH instructions and other output data are transmitted by the SLM to each SIOM, while PULL instructions and other input data are transmitted by each SIOM back to the SLM.

---

### Twisted-pair cabling

The Serial I/O subsystem requires two twisted-pair cables with each pair individually shielded for the communication link. The maximum cable length per channel is 3000 meters (10,000 feet).

---

### ATTENTION

ATTENTION—Refer to the *Logic Manager Installation* manual for cabling information and rules regarding installation of terminating resistors.

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*Continued on next page*



## 2.4 Serial I/O Subsystem, Continued

### Serial I/O subsystem specifications

Specifications for the Logic Manager Serial I/O subsystem are listed in Table 2-2.

Table 2-2 Serial Input/Output Subsystem Specifications

| Parameter                | Specification                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Transfer Medium     | Serial 4-wire full duplex                                                                                                                                                                                        |
| Data Rate                | 115.2 k baud                                                                                                                                                                                                     |
| Electrical Format        | EIA-422                                                                                                                                                                                                          |
| Serial Channel Scan Rate | Asynchronous to the processor scan. Dependent on the number of card files per link, number of card files containing PUSH/PULL modules, and the number of PUSH/PULL modules updated per scan (all or 1 per scan). |
| Maximum Drops/Channel    | 16                                                                                                                                                                                                               |
| Maximum I/O Per Channel  | 2040                                                                                                                                                                                                             |
| Maximum Channel Length   | 1200 m/4000 ft. (Belden 9729, use 200 ohm terminating resistor *) or 3000 m/10,000 ft. (Belden 9182, use 300 ohm terminating resistor *)                                                                         |
| Error Checking           | Cyclical redundancy check plus data receive timeout                                                                                                                                                              |
| Operating Current        | Model 621-9938R 2.06 A (+5 Vdc typical)<br>Model 621-9939 2.60 A (+5 Vdc typical)<br>Model 621-9940 2.03 A (+5 Vdc typical)                                                                                      |

\* Supplied with the model 621-9939 module (SLM).

## 2.5 Parallel I/O Subsystem

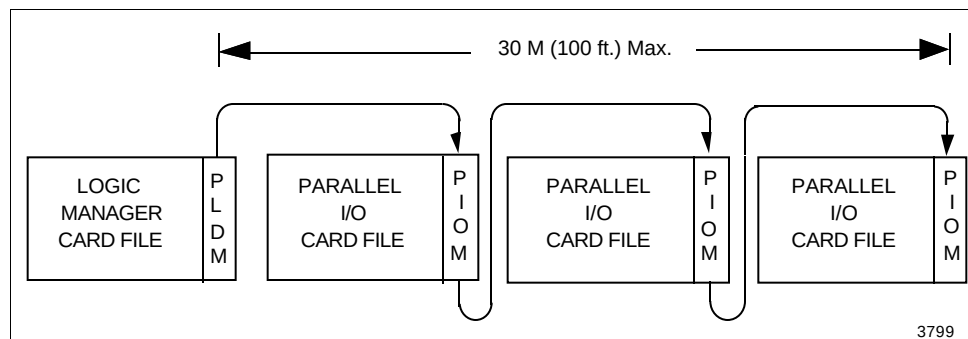
### Overview

The local Parallel I/O subsystem provides I/O update in real time. Up to 22 card files can be daisy chained in a Parallel I/O subsystem layout. A parallel transmission length of up to 30 meters (100 feet) is permitted.

### Parallel I/O link control modules

A typical parallel I/O layout is shown in Figure 2-5. The important interface modules in a parallel I/O configuration are the model 620-0088 Parallel Link Driver Module (PLDM) and the model 621-9937 Parallel I/O Module (PIOM). As shown in the Figure 2-5, the first I/O card file in the local Parallel I/O subsystem is connected to a D-type connector on the Parallel Link Driver Module. A second connector on the PIOM then connects to the next PIOM in the parallel I/O link.

Figure 2-5 Local Parallel I/O Network



### Parallel I/O cabling

The Parallel I/O Module, installed in the rightmost slot of each I/O card file in the parallel I/O configuration, operates as an interface to the PLDM in the Logic Manager Processor card file and to the other PIOMs in other parallel I/O card files. The PIOM has two D-type connectors. The **IN** connector (male) is positioned at the upper portion of the module's front plate, and the **OUT** connector (female) is positioned at the lower portion of the card file.

#### ATTENTION

ATTENTION—Refer to the *Logic Manager Installation* manual for cabling information that includes model numbers for standard cables.

## 2.6 Redundancy Control

---

### Overview

The Logic Manager's redundancy control provides backup for the processors in the Logic Manager subsystem. The redundancy consists of two identically configured Logic Manager Processor card files, each containing a model 620-0059 Redundancy Control Module (RCM) module. The RCMs are linked by a parallel data cable and initiate an orderly switchover of subsystem I/O from the lead (primary) processor to the backup (secondary) processor if a critical processor or I/O communications link fault occurs.

---

### Redundant processors

Both processors are connected to a common I/O subsystem that may be parallel I/O, serial I/O, or a combination of the two. Only serial I/O subsystems are recommended. An example is shown in Figure 2-6.

---

### Redundancy control modules

Serial I/O redundancy is accomplished through the model 621-9938R redundant Serial I/O Module (SIOM) and the model 621-9939 Serial Link Module (SLM). The optional model 621-9928R Serial Link Selector (SLS) module is available for switching the MS-DOS Loader communication link to either processor.

---

### ATTENTION

ATTENTION—Refer to the *Logic Manager Installation* manual for detailed cabling information and for rules covering the installation of terminating resistors for the redundant subsystem switchover communication link.

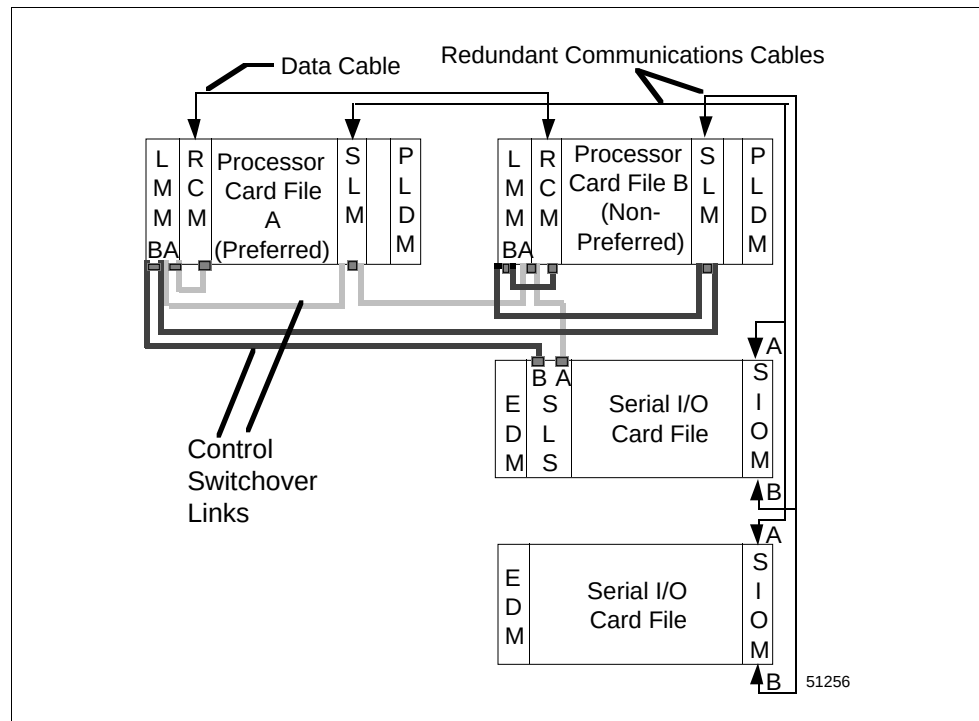
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## 2.6 Redundancy Control, Continued

### Redundancy control configuration

Figure 2-6 Redundant Serial I/O Configuration



## 2.7 Equipment Overview

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### Introduction

A Logic Manager subsystem, which consists of Logic Manager Processor and I/O card files, can have multiple physical configurations. The subsystem may reside in a single cabinet and the cabinet may be single access or dual access. The subsystem may also reside in a complex of up to four dual access cabinets. However, part of the subsystem may reside in the cabinet or cabinets, while the remainder of the subsystem, the I/O portion may be positioned externally and/or remotely in other cabinets or panel mounted. In any case, redundant Logic Manager Processor card files must reside together.

---

### 2.7.1 Processor Card File

---

#### Description

The Logic Manager Processor card file is a 19-inch chassis that can be frame or panel mounted. The card file includes the chassis framework, backplane, processor modules, I/O interface modules, a Power Supply Module, and front cover plates for the unused optional module slots. There are fourteen module slots, identified as A through N, with card slot A on the left when facing the card file. The Logic Manager Processor modules are vertically positioned in the card file with the component side toward the left. The backplane connectors are offset to prevent inserting a module upside down.

---

#### Processor card file configurations

The Logic Manager Processor card file has two configurations, nonredundant and redundant. The standard modules are required for both configurations. Other modules are optional and depend on the subsystem requirements. In either case, slot positioning of the modules is defined for the configuration.

---

#### Configurations defined

The following subsections define the two configurations of Logic Manager Processor card file and the additional option modules available for card slots C and D.

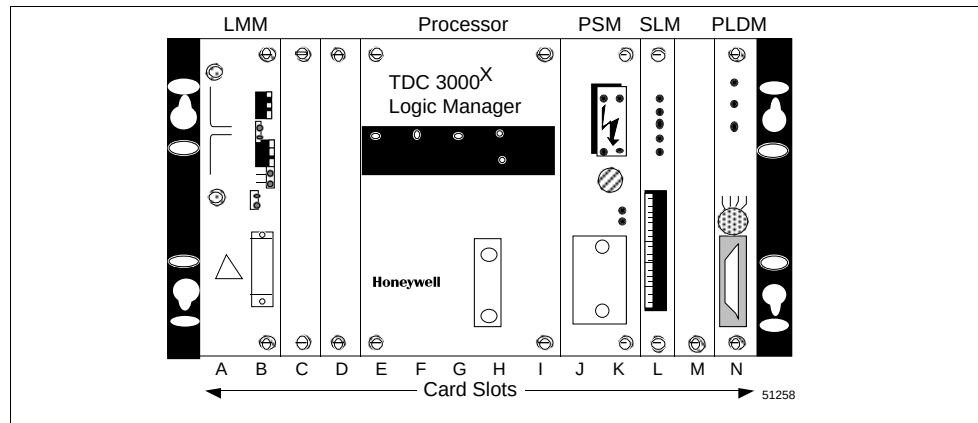
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## 2.7.1.1 Nonredundant Processor Card File

### Overview

The basic nonredundant Logic Manager Processor card file contains the standard modules shown in Figure 2-7. Figure 2-7 is a front view of the card file.

Figure 2-7 Nonredundant Processor Card File Front View



*Continued on next page*

## 2.7.1.1 Nonredundant Processor Card File, Continued

### Module slot assignment list

Table 2-3 defines the module slot assignments for the nonredundant Processor card file.

Table 2-3 Nonredundant Processor Card File Slot Assignments

| Card Slot | Module Type                           | Model Number | Notes      |
|-----------|---------------------------------------|--------------|------------|
| A and B   | Logic Manager Module (LMM)            | 620-0071     | 6, 8       |
| C         | Blank Metal Cover Plate               | 629-4004     | 1, 2, 4    |
| D         | Blank Metal Cover Plate               | 629-4004     | 1, 2, 4, 9 |
| E         | 24 K Memory Module                    | 620-0024     | 8          |
| E         | 4 K x 4 K Register Module             | 620-0056     |            |
| G         | System Control Module                 | 620-0054     |            |
| H         | Processor Module                      | 620-0080     |            |
| I         | 2 K I/O Control Module                | 620-0085     |            |
| J and K   | Power Supply Module (PSM) 115/230 Vac | 620-0036     | 6          |
| J and K   | Power Supply Module (PSM) 24 Vdc      | 620-0047     | 6          |
| L         | Serial Link Module (SLM)              | 621-9939     | 8          |
| M         | Blank Plastic Cover Plate             | 621-9900     | 2, 3, 5    |
| N         | Parallel Link Driver Module (PLDM)    | 620-0088     | 7, 8       |

Notes:

1. This slot is for optional modules.
2. Check DIP switch settings for option modules.
3. This slot is for the optional second model 621-9939 Serial Link Module.
4. Install the model 629-4004 blank metal cover plate if this slot is empty.
5. Install the model 621-9900 blank plastic cover plate if this slot is empty.
6. The module is double width.
7. The model 620-0088 Parallel Link Driver Module is required even when serial I/O is used exclusively because the PLDM performs some subsystem control functions in addition to supporting parallel I/O.
8. This module has its configuration DIP switches preset at the factory. See *Logic Manager Installation* manual for setting UCN address into DIP switch SW2.
9. The optional model 620-0027, 8 K Memory Module may be installed in this slot.

*Continued on next page*

## 2.7.1.1 Nonredundant Processor Card File, Continued

|                                          |                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model ML-LMSA35</b>                   | The card file is designated model ML-LMSA35 if it has an ac Power Supply Module in slots J and K.                                                                                                                                                                                                                                                                                            |
| <b>Model ML-LMSD35</b>                   | If it has a dc Power Supply Module in slots J and K, it is designated model ML-LMSD35.                                                                                                                                                                                                                                                                                                       |
| <b>Optional modules</b>                  | Optional modules listed in subsection 2.7.1.3 can be added to the card file as required.                                                                                                                                                                                                                                                                                                     |
| <b>Logic Manager Module</b>              | The Logic Manager Module has UCN drop cable connectors mounted on the front panel.                                                                                                                                                                                                                                                                                                           |
| <b>Optional module address selection</b> | DIP switch settings on optional modules that are added to the card file must be set at the time of installation. As two option slots are available, the optional modules must be set to a backplane address of 2 or 3. If two optional modules are implemented, one must be set to address 2 and the other set to address 3. The Logic Manager Module is preset at the factory to address 1. |

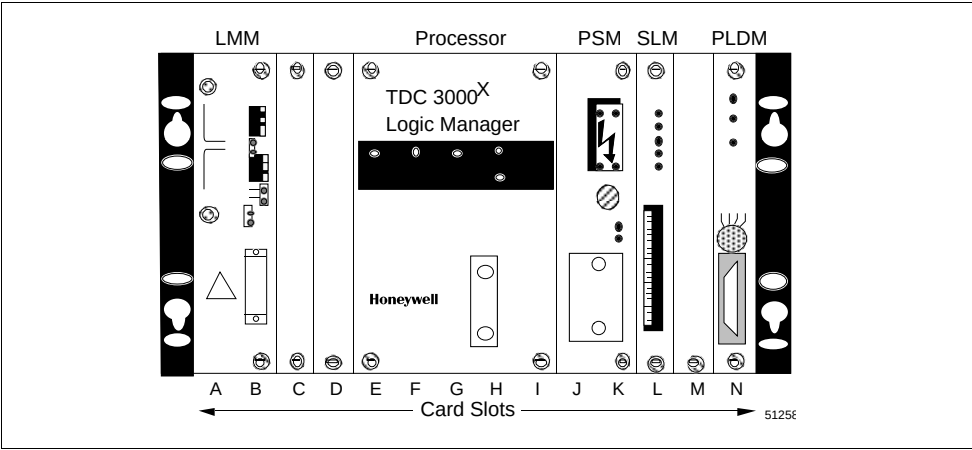


### 2.7.1.2 Redundant Processor Card File

**Overview**

The basic redundant Logic Manager subsystem has two processor card files with standard modules installed as shown by a front view of the card file in Figure 2-8.

Figure 2-8 Redundant Processor Card File Front View



*Continued on next page*

## 2.7.1.2 Redundant Processor Card File, Continued

**Module assignment list** Table 2-4 defines the module slot assignments for the redundant Processor card file.

Table 2-4 Redundant Processor Card File Slot Assignments

| Card Slot | Module Type                           | Model Number | Notes      |
|-----------|---------------------------------------|--------------|------------|
| A and B   | Logic Manager Module (LMM)            | 620-0071     | 6, 8       |
| C         | Redundancy Control Module (RCM)       | 620-0059     | 8          |
| D         | Blank Metal Cover Plate               | 629-4004     | 1, 2, 4, 9 |
| E         | 24 K Memory Module                    | 620-0024     | 8          |
| E         | 4 K x 4 K Register Module             | 620-0056     |            |
| G         | System Control Module                 | 620-0054     |            |
| H         | Processor Module                      | 620-0080     |            |
| I         | 2 K I/O Control Module                | 620-0085     |            |
| J and K   | Power Supply Module (PSM) 115/230 Vac | 620-0036     | 6          |
| J and K   | Power Supply Module (PSM) 24 Vdc      | 620-0047     | 6          |
| L         | Serial Link Module (SLM)              | 621-9939     | 8          |
| M         | Blank Plastic Cover Plate             | 621-9900     | 2, 3, 5    |
| N         | Parallel Link Driver Module (PLDM)    | 620-0088     | 7, 8       |

**Notes:**

1. This slot is for optional modules.
2. Check DIP switch settings for optional modules.
3. This slot is for the optional second model 621-9939 Serial Link Module.
4. Install the model 629-4004 blank metal cover plate if this slot is empty.
5. Install the model 621-9900 blank plastic cover plate if this slot is empty.
6. The module is double width.
7. The model 620-0088 Parallel Link Driver Module is required even when serial I/O is used exclusively because the PLDM performs some subsystem control functions in addition to supporting parallel I/O.
8. This module has its configuration DIP switches preset at the factory. See *Logic Manager Installation* manual for setting UCN address into DIP switch SW2.
9. The optional model 620-0027, 8 K Memory Module may be installed in this slot.

*Continued on next page*

## 2.7.1.2 Redundant Processor Card File, Continued

|                                      |                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Model ML-MLRA35</b>               | The card file is designated a model ML-LMRA35, if it has an ac Power Supply Module in slots J and K.                                                                                                                                                                                                                                                           |
| <b>Model ML-LMRA35</b>               | If a dc Power Supply Module is installed in slots J and K in the card file, the designation is model ML-LMRD35.                                                                                                                                                                                                                                                |
| <b>Redundancy Data Control cable</b> | A three meters (10 feet) long model 628-5000 Redundancy Data Control cable is also included to connect the Redundancy Control Modules in each card file. Optional modules listed in subsection 2.2.1.3 can be added to the card file as required.                                                                                                              |
| <b>Logic Manager Module</b>          | The Logic Manager Module has UCN drop cable connectors mounted on the front panel.                                                                                                                                                                                                                                                                             |
| <b>Module address selection</b>      | The Logic Manager Module is factory preset to address 1 at the factory. The Redundancy Control Module is preset to address 0 at the factory. DIP switch settings on optional modules added to the card file must be set at the time of installation. As only one optional slot is available, the optional module must be set to a backplane address of 2 or 3. |
| <b>ATTENTION</b>                     | <b>ATTENTION—Control switchover cables must be added per Figure 2-6.</b>                                                                                                                                                                                                                                                                                       |

## 2.7.1.3 Processor Card File Optional Modules

### Optional module assignment list

The optional modules listed in Table 2-5 can be installed in the optional card slots, C and D, of the nonredundant processor card file or the optional card slot D of the redundant processor card file as shown.

Table 2-5 Processor Card File Optional Modules Assignments

| Module Type                          | Model Number | Redundant Card Slot | Nonredundant Card Slot |
|--------------------------------------|--------------|---------------------|------------------------|
| 8 K Memory                           | 620-0027     | D                   | D                      |
| Control Network                      | 620-0038     | D                   | C or D                 |
| Modbus/RTU Computer Interface (CIM)  | 620-0043     | D                   | C or D                 |
| ABC Data Collection (DCM)            | 620-0048     | D                   | C or D                 |
| Dual Port ABC Data Collection        | 620-0052     | D                   | C or D                 |
| Operator Panel Interface             | AHM-4000-1   |                     | C or D                 |
| Operator Interface (Extended Memory) | AHM-4000-2   |                     | C or D                 |

### I/O communications modules

If an optional I/O communications module is used in a redundant Serial I/O subsystem, each of the redundant Processor card files must contain an identical optional module. Each optional module port must be linked with an additional model 620-9928 Serial Link Selector Module in a Serial I/O card file.

### Model 620-0048 or 620-0043 configuration

To utilize a model 620-0048 ABC Data Collection Module (DCM) or model 620-0043 Modbus Computer Interface Module (CIM) in a redundant Logic Manager subsystem, a DCM or CIM must be installed in both Processor card files and connected to an additional SLS in an I/O card file.

### Model 620-0052 configuration

To utilize a model 620-0052 Dual Port ABC Data Collection Module, an identical DCM must be installed in both Processor card files and connected to two additional SLS Modules in an I/O card file.

## 2.7.2 I/O Card File Assembly

---

**Description**

The Logic Manager I/O card file has two physical sizes, the full-size 19-inch and the half-size 12-inch chassis. Both sizes can be panel mounted, but only the 19-inch size is cabinet mounted. The card file includes the chassis, backplane, I/O interface modules, process I/O modules, a Power Supply Module, and front cover plates for the unused module slots.

The 19-inch I/O card file has fourteen module slots, identified as A through N, with slot A on the left side when facing the card file.

The half-size “special” 12-inch card file has eight module slots, identified as A through H. A maximum of twelve and six slots, respectively, are available for process I/O modules.

The I/O modules are vertically positioned in the card file with the component side toward the left. The backplane connectors are offset to prevent inserting a module upside down.

---

**Two configurations**

The Logic Manager I/O card file has two configurations, serial and parallel. Standard modules are required for both configurations. The other modules are optional and depend on the subsystem requirements. In either case, slot positioning of the modules is defined by the configuration.

---

**I/O card files defined**

The sections that follow define the different configuration types of Logic Manager I/O card files.

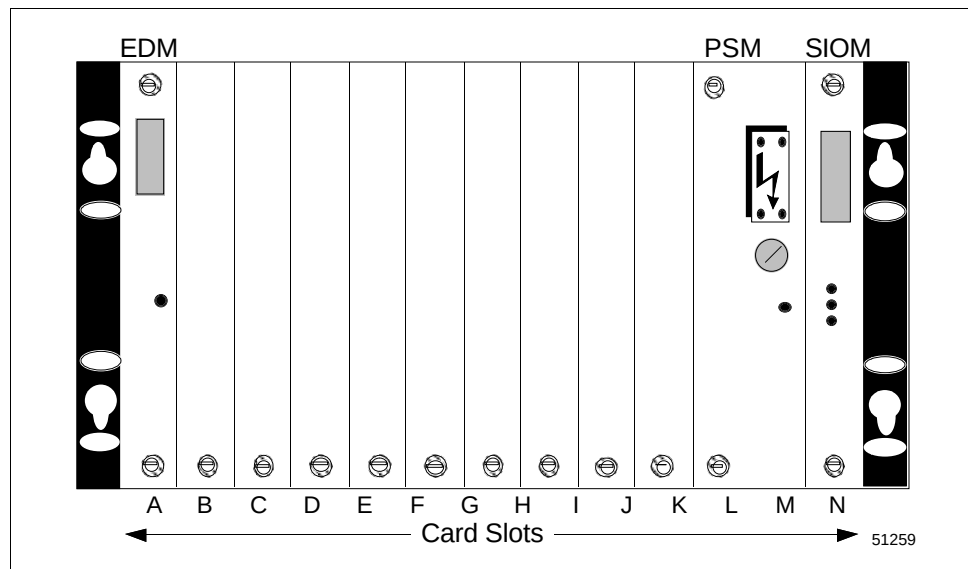
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## 2.7.2.1 Nonredundant Serial I/O Card File

### Full-size card file view

The basic full-size nonredundant Serial I/O card file contains the standard modules shown below and is designated model ML-LIOS01. Figure 2-9 is a view of the front of the card file.

Figure 2-9 Full-Size Nonredundant Serial I/O Card File Front View



*Continued on next page*

## 2.7.2.1 Nonredundant Serial I/O Card File, Continued

### Full-size slot assignment list

Table 2-6 defines the module slot assignments for the full-size nonredundant Serial I/O card file.

Table 2-6 Full-Size Nonredundant Serial I/O Card File Slot Assignments

| Card Slot                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Module Type                                   | Model Number | Notes      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|------------|
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3       |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| D                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| H                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| J                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| L                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4, 6 |
| L and M                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 5       |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2          |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2          |
| N                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Serial I/O Module (SIOM)                      | 621-9940     | 1, 3       |
| <b>Notes:</b><br><ol style="list-style-type: none"><li>1. This is a standard module slot.</li><li>2. This is an optional module slot.</li><li>3. Check and alter the switch settings, if applicable.</li><li>4. Install the model 629-9900 blank plastic cover plate if this slot is empty.</li><li>5. The module is double width.</li><li>6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.</li></ol> |                                               |              |            |

*Continued on next page*

## 2.7.2.1 Nonredundant Serial I/O Card File, Continued

### Half-size slot assignment list

The basic half-size nonredundant Serial I/O card file contains the standard modules shown in Table 2-7 and is designated model 621-9991. It is not a standard product, but a “special card file” and described for reference only.

Table 2-7 Half-Size Nonredundant Serial I/O Card File Slot Assignments

| Card Slot                                                                                                                                                                                                                                                                                                                                                                                                                                            | Module Type                                   | Model Number | Notes   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|---------|
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3    |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4 |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4 |
| D                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4 |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4 |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4 |
| F and G                                                                                                                                                                                                                                                                                                                                                                                                                                              | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 5    |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2       |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2       |
| H                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Serial I/O Module (SIOM)                      | 621-9940     | 1, 3    |
| Notes:<br><ol style="list-style-type: none"><li>1. This is a standard module slot.</li><li>2. This is an optional module slot.</li><li>3. Check and alter the switch settings, if applicable.</li><li>4. Install the model 629-9900 blank plastic cover plate if this slot is empty.</li><li>5. The module is double width.</li><li>6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.</li></ol> |                                               |              |         |

### Low and high level module slot assignment

All low level signal process I/O modules (analog, TTL, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

*Continued on next page*



## 2.7.2.1 Nonredundant Serial I/O Card File, Continued

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### Address assignment

All I/O card slots should be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and configured by the appropriate Serial I/O Module configuration DIP switches.

Each empty slot should be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion.

The only card slots to be assigned an address of zero are those card slots

- occupied by the left half of a double width module or
- both card slots occupied by a double width module that requires no addresses.

---

### Addresses 2040 through 2047

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial I/O module slot in a subsystem containing serial I/O.

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### Address assignments for serial diagnostics

NOTE—If serial I/O expanded diagnostics are being used, addresses 1992 through 2039 are also reserved for this feature. Expanded serial I/O diagnostics can be enabled if the Version/Revision of the SLM (621-9939) is at V.R.2.3 or higher and the Version/Revision of the Control Processor Module (620-0080) is at V.R.3.1 or higher.

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### Plastic cover plate for an empty slot

All empty I/O card slots must be covered with a model 621-9900 plastic cover plate.

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### Power Supply Module selection

The Power Supply Module selected for the card file must be compatible with the power input, 115/230 Vac or 24 Vdc, and must supply enough current to the card file to satisfy the total load requirements. Refer to the I/O Power Load table in the *Logic Manager Installation* manual.

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### MiniCOP Module requirements

All subsystems configured with a MiniCOP Module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file optional card slots.

The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP Module without a DCM.

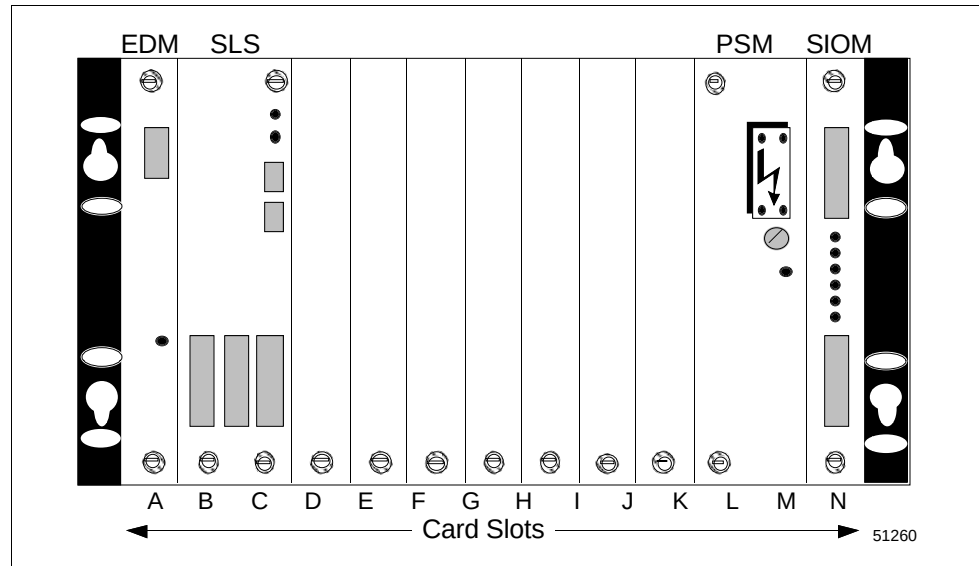
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## 2.7.2.2 Redundant Serial I/O Card File

### Full-size card file view

The basic redundant serial I/O card file contains the standard modules shown in Figure 2-10. Figure 2-10 is a view of the front of the card file.

Figure 2-10 Redundant Serial I/O Card File Front View



*Continued on next page*

## 2.7.2.2 Redundant Serial I/O Card File, Continued

**Card file assignment list** Table 2-8 defines the module slot assignments for the redundant serial I/O card file.

Table 2-8 Redundant Serial I/O Card File Slot Assignments

| Card Slot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Module Type                                   | Model Number | Notes      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|------------|
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3       |
| B and C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Serial Link selector Module (SLS)             | 621-9928R    | 2, 3, 4    |
| D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| J                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5, 7 |
| L and M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 6       |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2          |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2          |
| N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Serial I/O Module (SIOM)                      | 621-9938R    | 1, 3, 8    |
| Notes:<br>1. This is a standard module slot.<br>2. This is an optional module slot.<br>3. Check and alter the switch settings, if applicable.<br>4. At least one SLS is required for a redundant subsystem. Additional I/O card files can contain a SLS. Otherwise, the card slot can contain any valid model 621-xxxx I/O module.<br>5. Install the model 629-9900 blank plastic cover plate if this slot is empty.<br>6. The module is double width.<br>7. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.<br>8. The model 621-9938R SIOM has dual ports for processor redundancy. I/O card module models that have an "R" suffix can be removed while power is applied to the I/O card file. |                                               |              |            |

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## 2.7.2.2 Redundant Serial I/O Card File, Continued

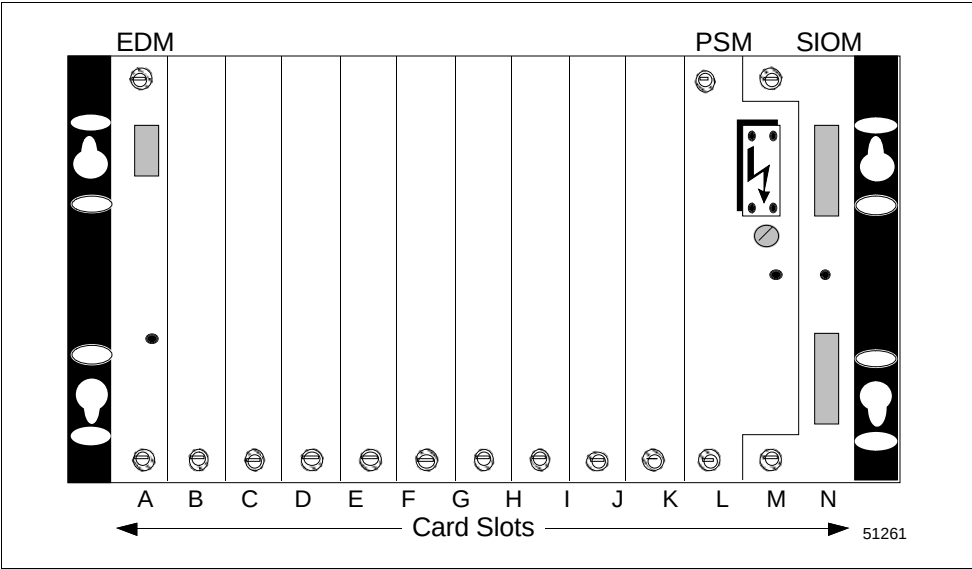
|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Low and high level module slot assignment</b>  | <p>All low level signal process I/O modules (analog, TTL, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal processor I/O modules must be grouped together, starting at the rightmost slot and proceeding left.</p>                                                                                                                                                                                                                                                                                                                                                           |
| <b>Address assignment</b>                         | <p>All I/O card slots should be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the card slot and configured by the appropriate Serial I/O Module configuration DIP switches.</p> <p>Each empty card slot should be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion.</p> <p>The only card slots to be assigned an address of zero are those card slots</p> <ul style="list-style-type: none"><li>• occupied by the left half of a double width module or</li><li>• both card slots occupied by a double width module that requires no addresses.</li></ul> |
| <b>Address 2040 through 2047</b>                  | <p>I/O addresses 2040 through 2047 must not be assigned to any parallel or serial card slot of a subsystem containing serial I/O.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Address assignments for serial diagnostics</b> | <p>NOTE—If serial I/O expanded diagnostics are being used, addresses 1992 through 2039 are also reserved for this feature. Expanded serial I/O diagnostics can be enabled if the Version/Revision of the SLM (621-9939) is at V.R.2.3 or higher and the Version/Revision of the Control Processor Module (620-0080) is at V.R.3.1 or higher.</p>                                                                                                                                                                                                                                                                                             |
| <b>Plastic cover plate for an empty slot</b>      | <p>All empty I/O card slots must be covered with a model 621-9900 plastic cover plate.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Power Supply Module selection</b>              | <p>The Power Supply Module selected for the card file must be compatible with the power input, 115/230 Vac or 24 Vdc, and must supply enough current to the card file to satisfy the total load requirements. Refer to the I/O Power Load table in the <i>Logic Manager Installation</i> manual.</p>                                                                                                                                                                                                                                                                                                                                         |
| <b>MiniCOP Module requirements</b>                | <p>All subsystems configured with a MiniCOP Module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file optional card slots.</p> <p>The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP Module without a DCM.</p>                                                                                                                                                                         |

### 2.7.2.3 Nonredundant Parallel I/O Card File

**Full-size parallel I/O card file view**

The basic full-size nonredundant parallel I/O Card File contains the standard modules as shown in Figure 2-11 and is designated model ML-LIOP01. Figure 2-11 for a front view of the card file.

Figure 2-11 Nonredundant Parallel I/O Card File Front View



*Continued on next page*

## 2.7.2.3 Nonredundant Parallel I/O Card File, Continued

### Full-size parallel I/O card file assignment list

Table 2-9 defines the module slot assignments for the nonredundant full-size parallel I/O card file.

Table 2-9 Nonredundant Parallel I/O Card File Slot Assignments

| Card Slot | Module Type                                   | Model Number | Notes      |
|-----------|-----------------------------------------------|--------------|------------|
| A         | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3       |
| B         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| C         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| D         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| E         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| F         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| G         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| H         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| I         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 5    |
| J         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| K         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| L         | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4, 6 |
| L and M   | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 5       |
| M         | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2          |
| M         | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2          |
| N         | Parallel I/O Module (PIOM)                    | 621-9937     | 1, 3       |

Notes:

1. This is a standard module slot.
2. This is an optional module slot.
3. Check and alter the switch settings, if applicable.
4. Install the model 629-9900 blank plastic cover plate if this slot is empty.
5. The module is double width.
6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.

*Continued on next page*

## 2.7.2.3 Nonredundant Parallel I/O Card File, Continued

### Half-size parallel I/O card file assignment list

The basic half-size nonredundant parallel I/O card file contains the standard modules as shown in Table 2-10 and is designated model 621-9991. It is not a standard product, but a “special I/O card file” and is described for reference only.

Table 2-10 Half-Size Nonredundant Parallel I/O Card File Slot Assignments

| Card Slot                                                                                                                                                                                                                                                                                                                                                                                                                                            | Module Type                                   | Model Number | Notes      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|------------|
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3       |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| D                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4, 6 |
| F and G                                                                                                                                                                                                                                                                                                                                                                                                                                              | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 5       |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2          |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2          |
| H                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Parallel I/O Module (PIOM)                    | 621-9937     | 1, 3       |
| Notes:<br><ol style="list-style-type: none"><li>1. This is a standard module slot.</li><li>2. This is an optional module slot.</li><li>3. Check and alter the switch settings, if applicable.</li><li>4. Install the model 629-9900 blank plastic cover plate if this slot is empty.</li><li>5. The module is double width.</li><li>6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.</li></ol> |                                               |              |            |

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### 2.7.2.3 Nonredundant Parallel I/O Card File, Continued

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Low and high level module slot assignment</b>  | All low level signal process I/O modules (analog, TTL, etc.) must be grouped together, starting at the leftmost card slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost card slot and proceeding left.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Address assignment</b>                         | <p>All I/O card slots should be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the card slot and configured by the appropriate Parallel I/O Module configuration DIP switches.</p> <p>Each empty card slot should be assigned a quantity of 16 addresses to accommodate spares that allows for future expansion.</p> <p>The only card slots to be assigned an address of zero are those card slots:</p> <ul style="list-style-type: none"><li>• occupied by the left half of a double width module</li><li>• both card slots occupied by a double width module that requires no addresses.</li></ul> |
| <b>Addresses 2040 through 2047</b>                | I/O addresses 2040 through 2047 must not be assigned to any parallel or serial I/O card slot of a subsystem containing serial I/O.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Address assignments for serial diagnostics</b> | NOTE—If serial I/O expanded diagnostics are being used, addresses 1992 through 2039 are also reserved for this feature. Expanded serial I/O diagnostics can be enabled if the Version/Revision of the SLM (621-9939) is at V.R.2.3 or higher and the Version/Revision of the Control Processor Module (620-0080) is at V.R.3.1 or higher.                                                                                                                                                                                                                                                                                                      |
| <b>Plastic cover plate for an empty slot</b>      | All empty I/O card slots must be covered with a model 621-9900 plastic cover plate. nonredundant Parallel I/O:empty card slot:plastic cover plate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Power Supply Module selection</b>              | The Power Supply Module selected for the card file must be compatible with the power input, 115/230 Vac or 24 Vdc, and must supply enough current to the card file to satisfy the total load requirements. Refer to the I/O Power Load table in the <i>Logic Manager Installation</i> manual.                                                                                                                                                                                                                                                                                                                                                  |
| <b>MiniCOP Module requirements</b>                | <p>All subsystems configured with a MiniCOP Module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file optional card slots.</p> <p>The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP Module without a DCM.</p>                                                                                                                                                                           |

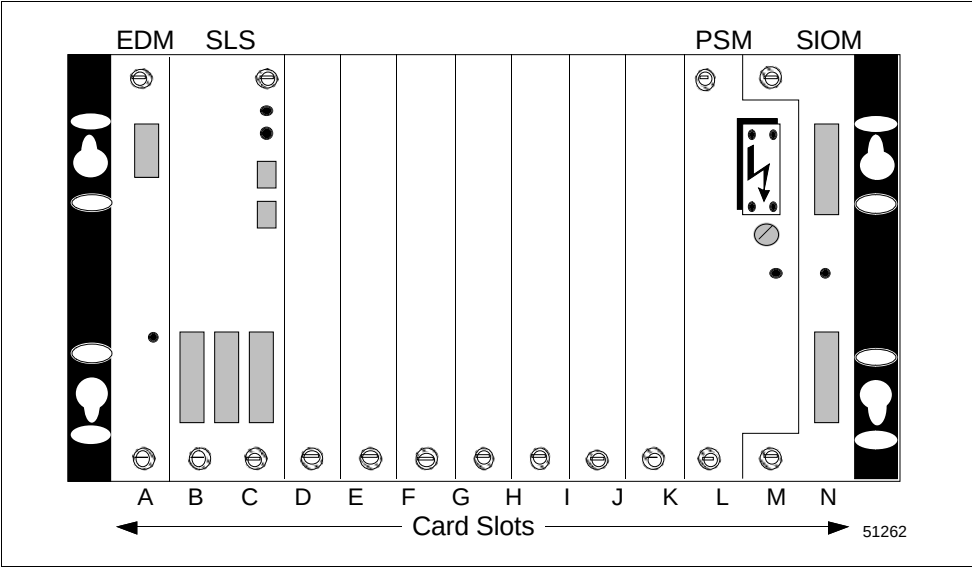


### 2.7.2.4 Redundant Parallel I/O Card File

**Redundant parallel I/O  
card file view**

The basic redundant Parallel I/O card file contains the standard modules as shown in Figure 2-12. Figure 2-12 is a front view of the card file. It is not a standard product, but a “special I/O card file” and is described for reference only.

Figure 2-12 Redundant Parallel I/O Card File Front View



*Continued on next page*

## 2.7.2.4 Redundant Parallel I/O Card File, Continued

### Redundant parallel I/O card file assignment list

Table 2-11 defines the module slot assignments for the redundant parallel I/O card file.

Table 2-11 Redundant Parallel I/O Card File Slot Assignments

| Card Slot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Module Type                                   | Model Number | Notes      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|------------|
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enhanced Diagnostic Module (EDM)              | 621-0021R    | 1, 3       |
| B and C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Serial Link Selector Module (SLS)             | 621-9928R    | 2, 3, 5, 7 |
| D and E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Parallel Bus Selector Module (PBSM)           | 621-9929     | 2, 3, 5, 7 |
| F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| J                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4    |
| L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Any valid 621-xxxx I/O Module                 |              | 2, 3, 4, 6 |
| L and M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Power Supply Module (PSM)<br>115/230 Vac 15 A | 621-9933     | 2, 5       |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Power Supply Module (PSM)<br>115/230 Vac 8 A  | 621-9934     | 2          |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Power Supply Module (PSM) 24 Vdc 8 A          | 621-9932     | 2          |
| N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Parallel I/O Module (PIOM)                    | 621-9937     | 1, 3       |
| Notes:<br>1. This is a standard module slot.<br>2. This is an optional module slot.<br>3. Check and alter the switch settings, if applicable.<br>4. Install the model 629-9900 blank plastic cover plate if this slot is empty.<br>5. The module is double width.<br>6. Not available for an I/O module if occupied by the double width model 621-9933 Power Supply Module.<br>7. At least one SLS Module and one PBSM is required for a redundant subsystem. Additional I/O card files can contain an SLS; otherwise, the card slots can contain any valid 621-xxxx I/O module. |                                               |              |            |

*Continued on next page*

## 2.7.2.4 Redundant Parallel I/O Card File, Continued

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Low and high level module slot assignment</b>  | <p>All low level signal process I/O modules (analog, TTL, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal process I/O modules must be grouped together, starting at the rightmost slot and proceeding left.</p>                                                                                                                                                                                                                                                                                                                                                               |
| <b>Address assignment</b>                         | <p>All I/O card slots should be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the card slot and configured by the appropriate Parallel I/O Module configuration DIP switches.</p> <p>Each empty card slot should be assigned a quantity of 16 addresses to accommodate spares, allowing for future expansion.</p> <p>The only card slots to be assigned an address of zero are those card slots</p> <ul style="list-style-type: none"><li>• occupied by the left half of a double width module or</li><li>• both card slots occupied by a double width module that requires no addresses.</li></ul> |
| <b>Addresses 2040 through 2047</b>                | <p>I/O addresses 2040 through 2047 must not be assigned to any parallel or serial card slot of a subsystem containing serial I/O.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Address assignments for serial diagnostics</b> | <p>NOTE—If serial I/O expanded diagnostics are being used, addresses 1992 through 2039 are also reserved for this feature. Expanded serial I/O diagnostics can be enabled if the Version/Revision of the SLM (621-9939) is at V.R.2.3 or higher and the Version/Revision of the Control Processor Module (620-0080) is at V.R.3.1 or higher.</p>                                                                                                                                                                                                                                                                                               |
| <b>Plastic cover plate for an empty card slot</b> | <p>All empty I/O card slots must be covered with a model 621-9900 plastic cover plate.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Power Supply Module selection</b>              | <p>The Power Supply Module selected for the card file must be compatible with the power input, 115/230 Vac or 24 Vdc, and must supply enough current to the card file to satisfy the total load requirements. Refer to the I/O Power Load table in the <i>Logic Manager Installation</i> manual.</p>                                                                                                                                                                                                                                                                                                                                           |
| <b>MiniCOP Module requirements</b>                | <p>All subsystems configured with a MiniCOP Module (double width model 627-1002R or triple width model 627-1003R) must contain a model 620-0048/0052 DCM (ABC Data Collection Module) installed in the Processor card file option card slots.</p> <p>The only exception to this rule is when the installer is fully knowledgeable of the complexity and the amount of relay ladder logic required to access information from the MiniCOP Module without a DCM.</p>                                                                                                                                                                             |

## 2.8 Process Card File Modules

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|                     |                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b> | The following subsections briefly describe the modules that are installed in the Process card file. Some of the modules are optional, depending on processor redundancy and the I/O configuration. |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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### 2.8.1 Logic Manager Card Module

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|                                |                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                | The model 620-0071 Logic Manager Module (LMM) is a four board, double wide module that must be installed in slots A and B at the left front side of the Processor card file.                                                                                                                                                                                                                                |
| <b>Description</b>             | The LMM obtains power and Logic Manager subsystem data signals through its backplane connectors. The LMM supports redundancy in the Logic Manager subsystem by monitoring inputs from the Redundancy Control Module (RCM) that activates one of two redundant LMMs connected to the Universal Control Network (UCN).                                                                                        |
| <b>Battery backup security</b> | The LMM's database information is secured by battery backup using the LMM's internal lithium battery which provides a minimum of six months data retention.                                                                                                                                                                                                                                                 |
| <b>Memory backup power</b>     | The LMM is shipped with a size 2/3 A lithium battery, model 629-3004, installed to provide backup power for the internal memory. The battery is accessed through a cover at the lower right of the module's front panel. An insulating wafer prevents battery discharge during shipment/storage and must be removed before the subsystem is operated. The LMM uses a pull strap for removal of the battery. |

---

### 2.8.2 Redundancy Control Module

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|                    |                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | The model 620-0059 Redundancy Control Module (RCMs) is the primary component for redundancy control and is responsible for transferring I/O data from the primary processor to the secondary processor. The module also monitors the status of the Logic Manager's Processor Modules. If this option is implemented, an RCM is installed in slot C of both Processor card files. |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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### 2.8.3 8 K Memory Module

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**Description**

The model 620-0027, 8 K Memory Module is an optional RAM addition to the standard 24 K Memory Module and is installed in slot D of the Processor card file. It contains 8 K words of memory and requires a mandatory starting memory address of 24 K. The total memory available to the processor is 32 K words when the module is installed, which is the maximum memory allowed.

---

### 2.8.4 24 K Memory Module

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**Description**

The model 620-0024, 24 K Memory Module is installed in slot E of the Processor card file. It contains a total of 24 K words of memory and requires a mandatory starting memory address of zero (0). The Memory Module stores the user's control program.

---

## 2.8.5 4 K x 4 K Register Module

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>              | The model 620-0056, 4 K x 4 K Register Module contains the Logic Manager's I/O Data Table and is installed in slot F of the Processor card file.                                                                                                                                                                                                                                                                            |
| <b>Register tables</b>       | The Data Table is divided into three areas; the I/O Status Table, System Status Table, and Register Table. The I/O Status Table contains 4096 (4 K) continuous single bit storage locations which have a starting address of zero (0). In the 4 K x 4 K Register Module, 2048 (2 K) locations are available for real I/O status or internal coils, and an additional 2048 (2 K) locations are available for internal coils. |
| <b>Register descriptions</b> | Timer and counter preset and accumulated values are stored in the register area. Other numerical data may be stored in the bit or register area. The register area is 16-bits wide, plus a sign bit, and is comprised of 4096 (4 K) registers. The 17th bit is normally used to indicate the sign of the data contained in the register. In certain operations it will indicate overflow conditions.                        |
| <b>System Status Table</b>   | The System Status Table stores processor diagnostic information that can be accessed by the Loader/Terminal or by the PULL instruction in the control program. The System Status Table consists of memory locations that are 8-bits wide. The table is divided into three sections; System Diagnostics, System Hardware Status, and System Identification.                                                                  |
| <b>Battery backup</b>        | The Data Table is battery backed by the Power Supply Module's battery to ensure data retention during subsystem power outages.                                                                                                                                                                                                                                                                                              |

## 2.8.6 System Control Module

|                    |                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | <p>The model 620-0054 System Control Module (SCM) is a functional extension of the Processor Module. It coordinates the interaction of all the modules that compose the Logic Manager's processor. The SCM also contains a single bit processor that executes all relay logic in the control program.</p> <p>The SCM is installed in slot G of the Processor card file.</p> |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2.8.7 Processor Module

### Description

The Processor Module (PM) executes the program stored in the Memory Module(s) and handles arithmetic computation and data movement instructions. The Processor Module is installed in slot H of the Processor card file.

A port connector, labeled **Loader/Terminal**, connects to the model 623-60 MS-DOS Loader and is protected by a removable front cover plate.

## 2.8.8 I/O Control Module

### Description

The model 620-0085 Input/Output Control Module (IOCM) coordinates communication and formats the data between the processor and the I/O subsystem. The IOCM also monitors individual I/O module fault diagnostics. The IOCM works in conjunction with the Parallel Link Driver Module (PLDM) and/or the Serial Link Module (SLM) to control data flow to and from the I/O subsystem.

The IOCM is installed in slot I of the Processor card file.

## 2.8.9 Power Supply Module

### Overview

The Power Supply Module (PSM) provides power to operate the Processor card file. The dual-card PSM is installed in slots J and K in the Processor card file. A choice of two models are available and are listed in Table 2-12.

Table 2-12 Process Card File Power Supply Modules

| Model Number | Input Voltage | Amperage            |
|--------------|---------------|---------------------|
| 620-0036     | 115/230 Vac   | 15 A (5 Vdc Output) |
| 620-0047     | 24 Vdc        | 15 A (5 Vdc Output) |

### Battery backup

A battery compartment, accessible through a PSM front cover, contains a D size lithium battery that provides backup power for the memory and register modules in the Logic Manager Processor card file. The PSM is shipped with the battery installed.

## 2.8.9.1 115/230 Vac Power Supply Module

|                                         |                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview                                | The model 620-0036 Power Supply Module provides a maximum of 15 A of $\pm 5$ Vdc power for use by the Logic Manager Processor card file modules. It also supplies 300 mA of +12 Vdc for optional modules that require the voltage.                                                                                                                     |
| Input voltage selection and connections | <p>Terminals on the front panel of the module, labeled <b>A</b> (L1 or Line), <b>B</b> (L2 or Common), and <b>GND</b> (Ground), provide ac power wiring connection points.</p> <p>The module operates with an input voltage of 115 Vac or 230 Vac. A shorting board, located inside the side cover at the top edge, selects the proper ac voltage.</p> |
| Fused power                             | The module contains a front accessible fuse holder. It is shipped with a 2 A slow blow barrel fuse installed for 115 Vac operation. A 1 A slow blow fuse for 230 Vac operation is also included with the module.                                                                                                                                       |
| <b>WARNING</b>                          | <b>WARNING</b> —This slow blow barrel fuse should not be used to remove ac power from the module. Physical damage to the fuse holder may result.                                                                                                                                                                                                       |

## 2.8.9.2 24 Vdc Power Supply Module

|                        |                                                                                                                                                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview               | The model 620-0047 Power Supply Module provides a maximum of 15 A of $\pm 5$ Vdc power for use by the Logic Manager card file modules. It also supplies 300 mA of +12 Vdc for optional modules requiring the voltage. |
| Input power connection | Input terminals on the front of the module are labeled +, –, and <b>GND</b> . The module operates with a nominal input voltage of 20 to 25 Vdc.                                                                       |
| Fused power            | The module has a front accessible fuse holder and is shipped with an 8 A slow blow fuse installed.                                                                                                                    |
| <b>WARNING</b>         | <b>WARNING</b> —This slow blow barrel fuse should not be used to remove ac power from the module. Physical damage to the fuse holder may result.                                                                      |



## 2.8.10 Serial Link Module

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                                              | The model 621-9939 Serial Link Module (SLM) controls serial I/O data transfer to and from the I/O. It is the intermediary between the Logic Manager's processor and the SIOM(s).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>SLM operation</b>                                         | The SLM collects input data from the SIOM(s). The processor reads data during its input status scan. Then, depending on its program instructions, the processor issues output commands to the SLM, which sends them to the SIOM(s). The SLM has its own microprocessor and data table and operates asynchronously to the processor.                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Addresses 2040 through 2047 contain serial I/O status</b> | Addresses 2040 through 2047 of the SLM provide serial I/O status information to the processor. This information includes the SIOM off-line flag, status of SIOM(s) on the redundant channel, the starting address of the off-line SIOM(s), and the number of SIOMs on-line for each channel.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Address assignments for serial diagnostics</b>            | NOTE—If serial I/O expanded diagnostics are being used, addresses 1992 through 2039 are also reserved for this feature. Expanded serial I/O diagnostics can be enabled if the Version/Revision of the SLM (621-9939) is at V.R.2.3 or higher and the Version/Revision of the Control Processor Module (620-0080) is at V.R.3.1 or higher.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Two serial channels supported</b>                         | Each SLM can control two serial channels; therefore, a processor card file with one SLM installed in slot L, supports a maximum of two channels. A processor card file with one SLM installed in slot L and one SLM installed in slot M supports a maximum of four channels. Each channel operates independently of the other serial channels in the Logic Manager subsystem.                                                                                                                                                                                                                                                                                                                           |
| <b>2040 I/O points supported</b>                             | A channel accommodates 1 to 16 drops (SIOMs), or a maximum of 2040 I/O points. However, the total number of I/O points must be within the I/O capacity of the Logic Manager subsystem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Serial I/O link terminating resistors</b>                 | <p>A kit of terminating resistors is provided with the SLM. These resistors are used for terminating the multidrop serial I/O link and for terminating the control switchover link in a redundant configuration. If an SLM serial I/O communications channel is at one end of the communications link, a resistor must be installed for that channel. Similarly, if an SLM is at the end of a redundant subsystem control switchover link, another 100 ohm terminating resistor must be installed. Refer to Figures 2-3, 2-4, and 2-6.</p> <p>Additional information on wiring and installation of terminating resistors can be found in Section 5 of the <i>Logic Manager Installation</i> manual.</p> |

*Continued on next page*

## 2.8.10 Serial Link Module, Continued

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|                                   |                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Front panel terminal block</b> | The front panel of the SLM is permanently attached and contains a 19-contact terminal block. The terminal block includes a two piece connector to which channels 1 and 2 and RCM control are wired. The terminal block is removable so that channel wiring can be easily disconnected from the module if the module requires servicing. |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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## 2.8.11 Parallel Link Driver Driver Module

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|                              |                                                                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | The model 620-0088 Parallel Link Driver Module (PLDM) operates with the I/O Control Module (IOCM) to coordinate communications between the parallel or serial I/O and the Logic Manager processor. The PLDM is installed in slot N of the processor card file. |
| <b>Mode Keyswitch use</b>    | The PLDM controls the operation of the processor with a 4-position Mode Keyswitch. The positions are labeled; <b>PROGRAM</b> , <b>DISABLE</b> , <b>RUN/PROG</b> , and <b>RUN</b> .                                                                             |
| <b>Front panel connector</b> | A D-type connector at the PLDM front panel interfaces to the PIOM in the first parallel I/O card file.                                                                                                                                                         |

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## 2.9 I/O Card File Modules

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|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b> | The serial and parallel I/O card files use modules that are standard or optionally installed in both types I/O card files. In particular, the Power Supply Module, discrete, analog, and special function process I/O modules are shared. See subsections 2.7.2.1 through 2.7.2.4 for the acceptable Power Supply Modules and I/O interface modules. In subsections 2.10.1 through 2.10.5, the discrete, analog, and special function process I/O modules are referenced. |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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## 2.9.1 Enhanced Diagnostic Module

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|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>        | The model 621-0021R Enhanced Diagnostic Module (EDM) provides the Logic Manager subsystem with additional subsystem monitoring capability, such as the detection of a problem with data and control lines on the I/O bus. It is installed in slot A of both the serial and parallel I/O card files. The EDM can also detect a scan loss with its watchdog timer. When detecting a subsystem malfunction, the EDM deenergizes an internal relay that can be used to remove power from the I/O subsystem's process output modules. The relay provides an effective method of halting the controlled process. Connection to the relay is at a front panel mounted terminal block. |
| <b>Address assignment</b> | The Enhanced Diagnostic Module must be assigned eight I/O addresses and must always be installed in slot A of a serial or parallel I/O card file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ATTENTION</b>          | ATTENTION—If redundant Logic Managers are used, the delay time must be set to 4.5 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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## 2.9.2 Serial Link Selector Module

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|                               |                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>               | The model 621-9928R Serial Link Selector Module (SLS) is a double wide module that is installed in the I/O slots of a serial or parallel I/O card file in a redundant Logic Manager. The SLS serves as a general purpose switching device and switches serial communication devices, such as the MS-DOS Loader, shared by the redundant Logic Manager processors. |
| <b>Front panel connectors</b> | The SLS's only connections to the Serial I/O card file backplane are the +5 Vdc supply, ground, and the power fail signal. All other connections are made through the front panel connectors.                                                                                                                                                                     |
| <b>RCM control</b>            | The state of the SLS's bus selector circuitry is controlled by the Redundancy Control Modules (RCMs), which send switch select output signals to the SLS through the two 3-contact front panel connectors.                                                                                                                                                        |

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## 2.9.3 Parallel Bus Selector Module

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                            | <p>The model 621-9929 Parallel Bus Selector Module (PBSM) is a double wide module installed in slots D and E of one I/O card file per parallel I/O subsystem in a redundant configuration only. A model 621-9928R Serial Link Selector (SLS) must also be installed in slots B and C of the same card file. This is a “special I/O subsystem” because redundant parallel I/O subsystems are not recommended. The PBSM connects the Logic Manager’s primary processor to the parallel I/O subsystem and switches the parallel I/O subsystem, upon request, from the primary (lead) to the Logic Manager’s secondary (backup) processor.</p> |
| <b>Front panel connectors</b>              | <p>The PBSM’s only connections to the I/O card file backplane are the +5 Vdc supply, ground, and the power fail signal. All other connections are made through front panel connectors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>RCM control</b>                         | <p>The state of the PBSM’s bus selector circuitry is controlled by the Redundant Control Modules (RCMs), which sends a switch select output signal to the PBSM through the two 3-contact front panel connectors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Three D-type front panel connectors</b> | <p>Two 50-pin D-type connectors interface with the Parallel Link Driver Modules (PLDMs) in the processor card files. A third 50-pin D-type connector interfaces with the first Parallel I/O Module (PIOM) in the parallel I/O subsystem.</p>                                                                                                                                                                                                                                                                                                                                                                                               |

## 2.9.4 Power Supply Module

### Overview

A Power Supply Module (PSM) provides power to operate the serial or parallel I/O card file. Three models available and are listed in Table 2-13.

Table 2-13 I/O Card File Power Supply Modules

| Model Number | Input Voltage | Amperage            |
|--------------|---------------|---------------------|
| 621-9932     | 24 Vdc        | 8 A (5 Vdc Output)  |
| 621-9933     | 115/230 Vac   | 15 A (5 Vdc Output) |
| 621-9934     | 115/230 Vac   | 8 A (5 Vdc Output)  |

### 2.9.4.1 24 Vdc 8 A Power Supply Module

#### Description

The model 621-9932 single card I/O Power Supply Module (PSM) provides 8 A of +5 Vdc for the I/O logic circuitry in the card file. It also supplies 600 mA of  $\pm 15$  Vdc power. The module occupies slot M in the full-size I/O card file or slot G in the half-size card file. The maximum power consumption of the module is 96 VA.

#### Input power terminal connections

Terminals at the front of the module are labeled +, –, and **GND**. The module operates within an input voltage range of 20 to 28 Vdc. The power fail level is 19 Vdc with a power fail delay of 4 ms.

#### Input power fused

The module has a front panel accessible fuse holder and is shipped with a 8 A fast blow fuse.

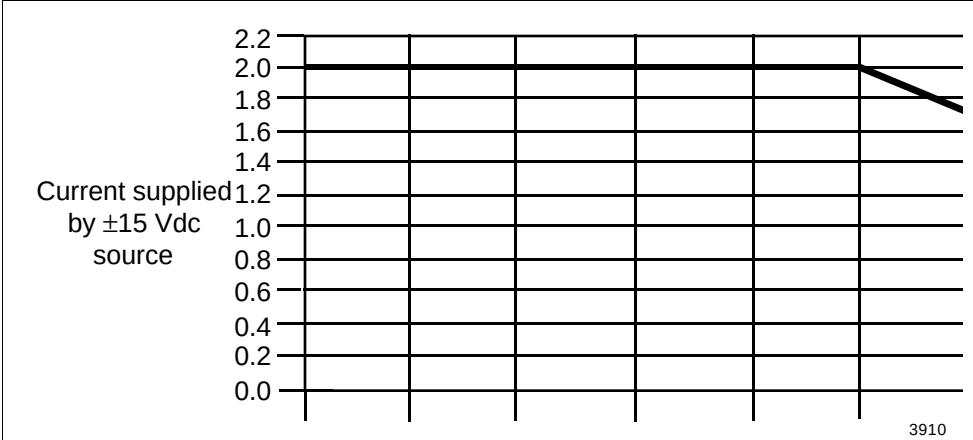
# 2.9.4.2 115/230 Vac 15 A Power Supply Module

## Overview

The model 621-9933 I/O Power Supply Module is a double wide module that provides 10 to 15 A of +5 Vdc for the logic circuitry in the I/O card file. It also supplies 1.2 to 2 A of  $\pm 15$  Vdc. The graph in Figure 2-13 shows the relationship between current loads for the  $\pm 15$  Vdc and the  $\pm 5$  Vdc sources.

If, for example, the  $\pm 15$  Vdc load requires 2 A, the  $\pm 5$  Vdc load cannot exceed 10 A. If the  $\pm 15$  Vdc load requires 1.2 A, the  $\pm 5$  Vdc load can not exceed 15 A. The maximum power consumption of the module is 110 VA.

Figure 2-13 Model 621-9933 Power Supply Current Relationship



## Card slot assignments

The Power Supply Module is installed in slots L and M of the full-size I/O card file or slots F and G of the half-size card file.

## 115 or 230 Vac input power selectable

The input is selectable for 115 or 230 Vac operation by the position of the toggle switch located under the component cover of the module. The 115 Vac selection permits a voltage range of 85 to 132 Vac. The 230 Vac selection permits a voltage range of 170 to 250 Vac operation. The frequency for both ranges is 47 to 63 Hz.

The power fail level is 83 volts for 115 Vac and 166 volts for 230 Vac with a delay of 11.5 milliseconds.

## Input power fused

The Power Supply Module has a front accessible fuse holder. It is shipped with a 4 A slow blow fuse installed for 115 Vac operation. A 2 A slow blow fuse for 230 Vac operation is also supplied with the module.

### 2.9.4.3 115/230 Vac 8 A Power Supply Module

|                                |                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                | The model 621-9934 I/O Power Supply Module provides 8 A of +5 Vdc for the I/O logic circuitry in the I/O card file. It also supplies 600 mA of $\pm 15$ Vdc power for special function I/O and optional module operation. The maximum power consumption of the module is 95 VA.                                                   |
| <b>Card slot assignments</b>   | The model 621-9934 Power Supply Module is installed in slot M of the full-size I/O card file or slot G of the half-size card file.                                                                                                                                                                                                |
| <b>Input power selectable</b>  | The voltage input is selectable for 115 or 230 Vac by the position of a selection board located under the component cover of the Power Supply Module. The 115 Vac selection allows a voltage range of 85 to 140 Vac. The 230 Vac selection allows 170 to 250 Vac operation. The frequency for both voltage ranges is 47 to 63 Hz. |
| <b>Input power fused</b>       | The Power Supply Module offers a front accessible fuse holder. It is shipped with a 1.6 A slow blow fuse installed for 115 Vac operation. A 0.8 A slow blow fuse for 230 Vac operation is also supplied with the module.                                                                                                          |
| <b>Power failure detection</b> | The power fail level is 83 volts for 115 Vac and 166 volts for 230 Vac operation. The power fail delay is 11.5 ms.                                                                                                                                                                                                                |

### 2.9.5 Serial I/O Module

|                             |                                                                                                                                                                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>             | The model 621-9938R or model 621-9940 Serial I/O Modules (SIOM) are installed in slot N of a serial I/O card file. They control the serial I/O card file and interface with a model 621-9939 SLM in the processor card file. |
| <b>Two SIOM models</b>      | The model 621-9940 SIOM is used in nonredundant serial I/O subsystems, while the model 621-9938R SIOM is used in a redundant serial I/O configuration, but can also be used for nonredundant applications.                   |
| <b>Single or dual ports</b> | The model 621-9938R SIOM has two serial ports and the model 621-9940 SIOM has one port. The model 621-9938R SIOM is used in redundant applications which require communication with redundant processors.                    |

*Continued on next page*

## 2.9.5 Serial I/O Module, Continued

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|                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Serial I/O link termination</b> | Two terminating resistors are required in a SIOM communications port if the port is at the end (either end) of a serial I/O communications link. The resistors are supplied in a kit that comes with a SLM. Use the 200 ohm resistors from the kit for installations using Belden 9729 cable and use the 300 ohm resistors, also supplied in the SLM resistor kit, for installations using Belden 9182 cable. |
| <b>References</b>                  | Refer to Figures 2-3 and 2-4 for examples showing terminating resistor requirements. Refer to Section 5 of the <i>Logic Manager Installation</i> manual for additional information on wiring and termination of serial I/O communication links.                                                                                                                                                               |

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## 2.9.6 Parallel I/O Module

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|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                      | The model 621-9937 Parallel I/O Module (PIOM) is the interface to the other parallel I/O card files or the Parallel Link Driver Module in the processor card file. The PIOM is installed in slot N of the full-size parallel I/O card file or slot H of the half-size parallel I/O card file.                                                                                                                                                               |
| <b>Number of I/O points</b>          | The PIOM permits any combination of 8, 16, or 32 point I/O modules in the parallel I/O card file, providing the number of I/O points stay within the boundaries permitted for the Logic Manager subsystem.                                                                                                                                                                                                                                                  |
| <b>Front panel D-type connectors</b> | The front panel of the PIOM has two 50-pin D-type connectors. The <b>IN</b> connector (male) is at the upper portion of the module's front panel, and the <b>OUT</b> connector (female) is at the lower portion.                                                                                                                                                                                                                                            |
| <b>Parallel I/O not recommended</b>  | Failure in a parallel I/O subsystem is potentially more hazardous than in a serial I/O subsystem because there is a greater risk of causing a total subsystem failure in the parallel I/O architecture. For this reason, a redundant Logic Manager with parallel I/O is not recommended by Honeywell. Only serial I/O is supported by a redundant Logic Manager subsystem. A redundant Logic Manager with parallel I/O is considered a "special subsystem." |

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## 2.10 Process I/O Modules

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### Introduction

The following subsections provide basic planning information for the use of process I/O modules which can reside in the serial and parallel I/O card files. The reader is advised to refer to the *Logic Manager Installation* manual for any additional information needed to plan for a particular I/O module. Topics of interest include:

- Terminal block set selection and installation
  - Address label selection and installation
  - Input module addressing considerations
  - Output module addressing considerations
  - Module jumper and DIP switch settings, if applicable
  - Cable selection and routing.
- 

### Module types

The module types listed in the following subsections include discrete input, discrete output, analog input, analog output and special function modules.

---

## 2.10.1 Discrete Input Modules

---

### Overview

Discrete input modules sense the ON or OFF voltage levels of various input devices. Among the more common devices are limit switches, proximity sensors, and pushbuttons. Once this ON/OFF status is obtained, the input modules convert it into logical data which is interpreted by the Logic Manager's processor.

---

### LED indicators

The LED indicators for each input circuit on 6-, 8-, and 16-point modules are on the field device side of the module's circuitry. The 32-point input modules have the LED indication for each circuit on the 5 V logic side of the module. All discrete inputs are optically isolated to 2500 Vdc.

---

### Physical construction

The modules are composed of two areas. A printed circuit board with a permanent front cover and label to identify each particular module, and a set of 8-, 16-, or 32-point terminal blocks comprise each complete I/O module. The printed circuit board is inserted into the desired slot in the I/O card file. An offset backplane connector makes it impossible to install a module upside down.

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*Continued on next page*

## 2.10.1 Discrete Input Modules, Continued

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### Terminal blocks

The removable 8- and 16-point double swing terminal blocks allow for easy module insertion, removal, and wiring. They are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fastener. The terminal blocks are keyed for the circuit board. The 32-point terminal blocks are inserted into block headers located at the front of the module. Field wiring connects to the terminal block from the front of the module and can be accessed without removing the terminal block.

---

### Sink or source dc modules

Each dc input module is either a sink or a source module. A sink input module provides the ground potential for the input device when the input device turns on. Current flows from the Power Supply Module through the field sensor input device, when it turns on, to ground in the sink input module. A source input module supplies the +Vdc potential to the field sensor input device when it turns on. Current flows from the Power Supply Module to the source input module through the field input sensor device to ground.

---

### Cable class observance

The proper cable class rule must be applied for each module model. Refer to subsection 3.6, Table 3-2 for an explanation of the cable classes. The number of point inputs offered are 6, 8, 16, or 32, depending on the module.

The module specifications, wiring diagrams, and related information are found in the *Logic Manager Installation* manual.

---

### Classes of wiring

IEEE Standard 518-1977 defines four classes of wiring that differ in signal level and noise susceptibility. See Section 3, Table 3-2 for details.

Table 2-14 is a list of the discrete input modules available.

---

### ATTENTION

ATTENTION—In Table 2-14 only the I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

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*Continued on next page*

## 2.10.1 Discrete Input Modules, Continued

### Discrete input module list

Table 2-14 Discrete Input Modules

| Module Description                   | Inputs | Class of Wiring | Model Number |
|--------------------------------------|--------|-----------------|--------------|
| 115 Vac/Vdc Input                    | 8      | Data 3          | 621-1100R    |
| 115 Vac/Vdc Isolated Input           | 6      | Data 3          | 621-1101R    |
| 115 Vac Input                        | 16     | Data 3          | 621-1160R    |
| 115 Vac Input                        | 32     | Data 3          | 621-1180R    |
| 230 Vac/Vdc Input                    | 8      | Data 3          | 621-1200R    |
| 230 Vac/Vdc Isolated Input           | 6      | Data 3          | 621-1201R    |
| 240 Vac Input                        | 16     | Data 3          | 621-1250R    |
| 240 Vac Isolated Input               | 16     | Data 3          | 621-1251R    |
| 24 Vac/Vdc Input                     | 8      | Data 2          | 621-1500R    |
| 24 Vac/Vdc Input                     | 16     | Data 2          | 621-1550R    |
| 5 Vdc Sink Input                     | 8      | Data 1          | 621-3300R    |
| 12 Vdc Sink Input                    | 16     | Data 1          | 621-3450R    |
| 12-24 Vdc Sink Input                 | 8      | Data 1, 2*      | 621-3500R    |
| 12-24 Vdc Sink Fast Response Input   | 8      | Data 1, 2*      | 621-3502R    |
| 24 Vdc Sink Fast Response Input      | 16     | Data 2          | 621-3552R    |
| 12-24 Vdc Sink Input                 | 16     | Data 2          | 621-3560R    |
| 12-24 Vdc Sink Input                 | 32     | Data 2          | 621-3580R    |
| 48 Vdc Sink Input                    | 8      | Data 2          | 621-3600R    |
| 48 Vdc Sink Input                    | 16     | Data 2          | 621-3660R    |
| 5 Vdc Source Input                   | 8      | Data 1          | 621-4300R    |
| 5 Vdc TTL Input                      | 16     | Data 1          | 621-4350R    |
| 12-24 Vdc Source Input               | 8      | Data 1, 2*      | 621-4500R    |
| 12-24 Vdc Source Fast Response Input | 8      | Data 1, 2*      | 621-4502R    |
| 24 Vdc Source Input                  | 16     | Data 2          | 621-4550R    |

\* The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

## 2.10.2 Analog Input Modules

### Overview

An analog input module accepts signals from field devices connected to the module's terminals and converts the signal data to 12-bit binary values (0 to 4095) through an analog-to-digital converter in the module's circuitry. The digital values are stored in a holding register on the card and can be accessed by the Logic Manager. The ladder logic PULL instruction reads from one to eight 16-bit data words at a time. These digital values can be read by the Logic Manager's processor.

### Analog input module list

Table 2-15 is a list of analog input modules. Data 1 cable class rules for the module must be observed. See Section 3, Table 3-2 for an explanation of the wiring class.

### ATTENTION

ATTENTION—In Table 2-15 only the I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

Table 2-15 Analog Input Modules

| Module Description            | Inputs | Class of Wiring | Model Number |
|-------------------------------|--------|-----------------|--------------|
| Universal Analog Input        | 16     | Data 1          | 621-0020R    |
| Isolated 0-20 mA Analog Input | 8      | Data 1          | 621-0022AR   |
| Isolated 0-10 V Analog Input  | 8      | Data 1          | 621-0022VR   |

### Reference for analog input modules

The module specifications, wiring diagrams, and related information are found in the *Logic Manager Installation* manual.

## 2.10.3 Discrete Output Modules

|                                  |                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                  | <p>The typical discrete output module offers the ability to control the on and off state of its end devices. Common discrete output devices are:</p> <ul style="list-style-type: none"><li>• motor starters</li><li>• solenoid valves</li><li>• indicator lamps.</li></ul>                                                             |
| <b>Source type Vdc modules</b>   | <p>All dc output modules are source type modules. This means they provide the +Vdc potential to the field output device when the circuit is turned on. Current flows from the external power supply to the source output card to the field output device.</p>                                                                          |
| <b>Zero crossing Vac modules</b> | <p>All ac output modules incorporate zero crossing. The output is turned on at the zero crossing of the ac wave form.</p>                                                                                                                                                                                                              |
| <b>LED indicators</b>            | <p>An LED indicator for each output circuit on the 8- and 16-point modules is located on the load side of the output, except isolated outputs and reed relays. The 32-point output modules have an LED indication for each circuit on the 5 Vdc logic side of the module. All discrete outputs are optically isolated to 2500 Vdc.</p> |
| <b>Front panel fault LED</b>     | <p>The output modules provide self-diagnostic assurance, since each module's front plate has an LED to indicate a fault, thus reducing an otherwise lengthy troubleshooting procedure.</p>                                                                                                                                             |
| <b>Module components</b>         | <p>Each discrete output module consists of three components: a printed circuit board with an permanent front plate, address labels, and a set of 8-, 16-, or 32-point terminal blocks.</p>                                                                                                                                             |
| <b>ATTENTION</b>                 | <p>ATTENTION—Terminal blocks are included with modules and address labels are included with logic processors (8-, 16-, or 32-point ).</p>                                                                                                                                                                                              |

*Continued on next page*

## 2.10.3 Discrete Output Modules, Continued

### Terminal block applications

The module's printed circuit board is inserted into the desired slot in the I/O card file. Offset keyed backplane connectors prevent installing a module upside down. The removable 8- and 16-point double swing terminal blocks allow easy module insertion, removal, and wiring. They are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fasteners.

The 32-point terminal blocks are inserted into terminal block headers located at the front of the module. Field wiring connects to the 32-point terminal block from the front of the module and can be accessed without removing the terminal block.

### 8-point terminal blocks

Discrete 8-point modules use terminals **T1** and **B1** to supply voltage connections for circuits 1 through 4. Terminals **T2** and **B2** supply voltage connections for circuits 5 through 8. Separate voltage sources may power circuits 1 through 4 and 5 through 8 by removing the factory installed jumpers across terminals **T1** and **T2** and terminals **B1** and **B2**. The 8-point modules have the potential for two common returns per module.

### 16-point terminal blocks

All discrete 16-point modules provide one common for all 16 outputs, requiring that all 16 circuits be powered from the same voltage source. The factory installed jumpers across terminals **T1**, **T2**, and **T3**, and terminals **B1**, **B2**, and **B3** distribute the load current to all outputs. For proper operation, do not remove these jumpers.

### Reference for discrete output modules

The module specifications, wiring diagrams, and related information are found in the *Logic Manager Installation* manual.

### Classes of wiring

IEEE Standard 518-1977 defines four classes of wiring that differ in signal level and noise susceptibility. See Section 3, Table 3-2 for details.

Table 2-16 is a list of the discrete output modules available.

### ATTENTION

ATTENTION—In Table 2-16 only the I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

*Continued on next page*

## 2.10.3 Discrete Output Modules, Continued

### Discrete output module list

Table 2-16 Discrete Output Modules

| Module Description                     | Inputs | Class of Wiring | Model Number |
|----------------------------------------|--------|-----------------|--------------|
| Reed Relay Output                      | 6      | Data 1, 2, 3    | 621-0007R    |
| 115 Vac Output                         | 8      | Data 3          | 621-2100R    |
| 115 Vac Isolated Output                | 6      | Data 3          | 621-2101R    |
| 115 Vac Self-Protected Output          | 8      | Data 3          | 621-2102R    |
| 115 Vac Output                         | 16     | Data 3          | 621-2150R    |
| 115 Vac Output                         | 32     | Data 3          | 621-2175R    |
| 230 Vac Output                         | 8      | Data 3          | 621-2200R    |
| 230 Vac Isolated Output                | 6      | Data 3          | 621-2201R    |
| 24 Vac Output                          | 8      | Data 2          | 621-2500R    |
| 24 Vac Output                          | 16     | Data 2          | 621-2550R    |
| 5 Vdc Source Output                    | 8      | Data 1          | 621-6300R    |
| 5 Vdc TTL Source Output                | 16     | Data 1          | 621-6350R    |
| 12 Vdc Source Output                   | 16     | Data 1          | 621-6450R    |
| 12-24 Vdc Source Output                | 8      | Data 1, 2*      | 621-6500R    |
| 12-24 Vdc Source Self-Protected Output | 8      | Data 1, 2*      | 621-6503R    |
| 24 Vdc Source Output                   | 16     | Data 2          | 621-6550R    |
| 24 Vdc Source Low Power Output         | 16     | Data 2          | 621-6551R    |
| 24 Vdc Source Output                   | 32     | Data 2          | 621-6575R    |
| 48 Vdc Source Output                   | 8      | Data 2          | 621-6600R    |
| 48 Vdc Source Output                   | 16     | Data 2          | 621-6650R    |
| 120 Vdc Source 0.5 A Output            | 8      | Data 3          | 621-6700R    |
| 120 Vdc Source 2 A Output              | 8      | Data 3          | 621-6701R    |

\* The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

## 2.10.4 Analog Output Modules

### Overview of analog output modules

An analog output module accepts ladder logic numerical data. The numeric values are translated through 12-bit, digital-to-analog (D/A) converters contained in the module's circuitry. The module's D/A converts the data into equivalent analog voltage or current values to control the field devices that are connected to the module's output terminals.

### Analog-to-digital converters

The analog-to-digital converters are addressed by the processor through a PUSH instruction. Once addressed, a converter can translate the digital value to a corresponding analog voltage or current level.

#### ATTENTION

ATTENTION—Terminal blocks are included with modules.

### References

The module specifications, wiring diagrams, and related information are found in the *Logic Manager Installation* manual.

### Analog output module list

Table 2-17 is a list of available analog output modules. Data 1 cable class rules for the module must be observed. See Section 3, Table 3-2 for an explanation of the wiring class.

#### ATTENTION

ATTENTION—In Table 2-17 only the I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

### Analog output module list

Table 2-17 Analog Output Modules

| Module Description    | Inputs | Class of Wiring | Model Number |
|-----------------------|--------|-----------------|--------------|
| 4-20 mA Analog Output | 4      | Data 1          | 621-0010AR   |
| 0-10 V Analog Output  | 4      | Data 1          | 621-0010VR   |



## 2.10.5 Special Function Modules

### Overview

Special function modules provide a variety of control and communication functions for the Logic Manager's I/O subsystem. The special function modules use PUSH/PULL instructions to communicate data to or from the processor over the I/O bus. The PUSH/PULL instructions permit the modules to send or receive up to eight 16-bit registers of data at a time.

### Reference for special function module

The module specifications, wiring diagrams, special instructions and related information are found in the *Logic Manager Installation* manual.

### Classes of wiring

IEEE Standard 518-1977 defines four classes of wiring that differ in signal level and noise susceptibility. See Section 3, Table 3-2 for details.

The special function modules are listed in Table 2-18.

### ATTENTION

**ATTENTION**—In Table 2-18 only the I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

### Special function module list

Table 2-18 Special Function Modules

| Module Description                           | Class of Wiring | Model Number |
|----------------------------------------------|-----------------|--------------|
| BCD Conversion Module                        | N/A             | 621-0006     |
| Pulse Input Module                           | Data 1          | 621-0008*    |
| Input Simulator Module                       | N/A             | 621-0009R    |
| ASCII Communication Module                   | Data 1          | 621-0012R    |
| Absolute Encoder Module                      | Data 1          | 621-0018R    |
| Pulse/Quadrature Input Module                | Data 1          | 621-0019R    |
| Pulse Input Module                           | Data 1          | 621-0024R    |
| Resistance Temperature Detector Module       | Data 1          | 621-0025R    |
| High-Speed Counter Module                    | Data 1          | 621-0307R    |
| 24 Vdc 24-Sink Input, 8-Source Output Module | Data 2          | 621-0576R    |
| 3-Port MiniCOP Module                        | Data 1          | 627-1002R    |
| 4-Port MiniCOP Module                        | Data 1          | 627-1003R    |

\* Superseded by 621-0024R



## Section 3 – Process Wiring Guidelines

### 3.1 Overview

---

#### Section Contents

The topics covered in this section are:

|        | Topic                                                 | See Page |
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#### Introduction

This section presents the rules which must be observed when wiring the process equipment to the Logic Manager. Field cable cabinet entry, cable routing in conduit and trays, cable selection, and card file interconnection wiring are some of the topics discussed.

---

#### Process wiring

The process wiring is connected to the Logic Manager at removable terminal blocks that are mounted on the face plate of each I/O module.

---

## 3.2 Cable Cabinet Entry

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### Cable entry

The process signal cables may enter the cabinet through the top or bottom of the cabinet. For bottom entry, the cabinet floor has sliding plates that can be adjusted to provide several entrance slots. For top entry, the top is removed by removing the cabinet lifting eye-bolts. The top plate should be carried to a work bench where entry holes can be cut in the plate as required and installed back onto the top of the cabinet.

---

### ATTENTION

ATTENTION—Do not attempt to cut holes in the top plate while it is still mounted to the top of the cabinet. Metal filings, dropping down into the electronic modules, could result in electrical damage when power is applied.

---

### Cable clamping

For either top or bottom entry, the cables should be clamped firmly inside the cabinet. The clamping should be able to withstand a 45 kg (100 pound) pull. The Logic Manager cabinet includes a special cable clamp rail at the bottom. The rail can be remounted at the top, if required.

---

### Reference

See Sections 4 and 5 for cabinet footprint drawings showing the floor cable entrance points for dual and single access cabinets.

---

### 3.3 Cable and Conduit Routing

---

#### Two wiring categories

Proper cables for the type of Logic Manager connections should be selected. All wires and cables can be classified by two voltage categories:

- high-level conductors
  - low-level conductors.
- 

#### High-level voltage conductors

High-level conductors carry high voltage and/or high current. Some examples are:

- AC voltage I/O lines
- High-level dc voltage I/O lines connected to devices such as switches, relays, or solenoids.
- AC power lines.

These are defined in the IEEE Standard 518 Data 2, 3, and 4 classes. See the Cable Spacing, subsection 3.6, for more information.

---

#### Low-level voltage conductors

Low-level conductors carry low voltage and/or low current. Typical low-level conductors are:

- Serial communication cables carry data to and from a remote I/O chassis, intelligent I/O modules, programmable controllers, programming terminals, computers, or printers.
- DC I/O lines connected to solid-state devices such as proximity switches or photoelectric sensors.
- TTL signal lines
- Low level analog signal lines
- Encoder signal lines
- Motion control signal lines
- Parallel I/O communication cables.

These are defined in the IEEE Standard 518 Data 1 class. See the Cable Spacing, subsection 3.6, for more information.

---

## 3.4 Cable Considerations

### Cabling guidelines

---

The following guidelines should be observed when routing cables to Logic Manager card files in cabinets or remote enclosures. Special attention must be paid to separating data cables from power cables to avoid induced electrical noise problems.

- Maintain a 60° C or less ambient air temperature at the top of each Logic Manager card file assembly.
  - All wiring should be installed according to IEEE Standard 518-1977.
  - When using a wire duct less than 89 millimeters (3.5 inches) deep, allow at least 38 millimeters (1.5 inches) between it and the card file assembly. For a wire duct greater than 89 millimeters (3.5 inches) deep, allow for a clearance of 50 millimeters (2 inches).
  - Separate all 440 Vac (or higher) electrical supply lines as far from the Logic Manager components and communication lines as possible. See Tables 3-5, 3-6, and 3-7 for the recommended minimum distances.
  - All 440 Vac lines must be routed in separate conduits away from I/O wiring ducts.
  - Separate all ac and low-level dc wiring. See Tables 3-5, 3-6, and 3-7 for recommended minimum distances.
  - When 8-inch deep enclosures are used, door stiffeners are not permitted in front of the Logic manager card files.
  - Separate the data cable from ac and dc (above 12 V) wiring when connecting a cable between the MS-DOS Loader and the Logic Manager's processor.
  - Some data cables should be terminated at both ends. Terminate shields for communications signal at both ends (serial I/O, EIA-422, or EIA-232). For analog signals, terminate the shields to ground at the source device. If a termination is not made or is intermittent, noise immunity is lost even though the subsystem may transmit apparent valid data.
  - Data and power cables should cross at right angles rather than running in parallel for any distance.
  - Ensure that the entire communications network subsystem is grounded to chassis ground.
  - Shield continuity must be maintained. Ensure that shield leads are not broken.
  - Cables should be routed around rather than through high noise areas.
  - To ensure proper grounding of shield connections, all module faceplates must be securely fastened to the chassis through the thumbscrews on the modules.
-

## 3.5 Conduit Shielding

### Overview

Conduit or cable trays should be used to route the process wiring within the plant. The same conduit should not carry both data and power wiring due to electromagnetic interference (EMI). The choice of materials and routing of the cable conduit and/or trays determines the degree of interference protection.

### Raceway shielding

Table 3-1, listing raceway shielding, is extracted from IEEE Standard 518-1977 for the installation of electrical equipment. The table compares different types of conduits and raceways for magnetic field attenuation at 60 Hz and for electrostatic attenuation at 100 kHz.

Table 3-1 Raceway Shielding

| Raceway Type                         | Thickness<br>mm (in.) | 60 Hz Magnetic Field<br>Attenuation |      | 100 Hz Electrical Field<br>Attenuation |      |
|--------------------------------------|-----------------------|-------------------------------------|------|----------------------------------------|------|
|                                      |                       | Ratio                               | db   | Ratio                                  | db   |
| Free Air                             |                       | 1:1                                 | 0    | 1:1                                    | 0    |
| 2-inch aluminum conduit              | 3.91 (0.154)          | 1.5:1                               | 3.3  | 2,150:1                                | 66.5 |
| #16 gauge aluminum tray              | 1.52 (0.060)          | 1.6:1                               | 4.1  | 15,500:1                               | 83.9 |
| #16 gauge aluminum tray              | 1.52 (0.060)          | 3:1                                 | 9.4  | 20,000:1                               | 86.0 |
| #16 gauge galvanized ingot iron tray | 1.52 (0.060)          | 3.2:1                               | 10.0 | 22,000:1                               | 86.8 |
| 2-inch IPS copper pipe               | 3.96(0.156)           | 3.3:1                               | 10.2 | 10,750:1                               | 80.6 |
| #16 gauge aluminum tray              | 1.52 (0.060)          | 4.2:1                               | 11.5 | 29,000:1                               | 89.3 |
| #14 gauge aluminum tray              | 1.90 (0.075)          | 6:1                                 | 15.5 | 23,750:1                               | 87.5 |
| 2-inch metallic conduit              | 1.65 (0.065)          | 6.7:1                               | 16.5 | 3,350:1                                | 70.5 |
| 2-inch rigid galvanized conduit      | 3.91 (0.154)          | 40:1                                | 32.0 | 8,850:1                                | 78.9 |

## 3.6 Cable Spacing

### Overview

Honeywell recommends noise creating sources, such as time-varying voltage or current, be separated from data signal cables. Cables with similar levels of noise susceptibility should be grouped together, and those with similar levels of noise generation should be separately grouped in trays or conduits.

### Classes of wiring

IEEE Standard 518-1977 defines four classes of wiring that differ in signal level and noise susceptibility as shown in Table 3-2.

Table 3-2 IEEE Standard 518-1977 Signal Classes

| Data Class | Description                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data 1     | High noise susceptibility – Analog signals of less than 50 V and digital signals of less than 15 V<br><br>Examples: Model 628-2000 series parallel I/O cable<br>Belden 9182 and 9729 serial I/O cable   |
| Data 2     | Medium susceptibility – Analog signals greater than 50 V, regulating signals of 50 V with currents less than 20 amps, and ac feeders less than 20 amps<br><br>Example: The power cable to the processor |
| Data 3     | Low susceptibility – Switching signals greater than 50 V, analog signals of 50 V with currents less than 20 amps, and ac feeders less than 20 amps                                                      |
| Data 4     | Power ac and dc buses of 0-1000 V with currents of 20-800 amps.                                                                                                                                         |

### Data 3 and Data 4 subclasses

Data 3 and Data 4 classes have one subclass each (3A and 4A) for an application that requires special cabling and spacing. See IEEE Standard 518-1977 for complete information on wiring classes and subclasses. Examples of Data 3A and Data 4A are as follows:

- Signals from communicating fields and line resistors
- Signals from shunts to regulators
- Power greater than 1000 volts or greater than 800 amps, or both.



## 3.7 Tray Spacing

---

**Data 1 and Data 2 classes**

Trays containing Data 1 or Data 2 cables should have solid metal bottoms and covers for complete shielding. Ventilation slots or louvers may be used only in trays containing Data 3 cables.

---

**Spacing requirements**

Tables 3-3, 3-4, and 3-5 list the recommended minimum spacing distances between trays, trays and conduits, and conduits, respectively. The distance in the tables is between the top of one tray/conduit and the bottom of the tray/conduit above, or between sides of adjacent trays/conduits.

---

*Continued on next page*

## 3.7 Tray Spacing, Continued

Spacing requirements Table 3-3 Spacing between Trays (Centimeters/Inches)

| Cable Class                                                                                                                                                                                                                                                | Data 1  | Data 2  | Data 3  | Data 3A | Data 4  | Data 4A |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|---------|---------|---------|
| Data 1                                                                                                                                                                                                                                                     |         | Note 2  | 15 (6)  | 15 (6)  | 66 (26) | 66 (26) |
| Data 2                                                                                                                                                                                                                                                     | Note 2  |         | 15 (6)  | 15 (6)  | 46 (18) | 66 (26) |
| Data 3                                                                                                                                                                                                                                                     | 15 (6)  | 15 (6)  |         | 0 (0)   | Note 1  | 31(12)  |
| Data 3A                                                                                                                                                                                                                                                    | 15 (6)  | 15 (6)  | 0 (0)   |         | 20 (8)  | 46 (18) |
| Data 4                                                                                                                                                                                                                                                     | 66 (26) | 46 (18) | Note 1  | 20 (8)  |         | 0 (0)   |
| Data 4A                                                                                                                                                                                                                                                    | 66 (26) | 66 (26) | 31 (12) | 46 (18) | 0 (0)   |         |
| Notes:<br>1. DATA 3 and 4 cables may be run in a common tray, but should be separated by a barrier. This barrier does not have to be grounded.<br>2. Data 1 and 2 cables may be in a common tray, provided they are separated by a grounded steel barrier. |         |         |         |         |         |         |

Table 3-4 Spacing between Trays and Conduits (Centimeters/Inches)

| Cable Class | Data 1  | Data 2  | Data 3 | Data 3A | Data 4  | Data 4A |
|-------------|---------|---------|--------|---------|---------|---------|
| Data 1      |         | 2.5 (1) | 10 (4) | 10 (4)  | 46 (18) | 46 (18) |
| Data 2      | 2.5 (1) |         | 10 (4) | 10 (4)  | 31(12)  | 46 (18) |
| Data 3      | 10 (4)  | 10 (4)  |        | 0 (0)   | 0 (0)   | 20 (8)  |
| Data 3A     | 10 (4)  | 10 (4)  | 0 (0)  |         | 15 (6)  | 31(12)  |
| Data 4      | 46 (18) | 31(12)  | 0 (0)  | 15 (6)  |         | 0 (0)   |
| Data 4A     | 46 (18) | 46 (18) | 20 (8) | 31(12)  | 0 (0)   |         |

Table 3-5 Spacing between Conduits (Centimeters/Inches)

| Cable Class | Data 1  | Data 2  | Data 3 | Data 3A | Data 4 | Data 4A |
|-------------|---------|---------|--------|---------|--------|---------|
| Data 1      |         | 2.5 (1) | 8 (3)  | 8 (3)   | 31(12) | 31(12)  |
| Data 2      | 2.5 (1) |         | 8 (3)  | 8 (3)   | 23(9)  | 31(12)  |
| Data 3      | 8 (3)   | 8 (3)   |        | 0 (0)   | 0 (0)  | 15 (6)  |
| Data 3A     | 8 (3)   | 8 (3)   | 0 (0)  |         | 15 (6) | 23(9)   |
| Data 4      | 31(12)  | 23(9)   | 0 (0)  | 15 (6)  |        | 0 (0)   |
| Data 4A     | 31(12)  | 31(12)  | 15 (6) | 23(9)   | 0 (0)  |         |

## 3.8 Parallel I/O Subsystem Cables

### Standard model 628-2xxxx cables

Parallel I/O connections between the Logic Manager card files must be made through standard model 628-2xxx cables supplied by Honeywell. Cables range in length from 20 meters (66 feet) for the model 628-2000 cable to 30 meters (98.43 feet) for the model 628-2026 cable.

### Cable length list

The Logic Manager subsystem supports parallel I/O card files up to 30 meters (100 feet) from the Processor card file. See Table 3-6 for the compatible cable list.

#### ATTENTION

ATTENTION—Although parallel I/O is not supported for redundant Logic Manager's, this table listing is provided for approved "special subsystem" applications.

Table 3-6 Parallel I/O Cables

| Model Number | Length<br>(Meter/Feet) | Model Number | Length<br>(Meter/Feet) |
|--------------|------------------------|--------------|------------------------|
| 628-2000     | 0.20/0.66              | 628-2016     | 10/32.80               |
| 628-2001     | 0.35/1.15              | 628-2017     | 12/39.37               |
| 628-2002     | 1.2/3.94               | 628-2018     | 14/45.93               |
| 628-2003     | 1.8/5.90               | 628-2019     | 16/52.49               |
| 628-2004     | 3/9.84                 | 628-2020     | 18/59.06               |
| 628-2010     | 4/13.12                | 628-2021     | 20/65.62               |
| 628-2011     | 5/16.40                | 628-2022     | 22/72.18               |
| 628-2012     | 6/19.69                | 628-2023     | 24/78.74               |
| 628-2013     | 7/22.97                | 628-2024     | 26/85.30               |
| 628-2014     | 8/26.25                | 628-2025     | 28/91.86               |
| 628-2015     | 9/29.53                | 628-2026     | 30/98.43               |

#### ATTENTION

ATTENTION—Parallel I/O card files mounted vertically must provide at least 13.3 centimeters (5.25 inches) clearance to allow for heat dissipation.

## 3.9 Serial I/O Subsystem Cables

### Overview

The remote serial I/O subsystem requires a four-wire full duplex connection. Two types of cable are recommended.

### Twisted-pair cable list

Belden 9729 twisted-pair cable is recommended for cable runs up to 1219 meters (4000 feet). Belden 9182 twin axial cable is used for runs from 1219 to 3048 meters (4000 to 10,000 feet). Two 9182 cables are required for each serial channel to achieve four-wire communication. Please call Honeywell TAC in Ft. Washington, PA for exceptions or substitutions.

Table 3-7 lists specifications for both types of serial cable.

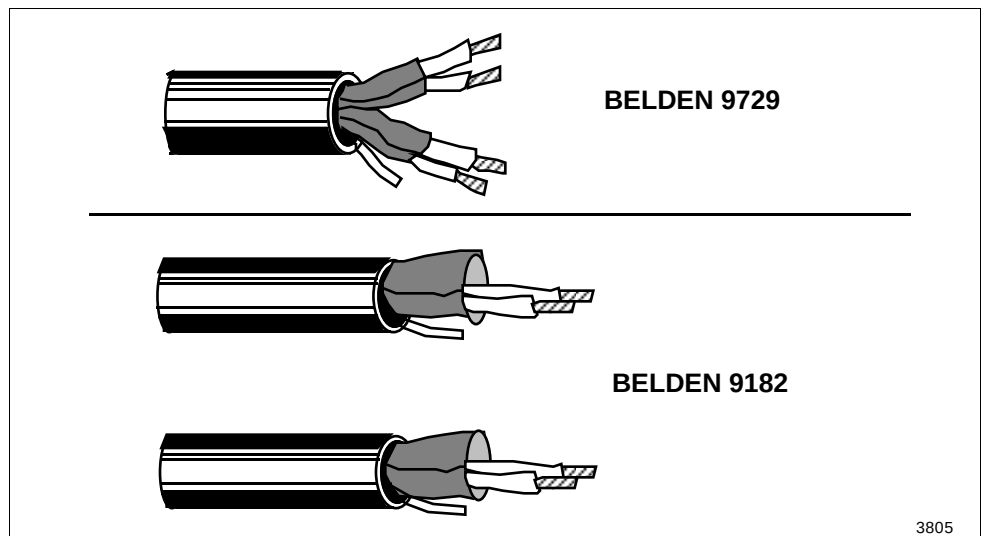
Table 3-7 Serial I/O Cable Specifications

|                 | <b>Belden Datalene 9182</b>                                              | <b>Belden Datalene 9729</b>                                 |
|-----------------|--------------------------------------------------------------------------|-------------------------------------------------------------|
| Standard Length | 500 and 1000 feet                                                        | 100, 500, and 1000 feet                                     |
| Wire Diameter   | 22 AWG (0.79 mm) tinned copper drain wire                                | 24 AWG (0.61 mm) tinned copper drain wire                   |
| Impedance       | 150 ohms nominal                                                         | 100 ohms nominal                                            |
| Propagation     | 1.3 nsec/ft. or 4.27 nsec/m<br>78% of the speed of light                 | 1.3 nsec/ft. or 4.27 nsec/m<br>(78% of the speed of light)  |
| Capacitance     | 28.9 pF/m (8.8 pF/ft.)                                                   | 41 pF/m (12.5 pF/ft.)                                       |
| Cable Type      | One pair with shield and tinned copper drain wire<br>(2 cables required) | Two shielded pairs with individual tinned copper drain wire |

Figure 3-1 is an illustration of both types of twisted-pair cables.

**Twisted-pair cable  
illustrations**

Figure 3-1 Serial I/O Subsystem Cable



## 3.10 Logic Manager Subsystem Cabling

---

|                              |                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Installation overview</b> | After the card files are in place and appropriate modules are installed, cabling connections are made between the modules in the Processor and I/O card files. The parallel I/O cables connect the Parallel Link Driver Module (PLDM) to the Parallel I/O Modules (PIOMs). The serial I/O cables connect to the Serial Link Module (SLM) to the Serial I/O Modules (SIOMs). |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

I/O terminal blocks must be installed and wired to the process equipment, and ac power cables must be connected to the Power Supply Modules.

---

|                           |                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Channel definition</b> | A channel is defined as a single data communication link between SLMs and SIOMs. SLM modules have dual channel functionality. |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|

---

### 3.10.1 Parallel I/O Subsystem

---

|                 |                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b> | Parallel I/O subsystem configurations are designed for applications where long distances between the I/O card files and the processor card file are not required. Parallel data cable is in the Data 1 class. The Logic Manager subsystem permits a maximum data cable length of 30 meters (100 feet) for a parallel I/O subsystem. Refer to Table 3-6 for the parallel I/O cable lengths available. |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parallel I/O not recommended</b> | Failure in a parallel I/O subsystem is potentially more hazardous than in a serial I/O subsystem, because there is a greater risk of causing a total subsystem failure in the parallel I/O architecture. For this reason, a redundant Logic Manager with parallel I/O is not recommended. Only serial I/O is supported by a redundant Logic Manager subsystem. A redundant Logic Manager with parallel I/O is considered a “special subsystem.” |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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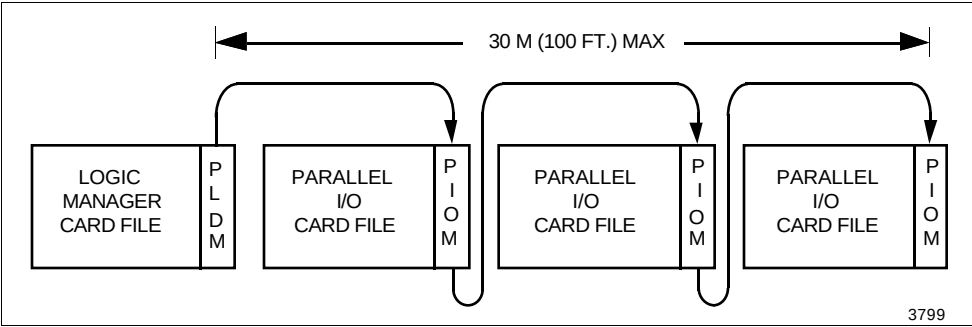
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### 3.10.1 Parallel I/O Subsystem, Continued

**Parallel I/O interconnections**

Figure 3-2 illustrates the wiring between the Parallel Link Drive Module and the PIOMs for a parallel I/O subsystem. Standard D-type connectors are used on the modules for the connections.

Figure 3-2 Parallel I/O Subsystem Interconnections



**Cable routing procedure**

The appropriate cable is routed from the Parallel Driver Link Module (PLDM) in the processor card file (slot N) to the Parallel I/O Module (PIOM) in the first parallel I/O card file in the subsystem. Another cable is then routed from the PIOM in the first parallel I/O card file to the PIOM in the second parallel I/O card file. The process is repeated until all the parallel I/O card files are connected in parallel with the PLDM.

## 3.10.2 Serial I/O Subsystem

### Overview

Remote serial I/O configurations are available for the Logic Manager. A serial I/O configuration has the advantage of permitting the I/O card files to be located near the process being controlled and thereby reducing the length of costly control circuit wiring. Serial I/O is accomplished with one or two Serial Link Modules (SLMs) which are located in the processor card file and a Serial I/O Module (SIOM) which is located in each of the remote serial I/O card files.

### Terminating resistor installation rules

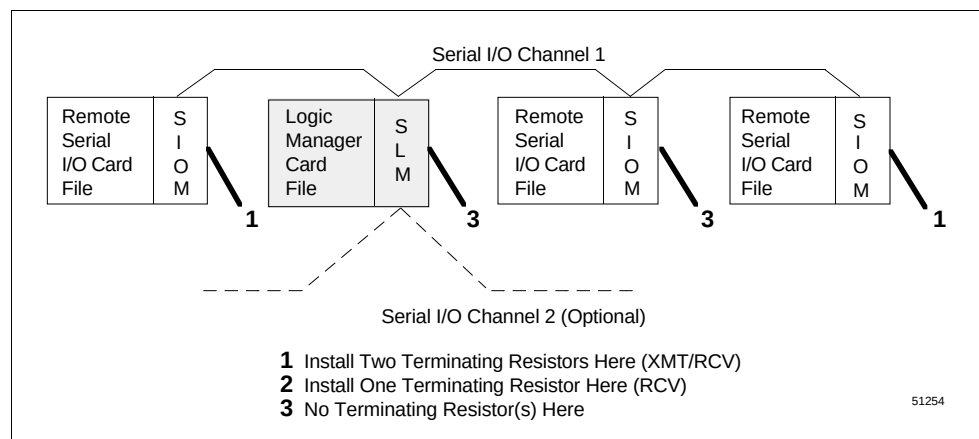
The subsystem is installed in a multidrop configuration as shown in either Figure 3-3 or 3-4. Channel terminating resistors are required at the ends of the multidrop communication lines. The rules for installing terminating resistors are as follows:

- If an SIOM is at the end (either end) of the communication line, two terminating resistors are installed in that SIOM. (See Note 1 on Figure 3-3.)

Terminating resistors are not installed in SIOMs at intermediate locations on the communication circuit. (See Figures 3-3 and 3-4 for examples.)

- A terminating resistor is not installed in a SLM channel if the SLM is at an intermediate location as shown in Figure 3-3.

Figure 3-3 Serial I/O Subsystem – Example 1



*Continued on next page*



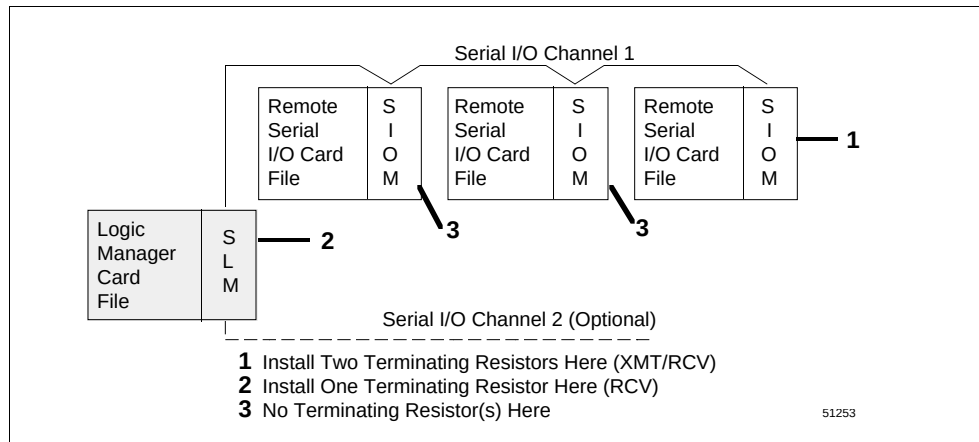
## 3.10.2 Serial I/O Subsystem, Continued

### Terminating resistor installation rules, continued

- If the SLM channel is at the end of a communication circuit, one terminating resistor is installed for that channel. There are two channels available in an SLM and each channel has a place for an optional terminating resistor.

Figure 3-4 is an example of an SLM channel requiring a terminating resistor.

Figure 3-4 Serial I/O Subsystem – Example 2



### ATTENTION

ATTENTION—Refer to the *Logic Manager Installation* manual for cabling information and rules regarding installation of terminating resistors.

*Continued on next page*

## 3.10.2 Serial I/O Subsystem, Continued

---

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>16 remote serial I/O card files permitted</b> | A processor card file can have two SLMs, and each SLM can support two communication channels. Each channel can connect to up to 16 remote serial I/O card files.                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Twisted-pair serial link cable used</b>       | A single cable, consisting of two individually shielded twisted-pairs, is used in configurations that require up to 1219 meters (4000 feet) of cable. Honeywell recommends using Belden Datalene 9729 cable for a configuration using this distance. Two twin-axial (single twisted-pair) cables are used in applications requiring up to 3048 (10,000 feet) maximum; Belden Datalene 9182 cable is recommended. The 9729 cable uses 200 ohm terminating resistors and the 9729 cable uses 300 ohm terminating resistors. Serial I/O data cable is classified as Data 1 wiring. |
| <b>SLM to SIOM cabling</b>                       | The cable is connected from the SLM in the processor card file to the SIOM in the first serial I/O card file. The first SIOM is then connected to the SIOM in the second serial I/O card file. Up to sixteen serial I/O card files can be installed in each serial link, providing the processor's addressing capacity (4095 maximum I/O address) is not exceeded.                                                                                                                                                                                                              |

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### 3.10.2 Serial I/O Subsystem, Continued

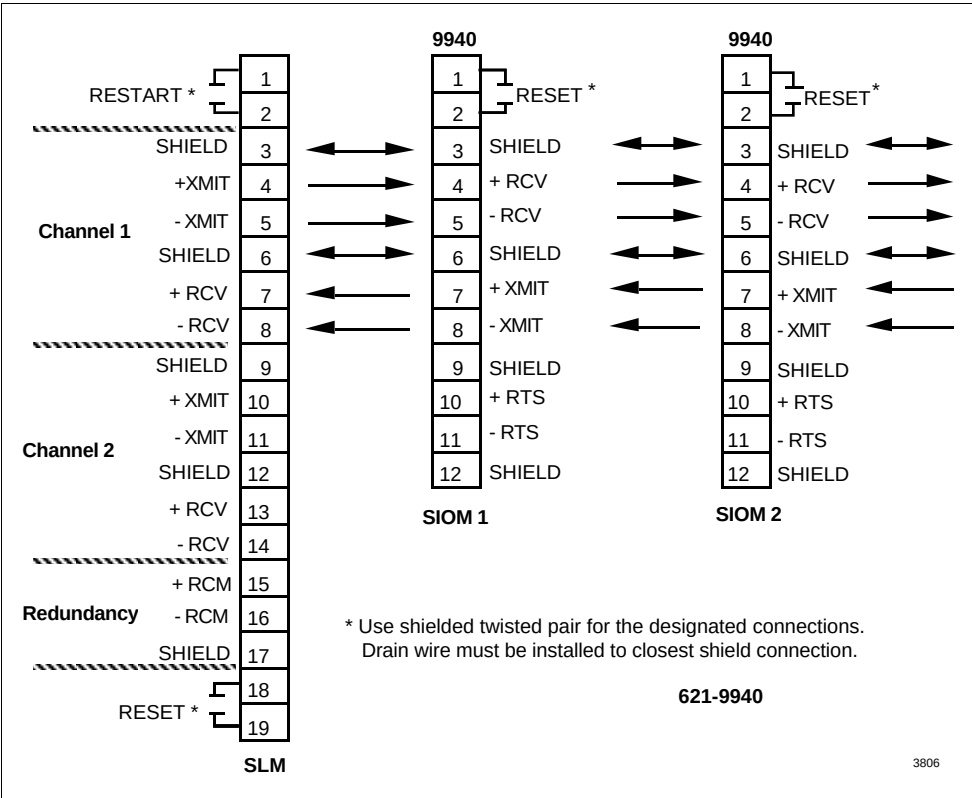
**SLM and SIOM cable connectors**

The model 621-9939 SLM contains a 19-pin terminal block for the cable connection. This model contains two communication channels. The terminal assignments and wiring connections are shown in Figure 3-6 or 3-7.

The model 621-9940 SIOM contains a 12-pin terminal block for the cable connection. This model has one serial I/O port. The terminal assignments and wiring connections are shown in Figure 3-5.

The model 621-9938R SIOM contains a two 12-pin terminal block for the cable connection. This model has two serial I/O ports. The terminal assignments and wiring connections are shown in Figure 3-6.

Figure 3-5 Single Channel SLM and SIOM Terminal Block Wiring Assignments

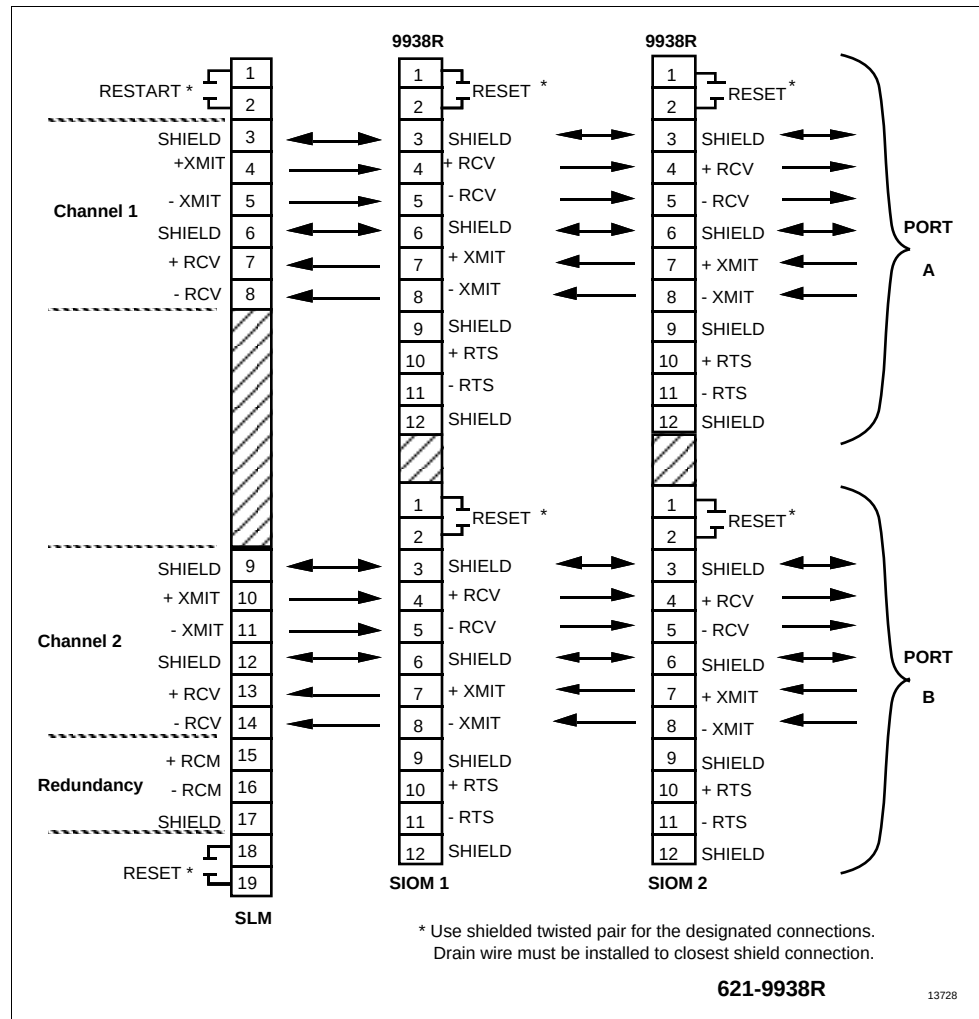


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## 3.10.2 Serial I/O Subsystem, Continued

SLM and SIOM cable connectors, continued

Figure 3-6 Dual Channel SLM and SIOM Terminal Block Wiring Assignments



### 3.10.3 Redundancy Control

---

#### Overview

The Logic Manager's redundancy control provides a means of incorporating a backup processor in the Logic Manager subsystem. The redundancy control consists of two identically configured processor card files, each containing a model 620-0059 Redundancy Control Module (RCM).

---

#### Terminology of redundancy

There are several terms that are used interchangeably when discussing redundancy control. Table 3-8 lists the terms that you may encounter.

Table 3-8 Terms Used in Processor Redundancy

| <b>Processor A *</b><br><b>(physically located in upper I/O rack)</b><br><b>Runs Process</b> | <b>Processor B*</b><br><b>(physically located in lower I/O rack)</b><br><b>Standby Status</b> |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Preferred                                                                                    | Non-Preferred                                                                                 |
| Primary                                                                                      | Secondary                                                                                     |
| Active                                                                                       | Inactive                                                                                      |
| Lead (or OK)                                                                                 | Non-Lead (or Backup)                                                                          |
| Master                                                                                       | Slave                                                                                         |

\* The roles (and terminology) of the processor(s) are reversed when a switchover occurs. When powering up, the initial role of the processors are determined by the setting of a DIP switch on each unit.

---

#### Redundancy cable

The RCMs exchange data through a model 628-5000 RCM Data cable that is supplied with the RCMs. The cable is 3 meters (10 feet) long and has 25-pin, D-type connectors at each end of the cable. The RCMs initiate an orderly switchover of the subsystem I/O from the lead (primary) to the backup (secondary) processor if a critical processor or serial link fault occurs.

---

#### Primary/secondary processor selection

One processor must be designated and configured as the primary or lead processor by the user, and the other processor as the secondary or backup processor. The configuration is implemented by a switch on each RCM.

---

#### Serial or parallel I/O subsystem

Both processors are connected to a common I/O subsystem that can be parallel I/O, serial I/O, or a combination of the two. Only serial I/O subsystems are recommended. Figure 3-8 shows the configuration of a redundant Logic Manager with a serial I/O subsystem.

---

*Continued on next page*

### 3.10.3 Redundancy Control, Continued

---

**Redundant serial I/O communication links**

A redundant Logic Manager with a serial I/O subsystem requires that the model 621-9938R Serial I/O Module (SIOM) be installed in each serial I/O card file. This version of the SIOM has dual communication ports that provide redundant serial I/O lines as shown in Figure 3-8.

---

**Loader/Terminal communications link**

When installed, the optional model 621-9928R Serial Link Selector Module (SLS) provides a serial communications link that is normally used by the Loader/Terminal. The link is always connected to the active processor (primary or secondary).

---

**RCM cabling**

Each RCM originates a control signal, called the Control Switchover Link, that is wired to corresponding switch control inputs on the LMMs, SLMs, and the SLSS, if used. These links are wired with shielded, twisted-pair cable that has a characteristic impedance greater than 60 ohms, such as the Belden 8761 cable.

The cable is wired starting at the RCM and follows the path shown in Figure 3-7. At the RCM, LMM, and optional SLS, connections are made at the 3-position terminal blocks. The three positions are labeled **+**, **-**, and **S**. The **+** and **-** terminals are for the twisted-pair connections, and the **S** terminal is for the shield drain wire connections.

The processor configured as the preferred processor is typically connected to the terminal blocks labeled **A**.

---

**SLM cabling**

At the SLM, the connections are made to positions **15 (+)**, **16 (-)**, and **17 (S)** on the 19-position terminal block as shown in Figure 3-7. When the cabling is complete, all of the **+** terminals should be connected together, all of the **-** terminals should be connected together, and the shield drain wires should be connected to the **S** terminals at the ends of each cable segment.

---

**Switchover link termination**

The last connection point in each link requires a 100 ohm, 1/2 watt terminating resistor. In Figure 3-7, this would be at the SLS. In the SLS, terminating resistors for the line are activated by a switch. If a SLS is not present, the SLM is at the end of the control switchover link. A 100 ohm resistor supplied in a terminating resistor kit with the SLM is installed in the SLM for this purpose. The maximum allowable length of each control switchover link is 6 meters (20 feet).

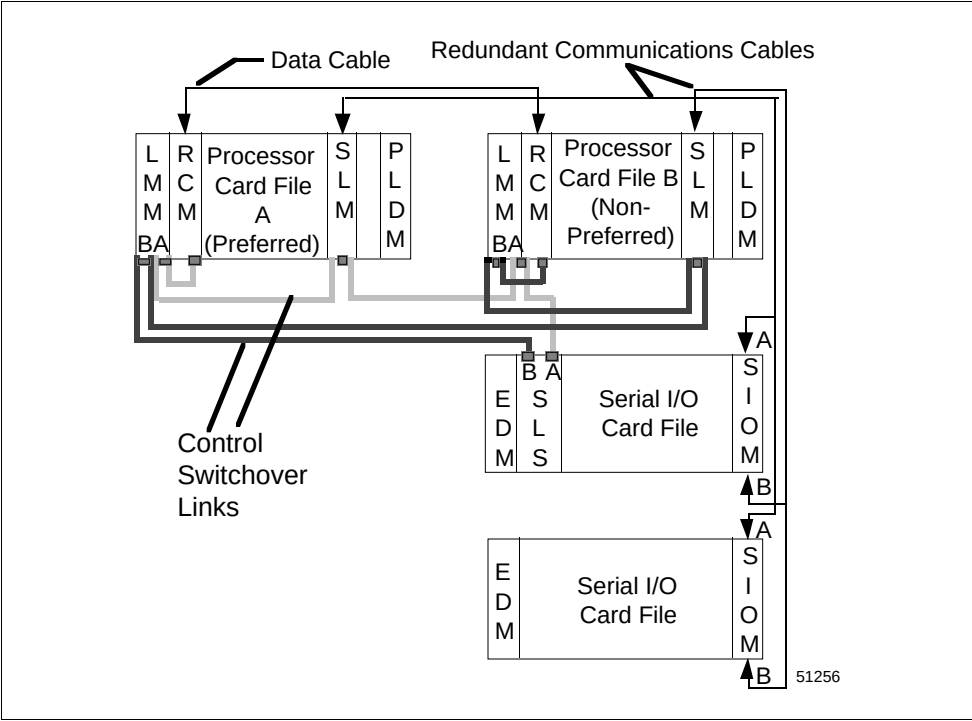
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*Continued on next page*

3.10.3 Redundancy Control, Continued

Switchover link termination, continued

Figure 3-7 Redundant Logic Manager with Serial I/O Subsystem (2 Channel)



## 3.11 I/O Terminal Blocks

### Description

An I/O module terminal block set, 8- or 16-point, consists of a top and bottom half. The halves are labeled **TOP** and **BOTTOM** to prevent improper installation. The **TOP** terminals are always labeled **L1** or **DC+**, and the **BOTTOM** terminals are always labeled **L2** or **DC-**.

**T1** and **T2**, and **B1** and **B2** on the 8-point terminal block models are jumpered at the factory to provide 12-amp capacity.

**T1**, **T2**, and **T3**, and **B1**, **B2**, and **B3** on the 16-point terminal block models are jumpered at the factory to provide 18-amp capacity.

The terminals are screw terminals with pressure pads.

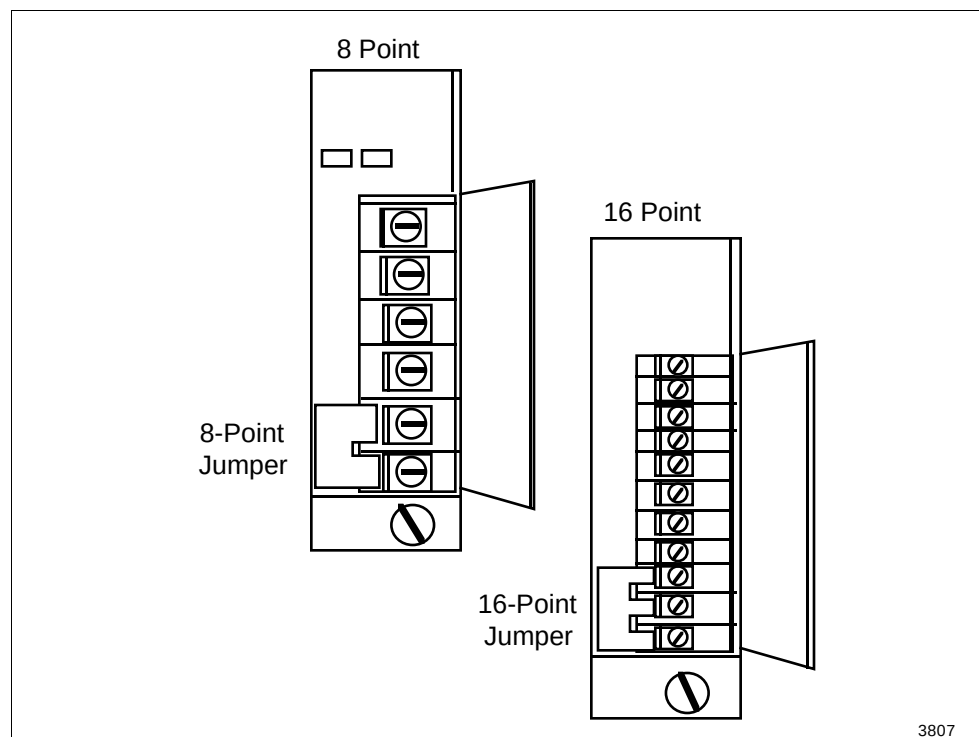
### Hinged coverplate provided

Each terminal block set includes a hinged terminal coverplate to protect the terminals and provide for placement of I/O address labels.

### Installing 8- and 16-point terminal blocks

The terminal block is inserted into the T-shaped slots at the top and bottom of the card file. A captive panel screw and two locating pins on each terminal block secure it to the I/O module and the card file. Figure 3-8 illustrates the 8-point and 16-point terminal blocks.

Figure 3-8 8-Point and 16-Point I/O Terminal Blocks



*Continued on next page*



## 3.11 I/O Terminal Blocks, Continued

### I/O card removal

The terminal blocks can be separated from the I/O module and locked in an open position without disconnecting the wiring. This allows removal and replacement of the I/O card with minimum effort.

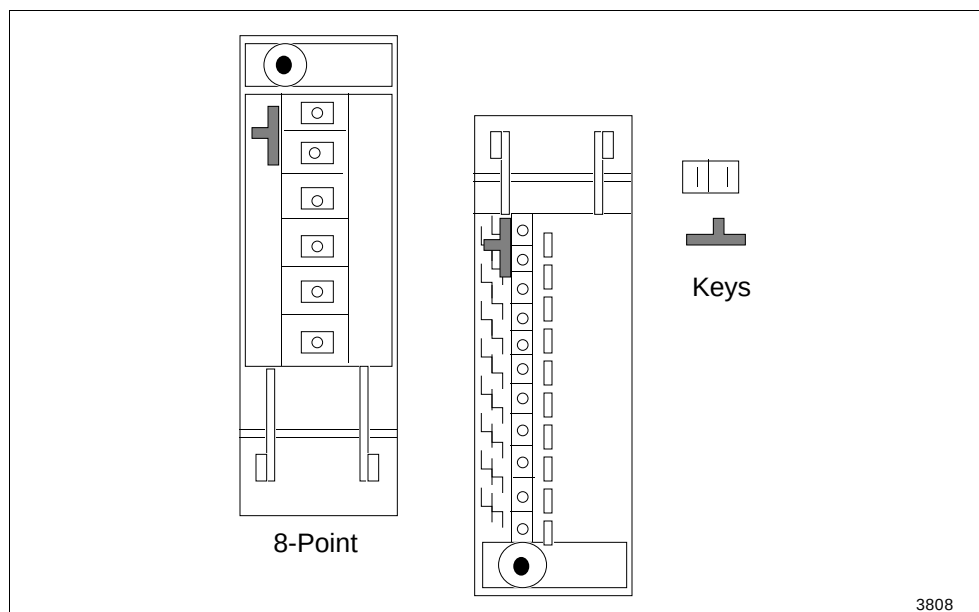
### Terminal block keying

The I/O circuit boards have slots which differentiate the various module voltage levels available in the Logic Manager subsystem.

To prevent damage to the replacement card, the user should install terminal block keys matching the circuit board's slot. This ensures that a card is replaced with the proper type of module. Four keys are supplied with each terminal block set.

See Figure 3-9 for an illustration of 8-point and 16-point keyed terminal blocks.

Figure 3-9 I/O Terminal Block Keys



*Continued on next page*

## 3.11 I/O Terminal Blocks, Continued

### Terminal block installation procedure

The 8-point and 16-point terminal blocks are installed using the following procedure.

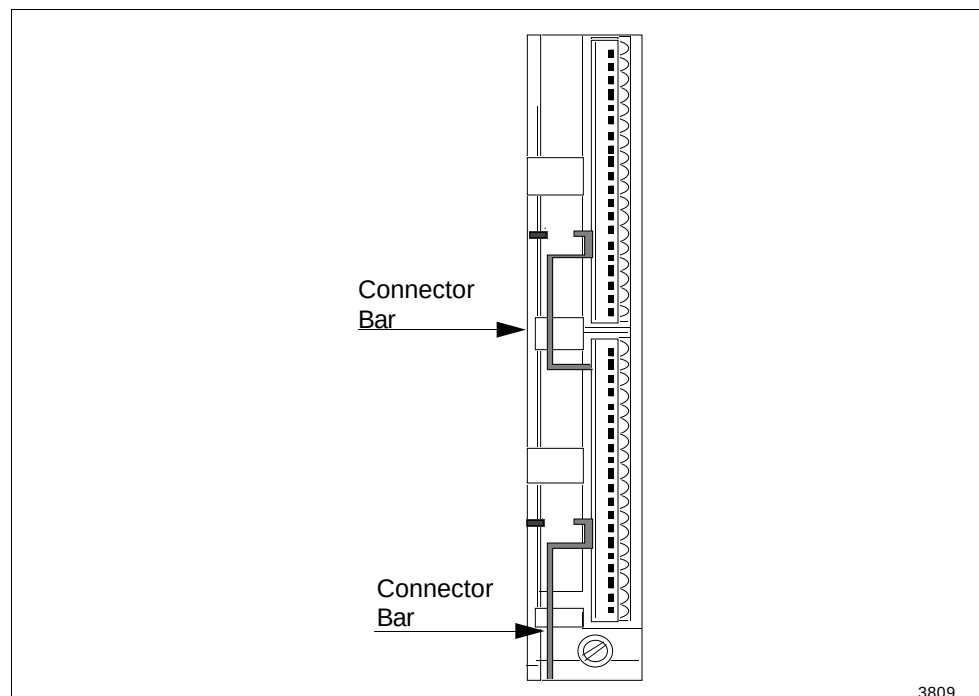
1. Keys are inserted in the terminal blocks to match the type of I/O card.
2. On the 8-point module, the tie wrap is inserted through the left hole in the terminal.
3. The terminal blocks are inserted into the T-shaped slots at the top and bottom of the card file.
4. The terminal blocks are slid forward and the **BOTTOM** terminal block is pulled down, while the **TOP** terminal block is pulled up over the I/O module's edge.
5. The locating pins and captive panel screw are aligned with their mating holes, and then the panel screw is tightened to secure the terminal blocks to the I/O module and card file.

### 32-point terminal block description

The 32-point I/O modules use two 19-position terminal blocks that are attached to the front of the I/O module. The field wiring is installed in the front of the module by set screws that are accessed without removing the connector from the module. If a connector must be removed, metal connector bars eject the connectors from the module.

Figure 3-10 is an illustration of the 16-point terminal block.

Figure 3-10 32-Point I/O Terminal Blocks



## 3.12 I/O Multidrop Application Terminating Resistors

---

### Modules using EIA-422/485 multidrop

The following Logic Manager products are equipped with transmitter/receiver components capable of operating on an EIA-422/485 serial multidrop communications network. Table 3-9 lists the associated user manual for each Logic Manager product.

Table 3-9 Logic Manager Products and User Manuals

| Model Number                                     | User Manual Form Number |
|--------------------------------------------------|-------------------------|
| 620-0048 and 620-0052<br>Data Collection Modules | 620-8980                |
| 621-0012<br>ASCII Communications Module          | 621-8999                |
| 627-1002 and 627-1003<br>MiniCOP Modules         | 627-8991                |

---

### Use biasing resistors

To ensure proper operation, there are several factors you should consider when installing a multidrop network. One key factor is biasing terminating resistors on the network to limit line noise that can cause erroneous processor interrupts.

---

### References

Specific information on biasing terminating resistors in a multidrop network can be found in user manuals for the models listed above. General information on biasing terminating resistors in a multidrop network can be found in the MiniCOP User Manual Appendix B (Form Number 627-8991, Dated 10/93).

---



## Section 4 – Model MU-CBSM01/MU-CBDM01 Cabinets

### 4.1 Overview

---

#### Section Contents

The topics covered in this section are:

|     | Topic                         | See Page |
|-----|-------------------------------|----------|
| 4.1 | Overview.....                 | 87       |
| 4.2 | Cabinet Description .....     | 90       |
| 4.3 | Card File Configurations..... | 94       |
| 4.4 | Card File Description.....    | 95       |
| 4.5 | Cabinet Floor Planning.....   | 96       |

---

#### Introduction

The Logic Manager model MU-CBSM01 and MU-CBDM01 cabinets provide single and dual access, respectively. Access is defined as having a single or dual entry, front and rear doors. The model MU-CBSM01 and MU-CBDM01 cabinets are also referred to as a Markhon cabinet because Markon is the manufacturer.

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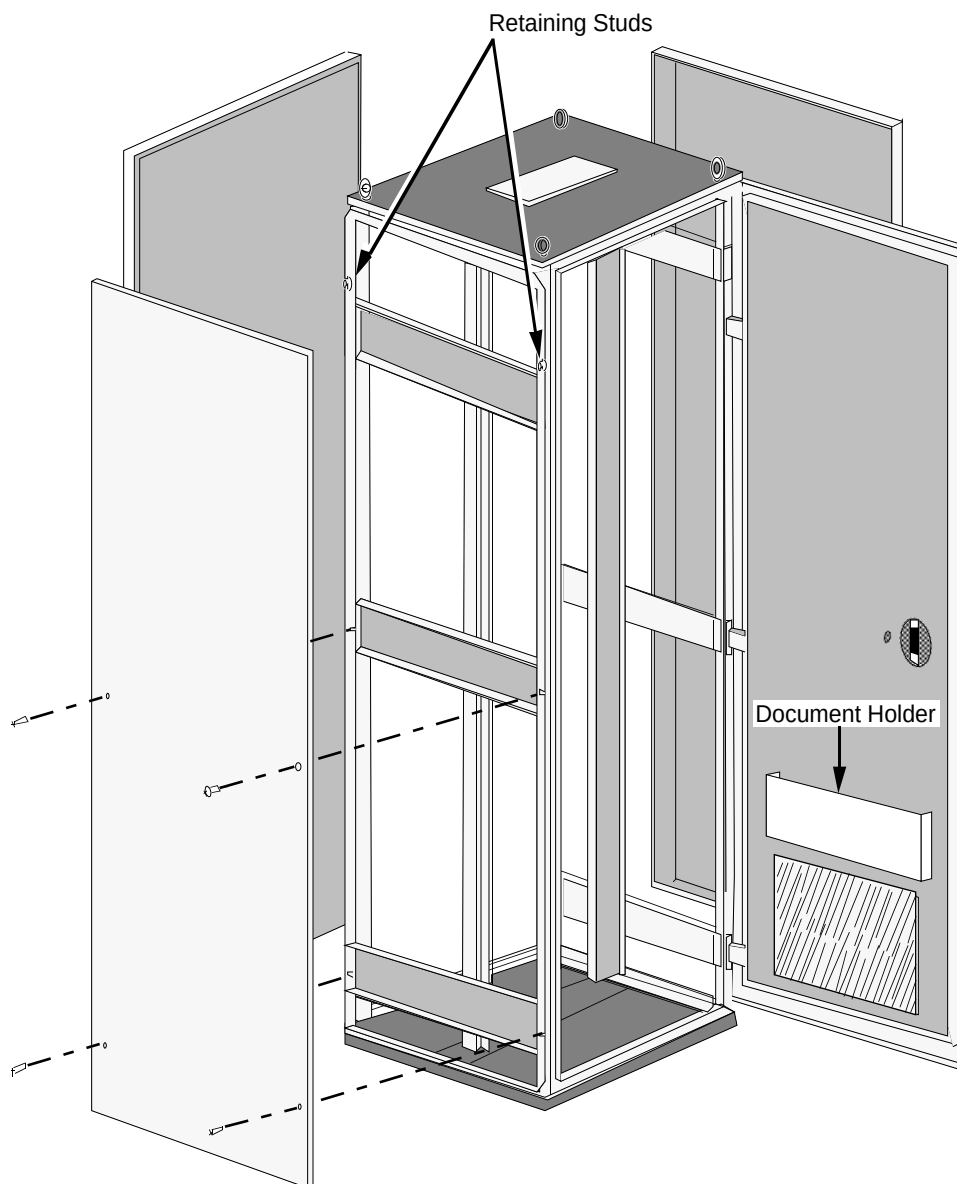
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## 4.1 Overview, Continued

### Single access cabinet description

The single access cabinet has one equipment entry point and that is at the front of the cabinet through a single door. The Logic Manager card files are mounted inside the single access door. The cabinet side panels can be removed for access. Figure 4-1 is an illustration of a single access cabinet.

Figure 4-1 Single Access Cabinet



| Dimension     | Width         | Depth       | Height         |
|---------------|---------------|-------------|----------------|
| Single Access | .8 m/31.5 in. | .55 m/21.75 | 2.1 m/81.5 in. |

3096

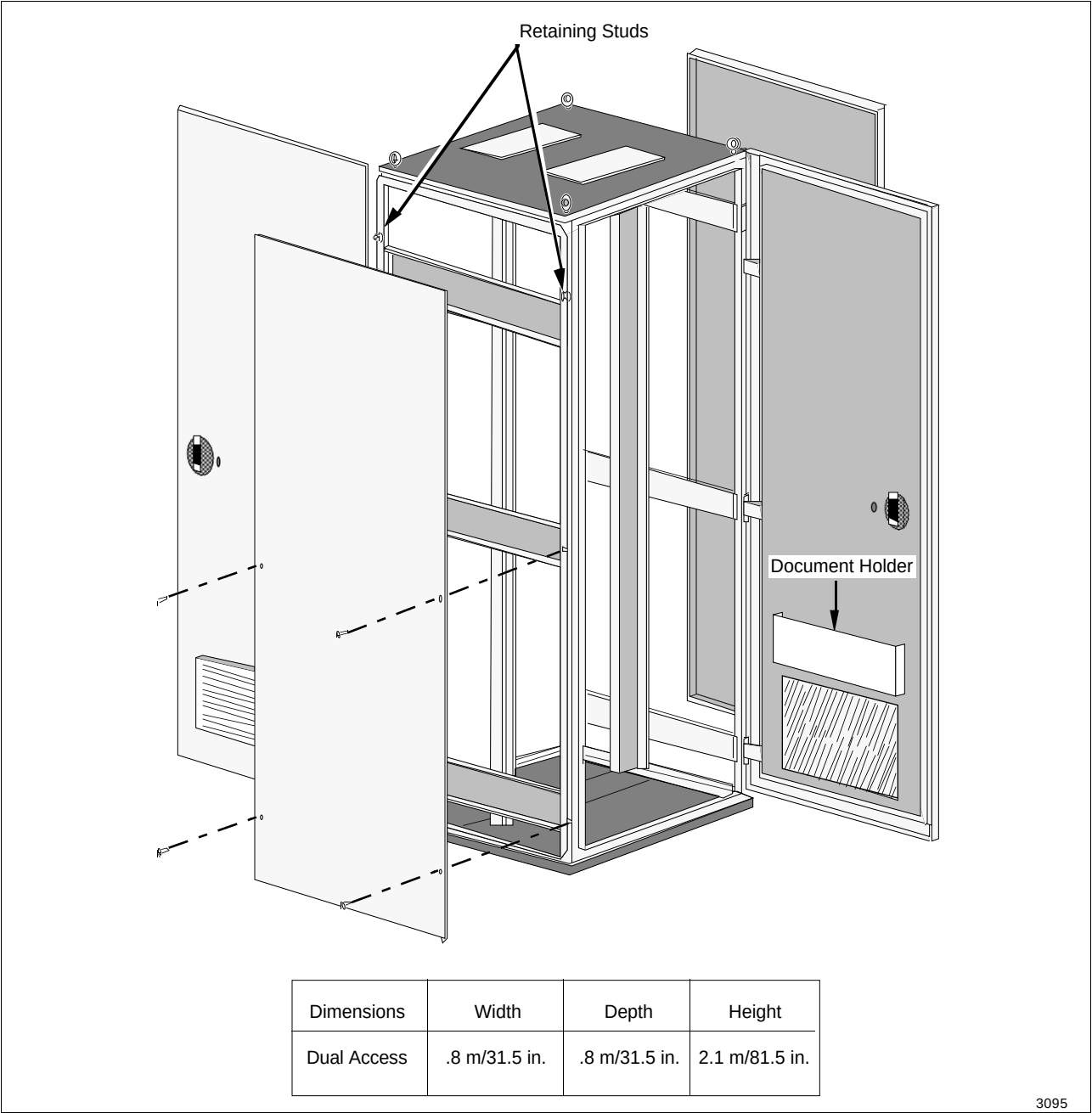
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# 4.1 Overview, Continued

**Dual access cabinet description**

The dual access cabinet has two equipment entry points. The Logic Manager card files are mounted inside either the front or rear access doors, or both. The cabinet side panels can be removed for access. Figure 4-2 is an illustration of a dual access cabinet.

Figure 4-2 Dual Access Cabinet



## 4.2 Cabinet Description

### Cabinet dimensions

The single access cabinet, with the front door and side and rear panels installed, has the following exterior dimensions.

- 0.8 meters wide x 0.55 meters deep x 2.1 meters high  
(31.5 inches x 21.75 inches x 81.5 inches)

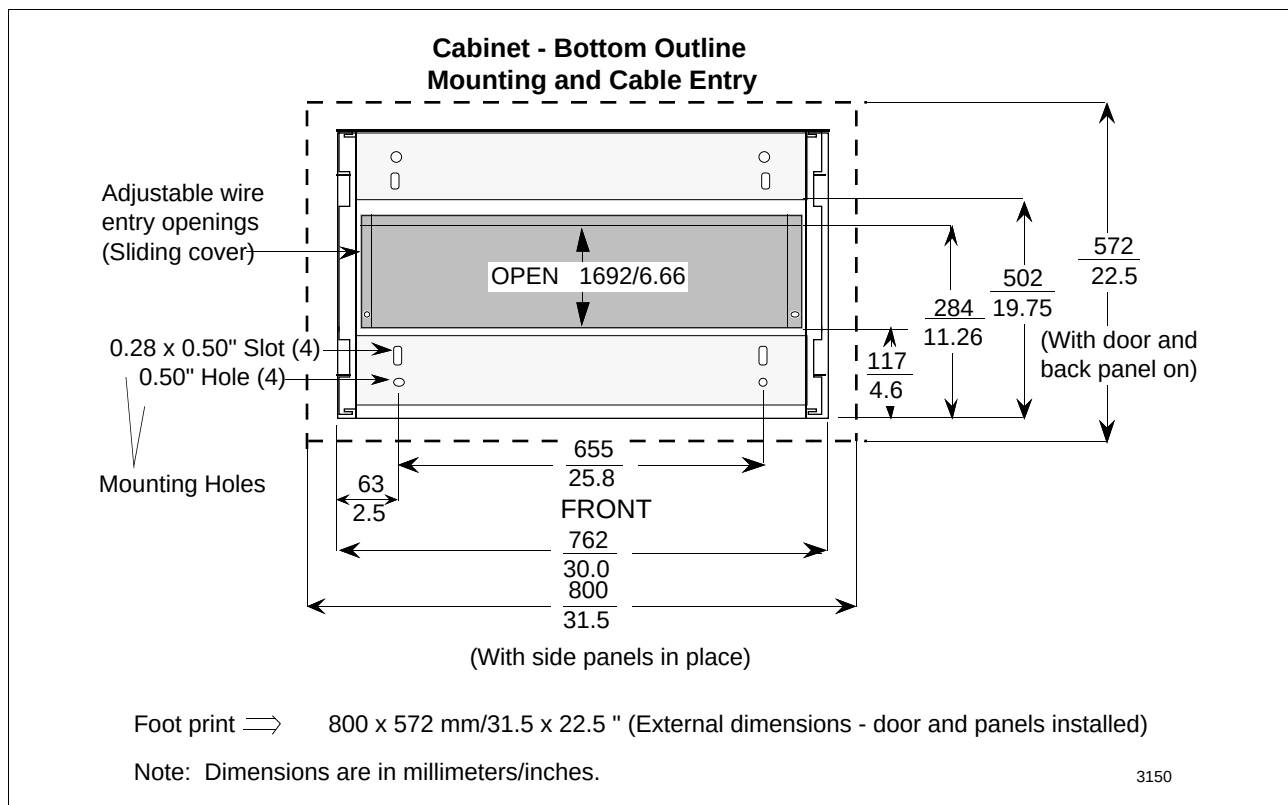
The dual access cabinet, with the front and rear doors and side panels installed, has the following exterior dimensions.

- 0.8 meters wide x 0.55 meters deep x 2.1 meters high  
(31.5 inches x 21.75 inches x 81.5 inches)

### Top and bottom cabinet entry

Both cabinets are NEMA 1 rated and support top and bottom entry for process control wiring. Top entry requires removal of the top plate(s), and/or associated fan assemblies, before entry holes can be punched in the top panel. Honeywell will prepunch the holes if specified when ordering the cabinet. Figures 4-3 and 4-4 are illustrations of the bottom cable entry slots for the single and dual access cabinets, respectively.

Figure 4-3 Single Access Cabinet Bottom Cable Entry Slots



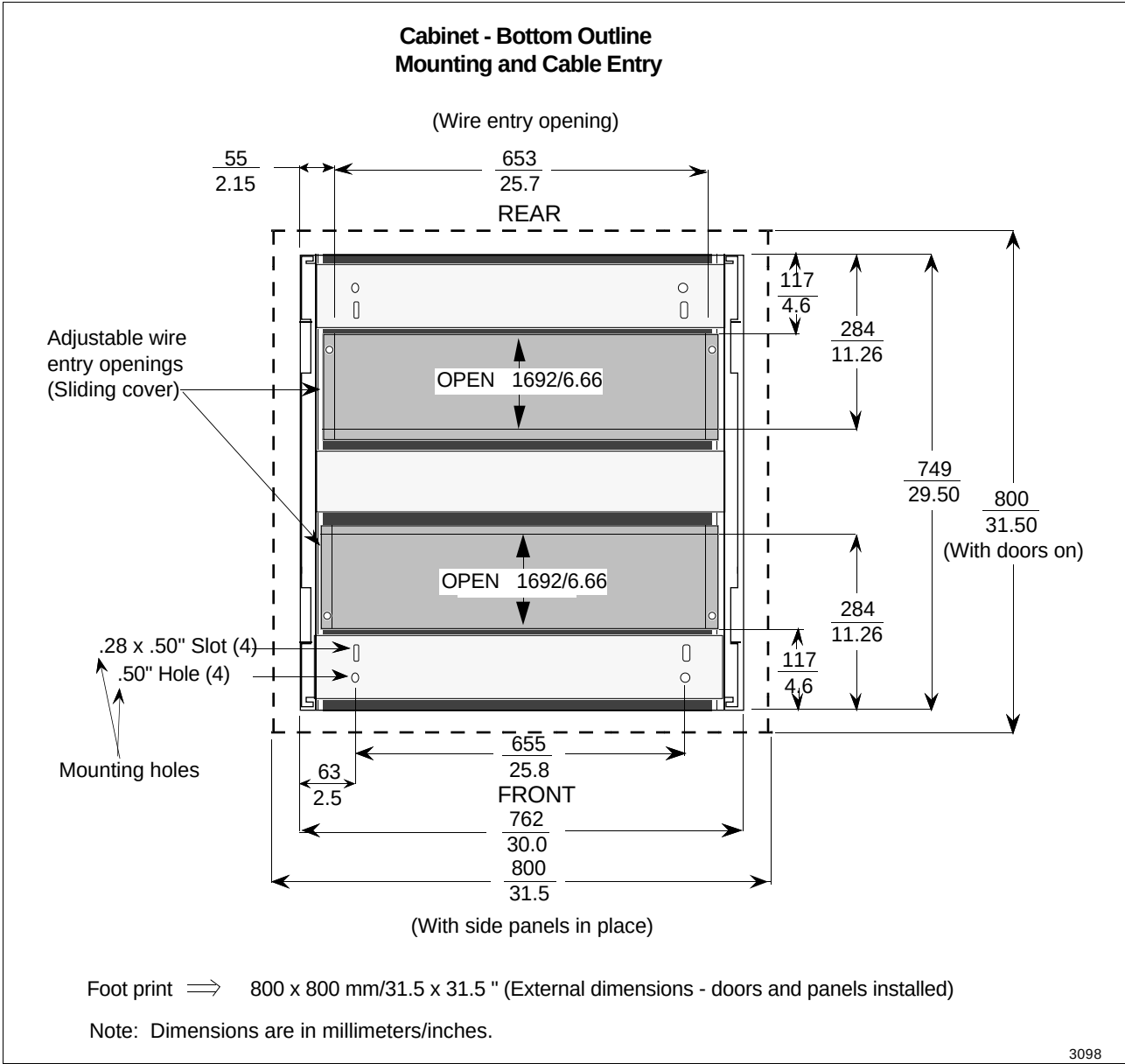
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## 4.2 Cabinet Description, Continued

### Top and bottom cabinet entry, continued

Figure 4-4 Dual Access Cabinet Bottom Cable Entry Slots



## 4.2 Cabinet Description, Continued

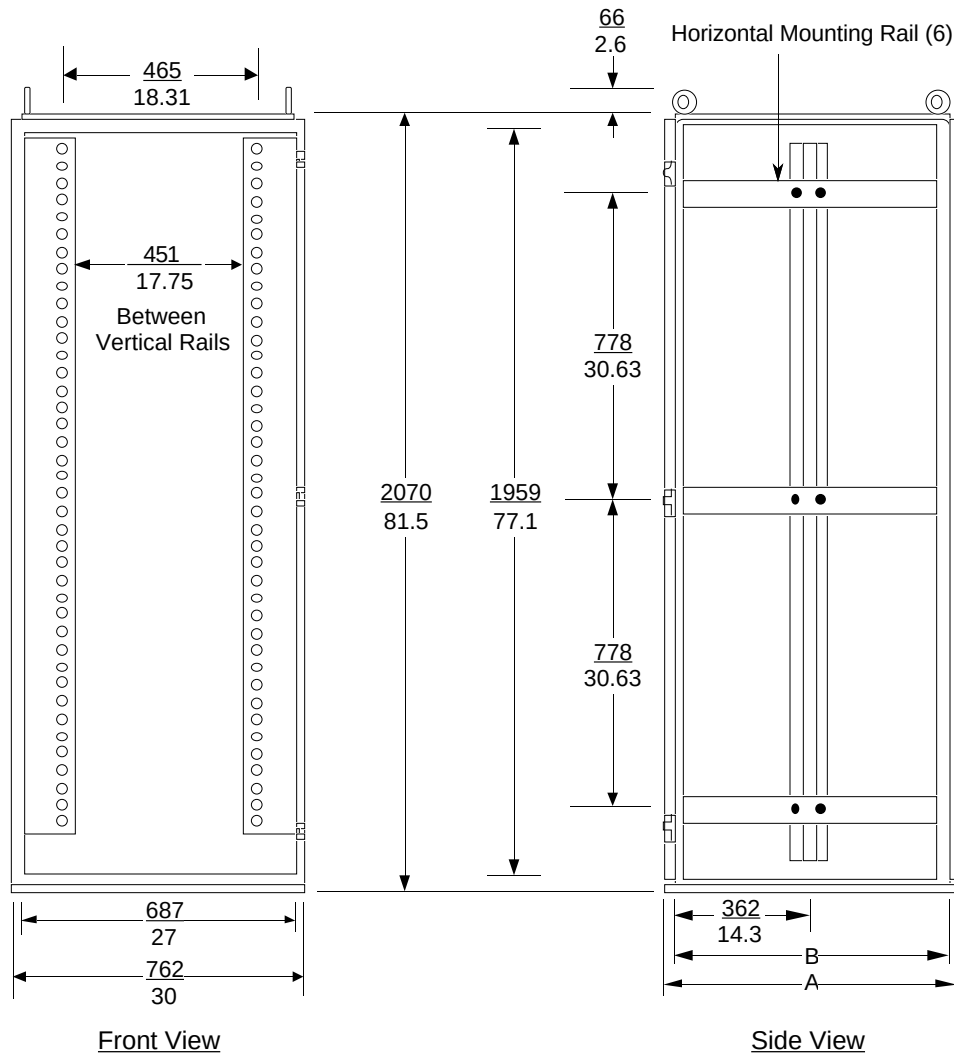
|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Independent cabinet entry</b>  | <p>The dual access cabinet has two independent entry sides. A single access cabinet has only one entry side. Logic Manager card files can be installed in either side of the cabinet without being constrained by the equipment installed in the other side of the cabinet.</p> <p>The cabinet can be ordered with doors hinged on either the left or right side. The door latch is recessed and lockable.</p>                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Cabinet complexing</b>         | <p>A maximum of four cabinets can be complexed together without intervening side panels by a complexing kit, Honeywell part number 51109532-200 (single access cabinet ), or 51109532-100 (dual access cabinet).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>NEMA 12</b>                    | <p>NEMA 12 rated cabinets can be ordered from Honeywell.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Cabinet cooling</b>            | <p>Cabinet cooling is accomplished with the use of one or two fan assemblies mounted over appropriate cutouts in the cabinet top. Use of a fan assembly is mandatory for a cabinet entry side containing card files. Because of power dissipation, the cabinet configuration determines if a fan assembly is required for a cabinet entry side that contains Logic Manager components. Fan assemblies are available for voltages and frequencies noted below.</p> <ul style="list-style-type: none"><li>• Cabinet Fan Assembly (240 Vac, 50/60 Hz)</li><li>• Cabinet Fan Assembly (120 Vac, 50/60 Hz)</li></ul> <p>A blank plate is available to cover the hole(s) in the top of the cabinet in the event a fan assembly is not needed. The fan opening cabinet top coverplate is Honeywell part number 51304098-200.</p> |
| <b>Cabinet internal structure</b> | <p>The cabinet is provided with an internal structure that is capable of accepting the full-size (19 inch) Logic Manager card file assemblies as shown in Figure 4-5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

*Continued on next page*

## 4.2 Cabinet Description, Continued

### Cabinet internal structure, continued

Figure 4-5 Cabinet Interior Dimensions



| Cabinet       | A                  | B                  |
|---------------|--------------------|--------------------|
| Dual Access   | <u>793</u><br>31.2 | <u>749</u><br>29.5 |
| Single Access | <u>552</u><br>21.7 | <u>508</u><br>20   |

Note: Dimensions are in millimeters/inches.

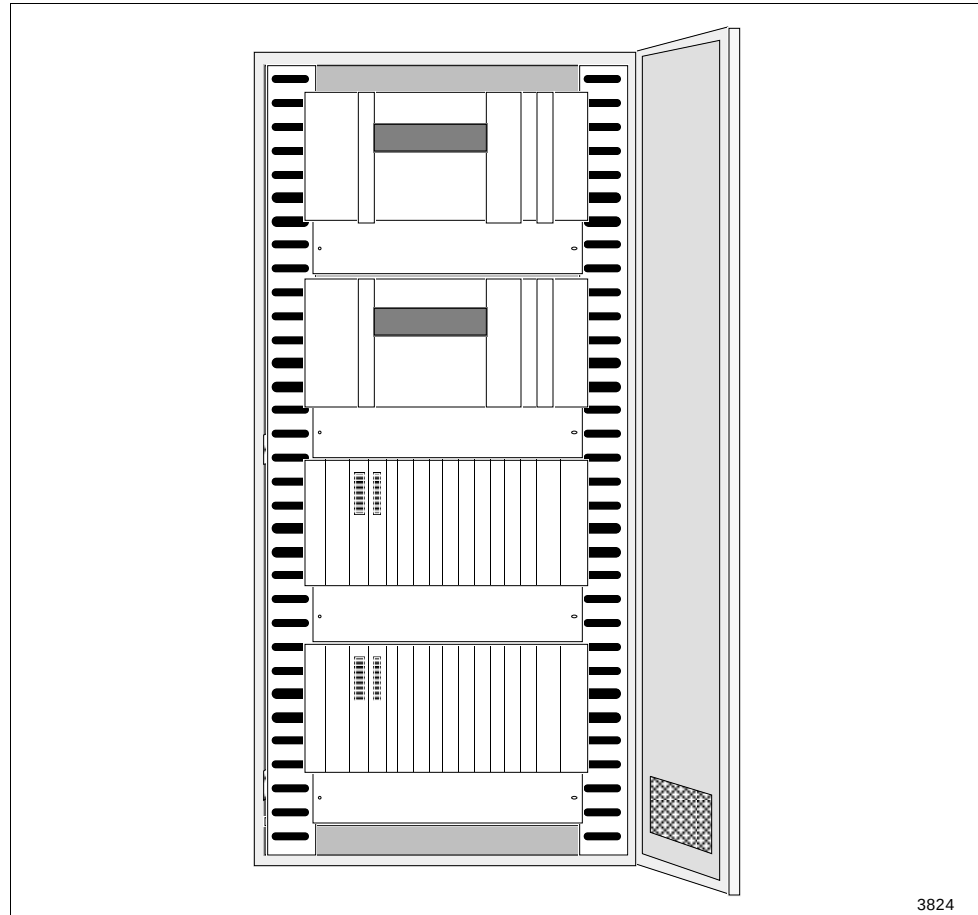
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## 4.3 Card File Configurations

### Cabinet layout

The Logic Manager card files install in the single or dual access cabinets as shown in Figure 4-6.

Figure 4-6 Cabinet Card File Layout

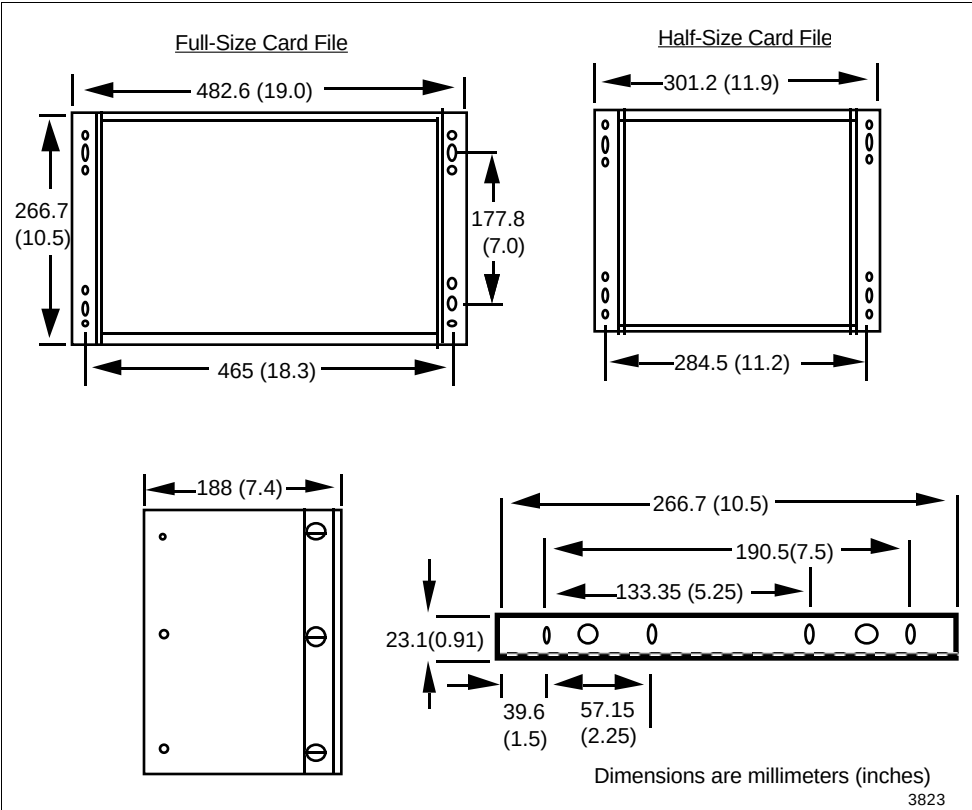


# 4.4 Card File Description

## Description

Both card file assemblies, Processor and I/O, are available as a full-size 19-inch assembly and install on a 19-inch RETMA mounting structure. The I/O card file is also available as a half-size, 12-inch assembly, but must be mounted external to the the Logic Manager cabinet. The dimensions of both card file sizes is shown in Figure 4-7.

Figure 4-7 Full and Half-Size Card File Dimensions

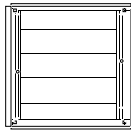


## 4.5 Cabinet Floor Planning

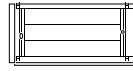
### Floor template

Figure 4-8 is a cabinet floor template that will help you plan your facility layout when installing Process Manager cabinets.

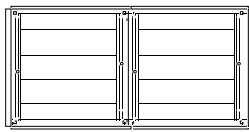
Figure 4-8 Cabinet Floor Planning Templates



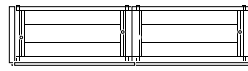
Single Dual Access Cabinet



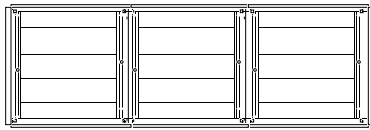
Single Single Access Cabinet



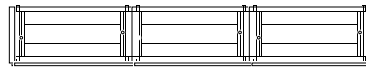
Dual Dual Access Cabinet Complex



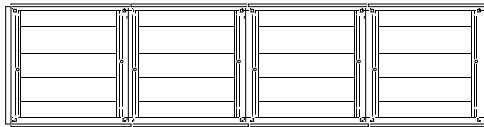
Dual Single Access Cabinet Complex



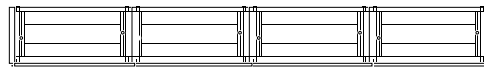
Triple Dual Access Cabinet Complex



Triple Single Access Cabinet Complex



Quad Dual Access Cabinet Complex



Quad Single Access Cabinet Complex

#### Note:

1. Cabinet scale is 1/4 inch = 1 foot.
2. A dual access cabinet is approximately 30 inches by 30 inches. The side and door panels add 1.5 inches to each cabinet dimension. When cabinets are complexed, 1.5 inches is added to the total complex dimensions.
3. A single access cabinet is approximately 30 inches wide and 21 inches deep. The side panels and the single door panel add 3/4 inch to the cabinet depth and 1.5 inches to the cabinet width. When cabinets are complexed, 3/4 inch is added to the complex depth and 1.5 inches is added to the total complex width.

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## Section 5 – Model MU-CBSX01/MU-CBDX01 Cabinets

### 5.1 Overview

---

#### Section Contents

The topics covered in this section are:

|     | Topic                         | See Page |
|-----|-------------------------------|----------|
| 5.1 | Overview.....                 | 97       |
| 5.2 | Cabinet Description .....     | 100      |
| 5.3 | Card File Configurations..... | 104      |
| 5.4 | Card File Description.....    | 105      |
| 5.5 | Cabinet Floor Planning.....   | 106      |

---

#### Introduction

The Logic Manager model MU-CBSX01 and MU-CBDX01 cabinets provide single and dual access, respectively. Access is defined as having a single or dual entry, front and rear doors. The model MU-CBSX01 and MU-CBDX01 cabinets are also referred to as a Rittal cabinet because Rittal is the manufacturer.

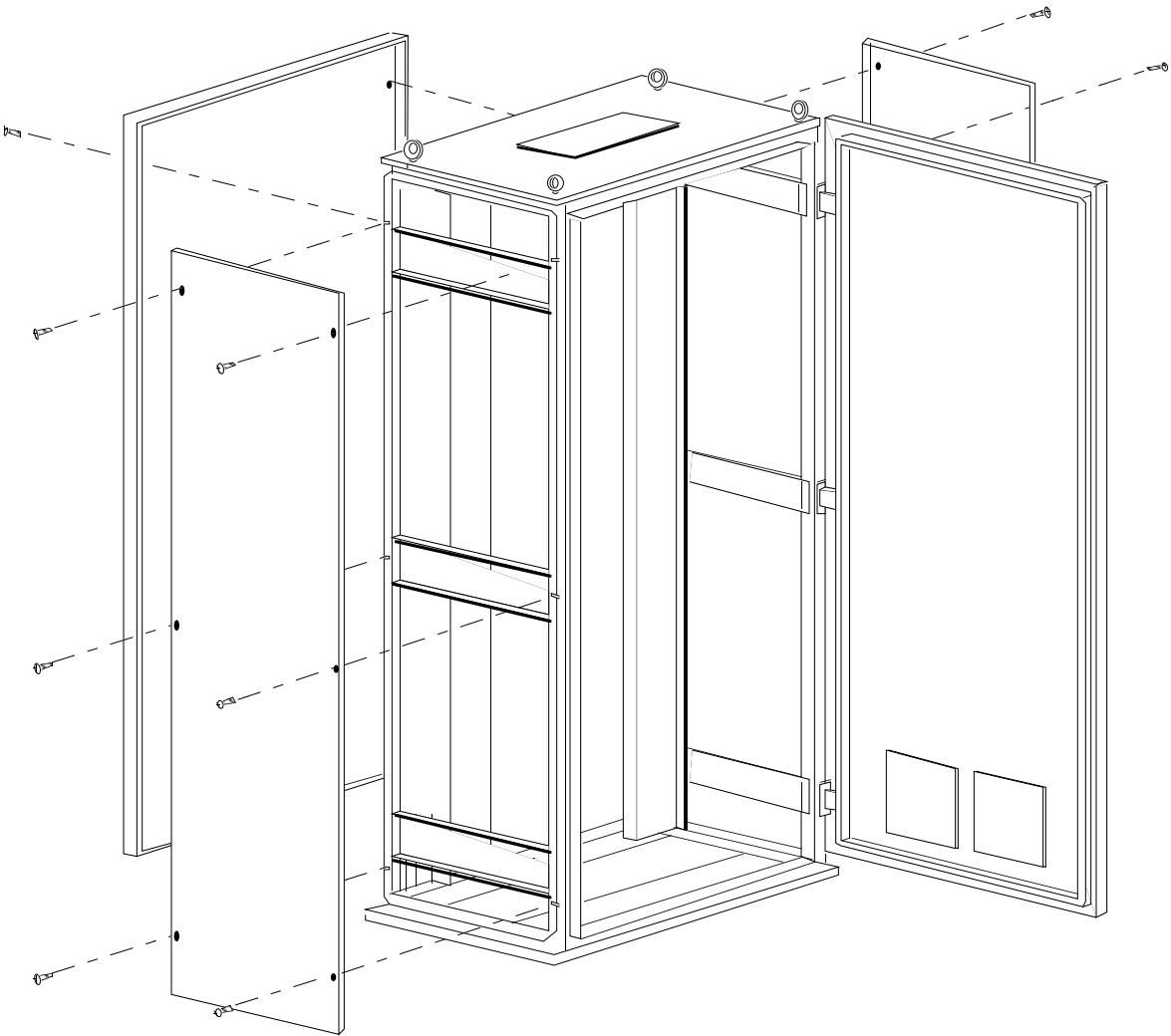
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# 5.1 Overview, Continued

**Single access cabinet description**      The single access cabinet has one equipment entry point and that is in the front of the cabinet through a single door. The Logic Manager card files are mounted inside the single access door. The cabinet side panels can be removed for access. Figure 5-1 is an illustration of a single access cabinet.

Figure 5-1      Single Access Cabinet



| Dimension     | Width          | Depth          | Height       |
|---------------|----------------|----------------|--------------|
| Single Access | .8 m/31.49 in. | .5 m/19.68 in. | 2 m/78.9 in. |

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Continued on next page

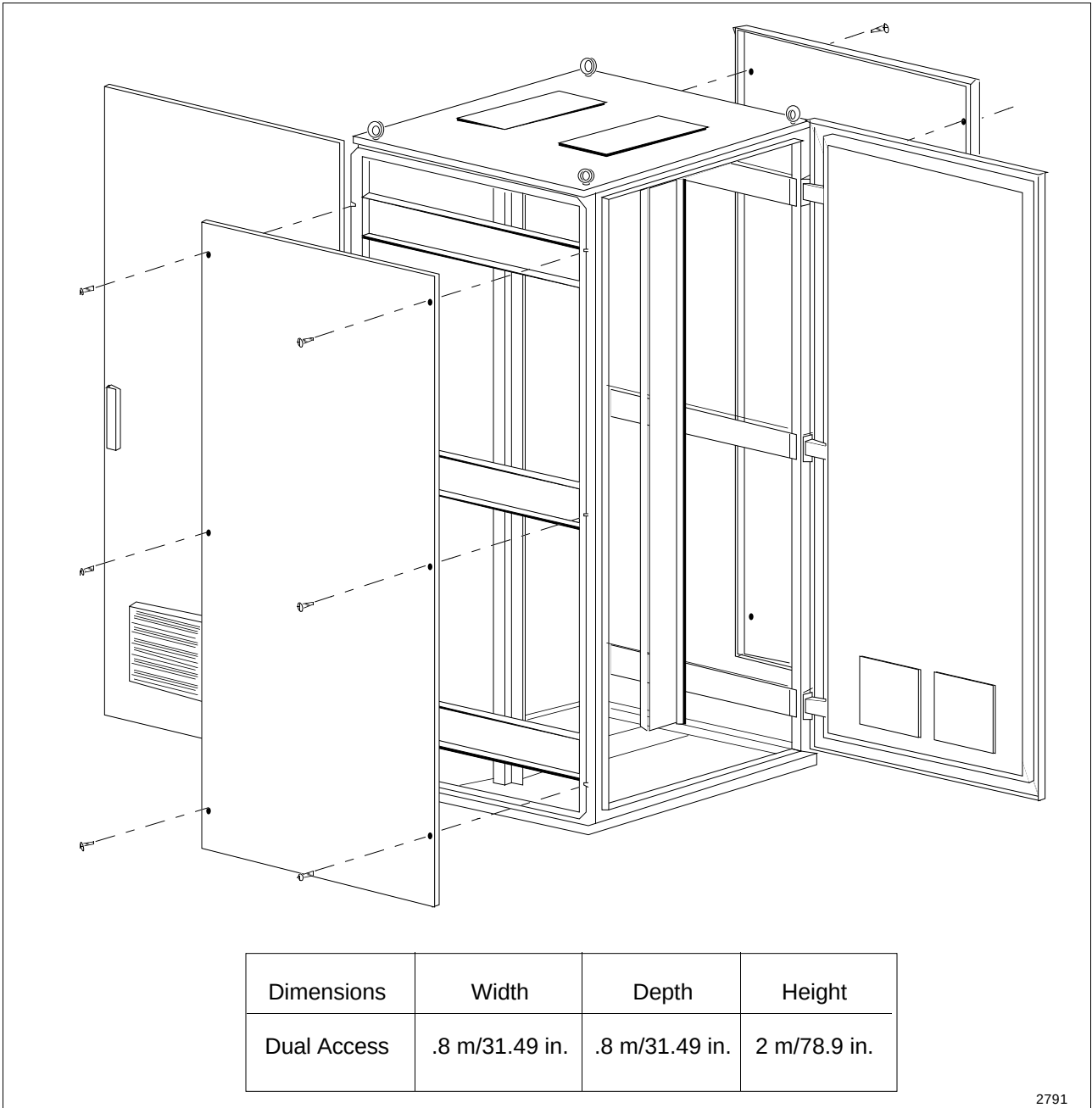


# 5.1 Overview, Continued

**Dual access cabinet description**

The dual access cabinet has two equipment entry points. The Logic Manager card files are mounted inside either the front or rear access doors, or both. The cabinet side panels can be removed for access. Figure 5-2 is an illustration of a dual access cabinet.

Figure 5-2 Dual Access Cabinet



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## 5.2 Cabinet Description

### Cabinet dimensions

The single access cabinet, with the front door and side and rear panels installed, has the following exterior dimensions.

- 0.8 meters wide x 0.5 meters deep x 2 meters high  
(31.5 inches x 19.7 inches x 78.9 inches)

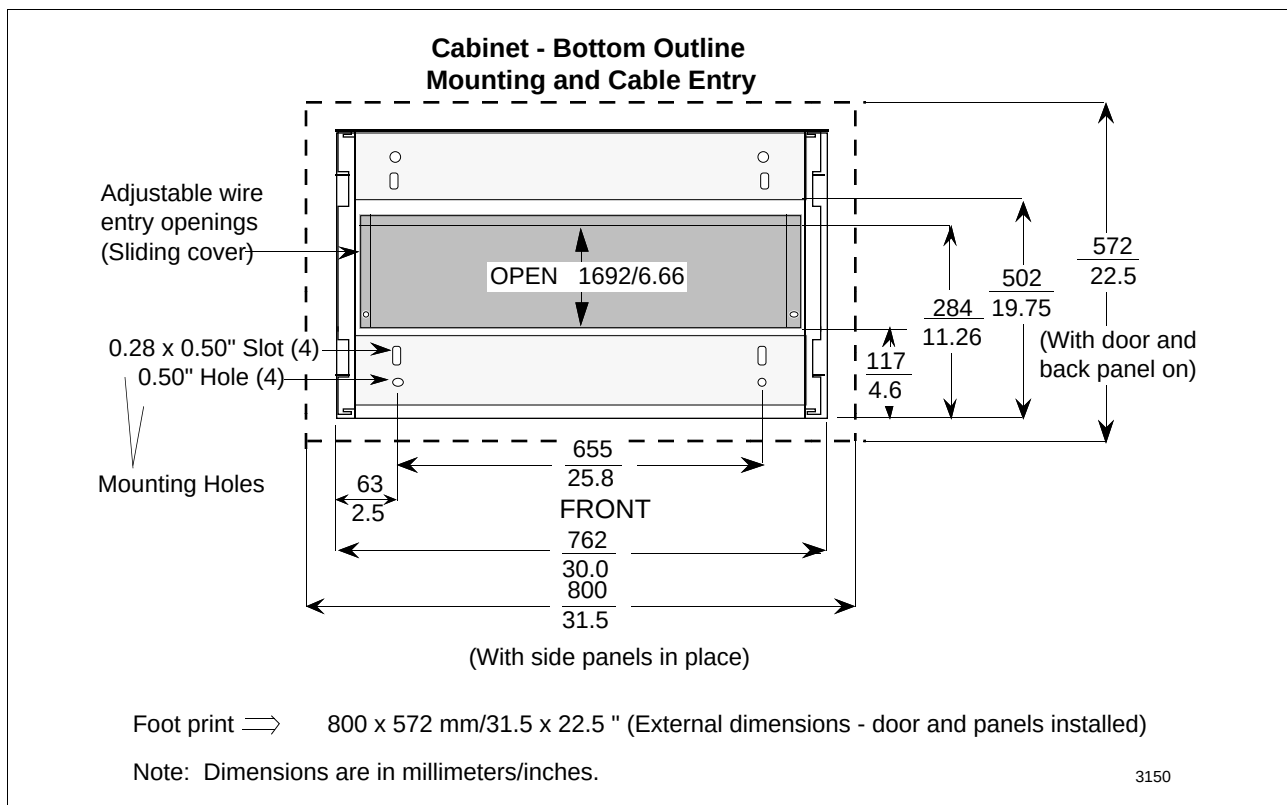
The dual access cabinet, with the front and rear doors and side panels installed, has the following exterior dimensions.

- 0.8 meters wide x 0.55 meters deep x 2.1 meters high  
(31.5 inches x 31.5 inches x 78.9 inches)

### Top and bottom cabinet entry

Both cabinets are NEMA 1 rated and support top and bottom entry for process control wiring. Top entry requires removal of the top plate(s), and/or associated fan assemblies, before entry holes can be punched in the top plate. Honeywell will prepunch the holes if specified when ordering the cabinet. Figures 5-3 and 5-4 are illustrations of the bottom cable entry slots for the single and dual access cabinets, respectively.

Figure 5-3 Single Access Cabinet Bottom Cable Entry Slots

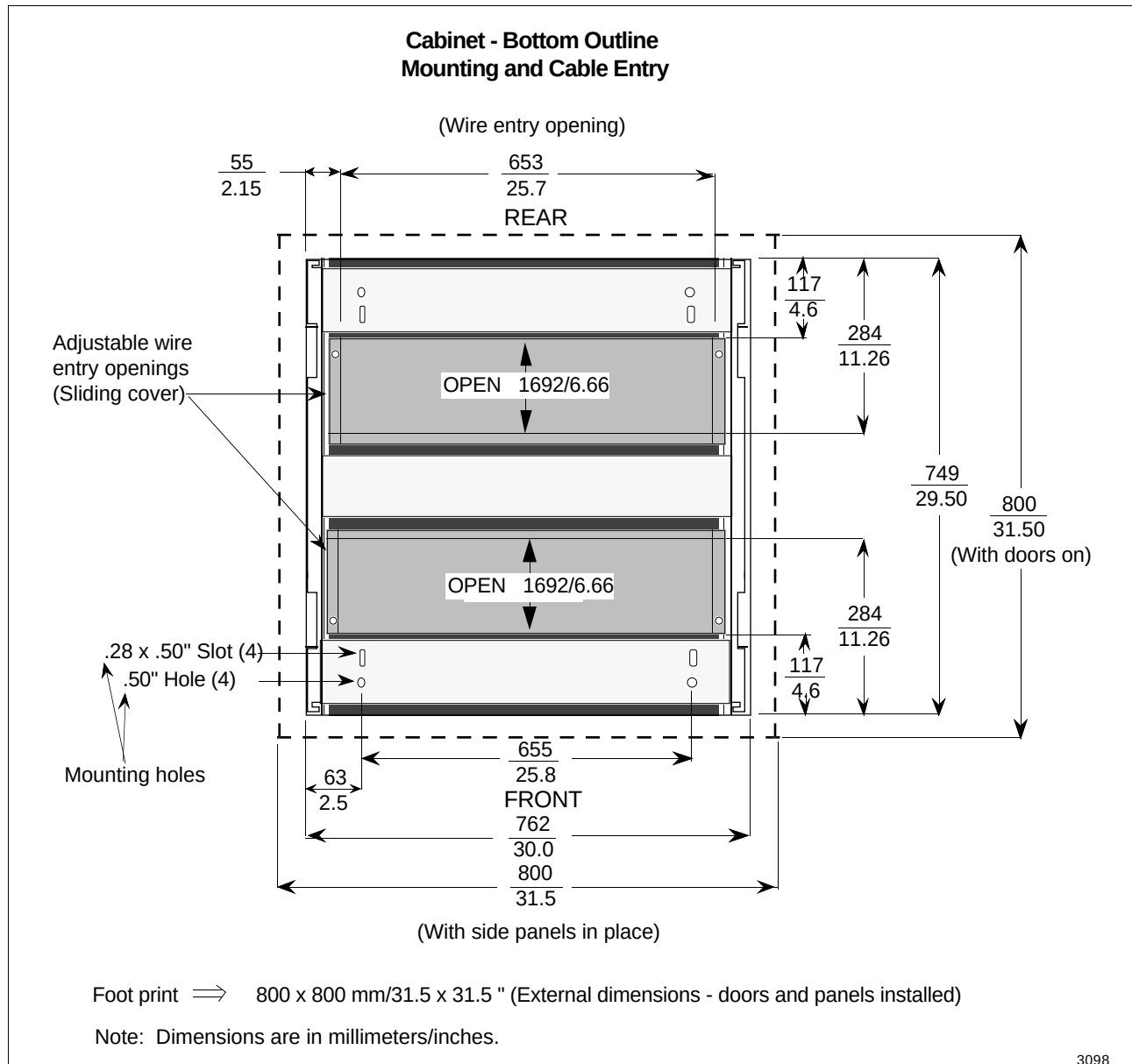


*Continued on next page*

## 5.2 Cabinet Description, Continued

### Top and bottom cabinet entry, continued

Figure 5-4 Dual Access Cabinet Bottom Cable Entry Slots



## 5.2 Cabinet Description, Continued

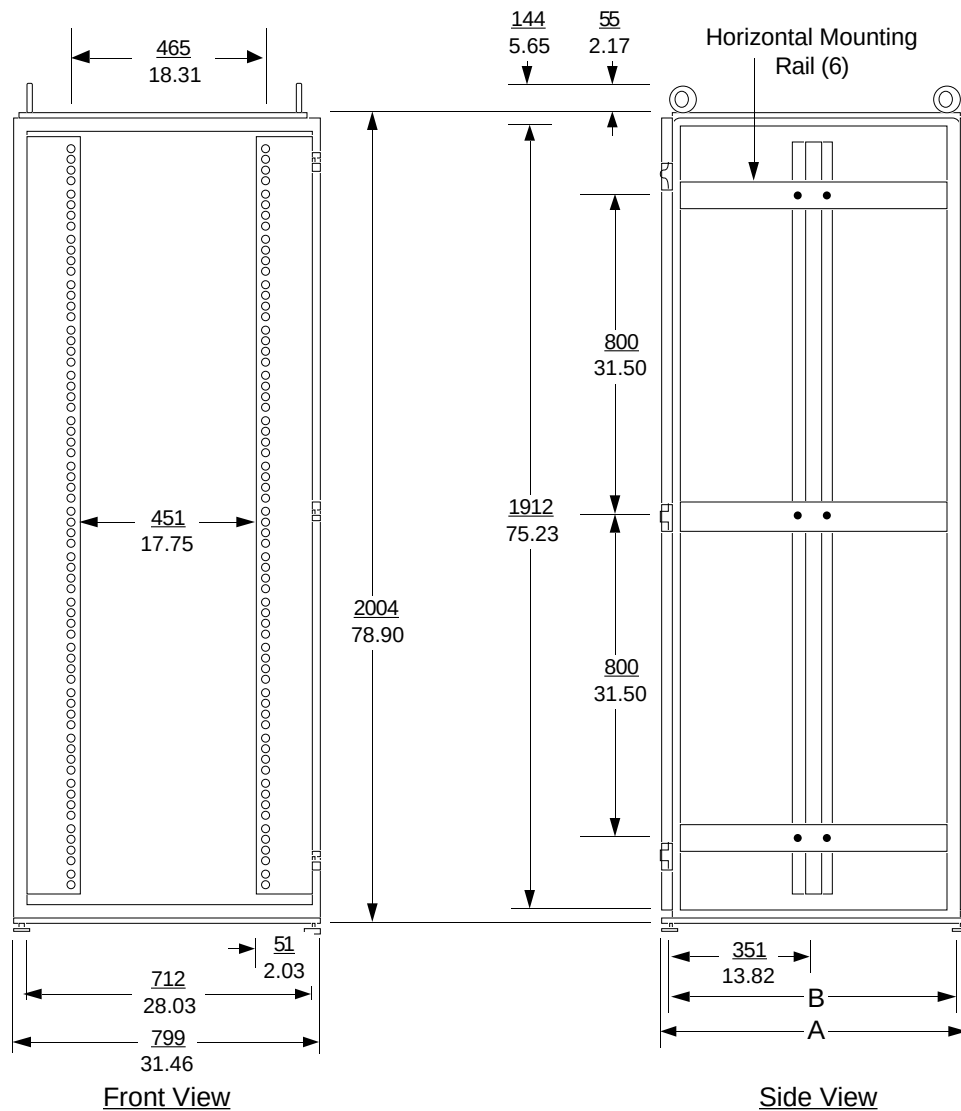
|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Independent cabinet entry</b>  | <p>The dual access cabinet has two independent entry sides. A single access cabinet has only one entry side. Logic Manager card files can be installed in either side of the cabinet without being constrained by the equipment installed in the other side of the cabinet.</p> <p>The cabinet can be ordered with doors hinged on either the left or right side. The door latch is lockable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cabinet complexing</b>         | <p>A maximum of four cabinets can be complexed together without intervening side panels by a complexing kit, Honeywell part number 51109524-200 (single access cabinet ), or 51109524-100 (dual access cabinet).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>NEMA 12</b>                    | <p>NEMA 12 rated cabinets can be ordered from Honeywell.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Cabinet cooling</b>            | <p>Cabinet cooling is accomplished with the use of one or more fan assemblies mounted over appropriate cutouts in the cabinet top. Use of a fan assembly is mandatory for a cabinet entry side containing card files. Because of power dissipation, the cabinet configuration determines if a fan assembly is required for a cabinet entry side containing Logic Manager components. Fan assemblies are available for voltages and frequencies noted below.</p> <ul style="list-style-type: none"><li>• Cabinet Fan Assembly (240 Vac, 50/60 Hz)</li><li>• Cabinet Fan Assembly (120 Vac, 50/60 Hz)</li></ul> <p>A blank plate is available to cover the hole(s) in the top of the cabinet in the event a fan assembly is not needed. The fan opening cabinet top coverplate is Honeywell part number 51304098-200.</p> |
| <b>Cabinet internal structure</b> | <p>The cabinet is provided with an internal structure that is capable of accepting the Logic Manager card file assemblies as shown in Figure 5-5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

*Continued on next page*

## 5.2 Cabinet Description, Continued

### Cabinet internal structure, continued

Figure 5-5 Cabinet Interior Dimensions



| Cabinet       | A                   | B                   |
|---------------|---------------------|---------------------|
| Dual Access   | <u>789</u><br>31.06 | <u>702</u><br>27.64 |
| Single Access | <u>489</u><br>19.25 | <u>402</u><br>15.83 |

Note: Dimensions are in millimeters/inches.

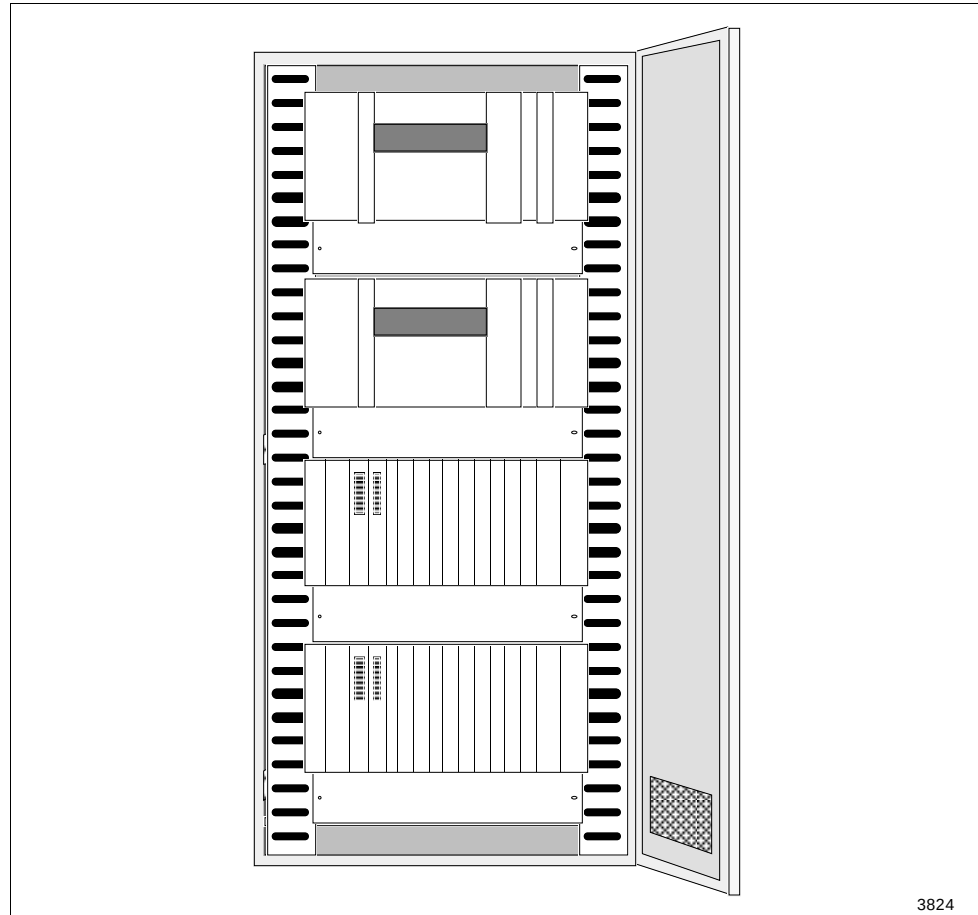
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## 5.3 Card File Configurations

### Cabinet layout

The Logic Manager card files install in the single or dual access cabinets as shown in Figure 5-6.

Figure 5-6 Cabinet Card File Layout

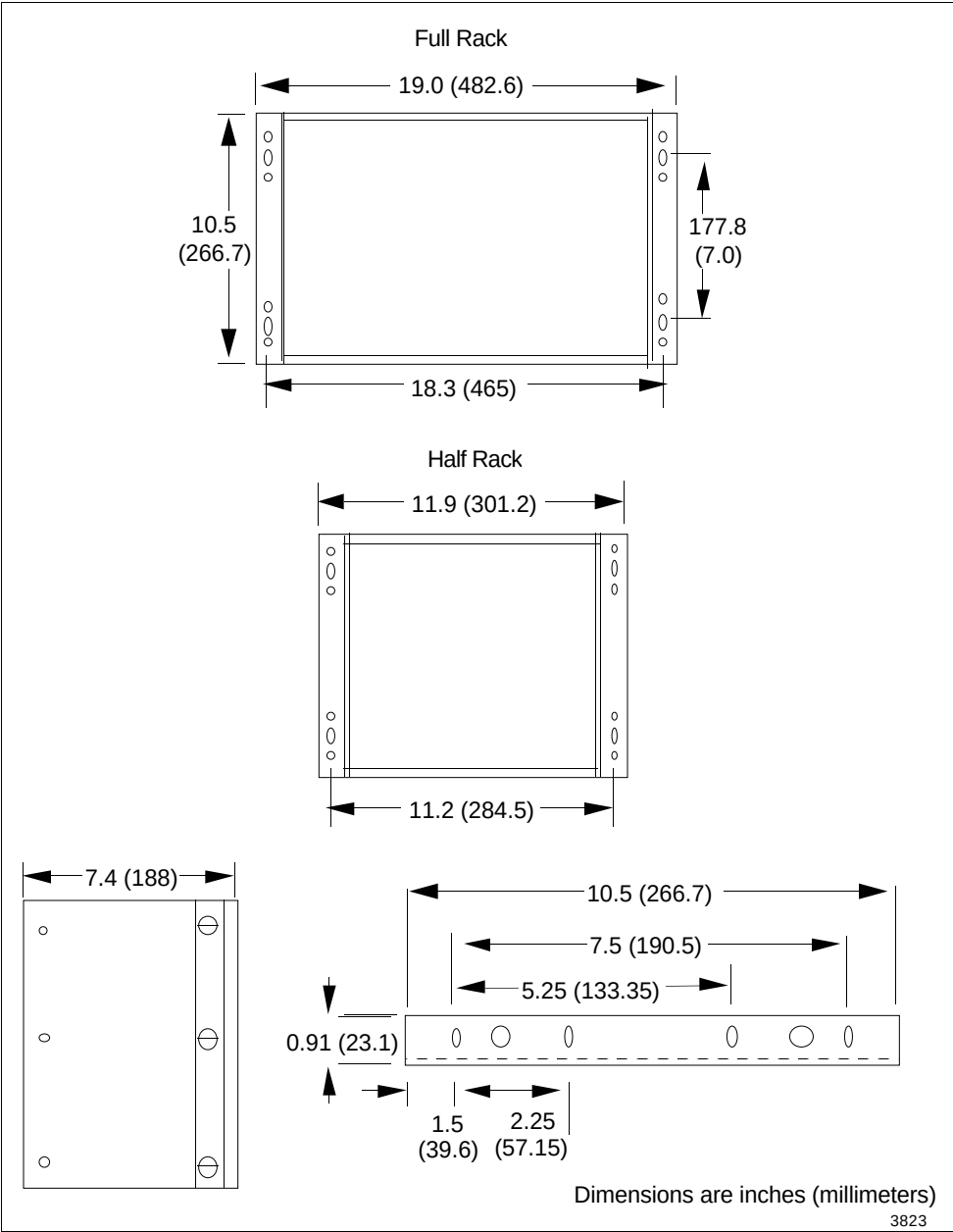


# 5.4 Card File Description

## Description

Both card file assemblies, Processor and I/O, are available as a full-size 19-inch assembly and install on a 19-inch RETMA mounting structure. The I/O card file is also available as a half-size, 12-inch assembly, but must be mounted external to the the Logic Manager cabinet. The dimensions of both card file sizes is shown in Figure 5-7.

Figure 5-7 Full- and Half-Size Card File Dimensions

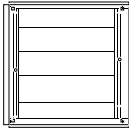


## 5.5 Cabinet Floor Planning

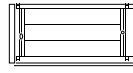
### Floor template

Figure 5-8 is a cabinet floor template that will help you plan your facility layout when installing Process Manager cabinets.

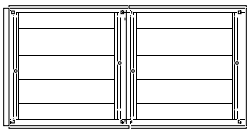
Figure 5-8 Cabinet Floor Planning Templates



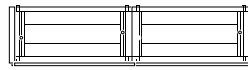
Single Dual Access Cabinet



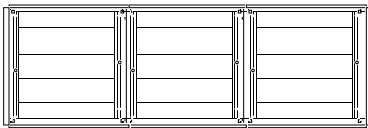
Single Single Access Cabinet



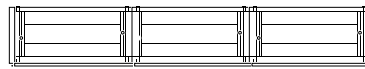
Dual Dual Access Cabinet Complex



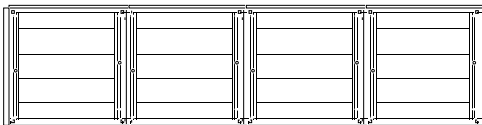
Dual Single Access Cabinet Complex



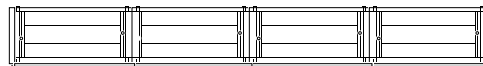
Triple Dual Access Cabinet Complex



Triple Single Access Cabinet Complex



Quad Dual Access Cabinet Complex



Quad Single Access Cabinet Complex

#### Note:

1. Cabinet scale is 1/4 inch = 1 foot.
2. A dual access cabinet is approximately 30 inches by 30 inches. The side and door panels add 1.5 inches to each cabinet dimension. When cabinets are complexed, 1.5 inches is added to the total complex dimensions.
3. A single access cabinet is approximately 30 inches wide and 21 inches deep. The side panels and the single door panel add 3/4 inch to the cabinet depth and 1.5 inches to the cabinet width. When cabinets are complexed, 3/4 inch is added to the complex depth and 1.5 inches is added to the total complex width.

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## Section 6 – Agency Approval Status

### 6.1 Overview

**Section Contents** The topics covered in this section are:

|     | Topic                     | See Page |
|-----|---------------------------|----------|
| 6.1 | Overview.....             | 107      |
| 6.2 | Agency Approval List..... | 107      |

### 6.2 Agency Approval List

#### Introduction

Table 6-1 is a list of Logic Manager components and there present approval status by Underwriters Laboratories, the Canadian Standards Association, and Factory Mutual.

The agency key, label key, and the file numbers under which the Logic Manager components are listed is defined below.

|            |     |                                                             |
|------------|-----|-------------------------------------------------------------|
| Agency Key | UL  | Underwriters Laboratories                                   |
|            | CSA | Canadian Standards Association<br>(OL = Ordinary locations) |
|            | FM  | Factory Mutual                                              |

|           |     |                 |
|-----------|-----|-----------------|
| Label Key | X   | Approved/Listed |
|           | S   | Submitted       |
|           | F   | Future          |
|           | N/P | Not Possible    |

N/A Not Applicable

File numbers under which Logic Manager components are listed.

|        |                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UL     | E92208                                                                                                                                                                        |
| CSA OL | LR57938                                                                                                                                                                       |
| FM     | Under one of the following file numbers<br>2J2A4.AX<br>1Q6A2.AX<br>1M0A3.AX<br>0N2A.AX<br>1M9A0.AX<br>2N1A7.AX<br>(Class I, Division 2) Class 3611<br>(Groups A, B, C, and D) |

*Continued on next page*

## 6.2 Agency Approval List, Continued

**Model numbers with the suffix “R”** Only I/O modules with a model number that has a suffix “R” can be inserted or removed from an I/O card file while power is applied in a serial I/O subsystem using the model 621-9938R Serial I/O Module (SIOM).

### Agency approval status list

Table 6-1 Agency Approval Status

| Model Number | Description                             | UL | CSA OL | FM |
|--------------|-----------------------------------------|----|--------|----|
| 620-0024     | 24 K Memory Module                      | X  | X      | X  |
| 620-0027     | 8 K Memory Module                       | X  | X      | X  |
| 620-0036     | 115/230 Vac 15 A Processor Power Supply | X  | X      | X  |
| 620-0038     | Control Network Module                  | X  | X      | X  |
| 620-0043     | Communications Interface Module         | X  | X      | X  |
| 620-0047     | 24 Vdc 15 A Processor Power Supply      | F  | X      | X  |
| 620-0048     | Data Collection Module                  | X  | X      | X  |
| 620-0052     | Dual Port Data Collection Module        | X  | X      | X  |
| 620-0054     | System Control Module                   | X  | X      | X  |
| 620-0056     | 4 K x 4 K Register Module               | X  | X      | X  |
| 620-0059     | Redundancy Control Module               | X  | X      | X  |
| 620-0071     | Logic Manager Module                    | X  | X      | X  |
| 620-0072     | Loop Processor Module                   | X  | X      | X  |
| 620-0073     | Ethernet Loop Processor Module          | X  | X      | S  |
| 620-0080     | Processor Module                        | X  | X      | X  |
| 620-0081     | Hiway Interface Module                  | X  | X      | X  |
| 620-0083     | Power Supply Module                     | X  | X      | X  |
| 620-0085     | 2 K I/O Control Module                  | X  | X      | X  |
| 620-0088     | Parallel Link Driver Module             | X  | X      | X  |
| 620-0089     | Ethernet Interface Module               | X  | X      | S  |
| 620-1200     | 620-12 Processor                        | X  | X      | X  |
| 620-1633     | 620-16 Processor                        | X  | X      | X  |
| 620-3590     | Processor Card File                     | X  | X      | X  |

*Continued on next page*

## 6.2 Agency Approval List, Continued

Agency approval  
status list, continued

Table 6-1 Agency Approval Status, continued

| Model Number | Description                          | UL | CSA OL | FM |
|--------------|--------------------------------------|----|--------|----|
| 620-3632     | 620-36 Processor                     | X  | X      | X  |
| 621-0006R    | BCD Conversion Module                | X  | X      | X  |
| 621-0007R    | Reed Relay Output 6-Point Module     | X  | F      | F  |
| 621-0009R    | Input Simulator 8-Point Module       | X  | X      | X  |
| 621-0010AR   | 4-20 mA Analog Output Module         | X  | X      | X  |
| 621-0010VR   | 0-10 V Analog Output Module          | X  | X      | X  |
| 621-0012R    | ASCII Communications Module          | X  | X      | X  |
| 621-0018R    | Absolute Encoder Module              | X  | X      | X  |
| 621-0019R    | Pulse/Quadrature Input Module        | X  | X      | X  |
| 621-0020R    | Universal Analog Input Module        | X  | X      | X  |
| 621-0021R    | Enhanced Diagnostic Module           | X  | X      | X  |
| 621-0022AR   | 4-20 mA Isolated Analog Input Module | X  | X      | X  |
| 621-0022VR   | 0-10 V Isolated Analog Input Module  | X  | X      | X  |
| 621-0024R    | Pulse Input Module                   | X  | X      | X  |
| 621-0025R    | Resistance Temperature Device Module | X  | S      | X  |
| 621-0307R    | High-Speed Counter Module            | X  | X      | X  |
| 621-0576R    | 24 Vdc 24 Sink Input/8 Source Output | F  | F      | F  |
| 621-1100R    | 115 Vac/Vdc Input 8-Point            | X  | X      | X  |
| 621-1101R    | 115 Vac/Vdc Isolated Input 6-Point   | X  | X      | X  |
| 621-1160R    | 115 Vac Input 16-Point               | X  | X      | X  |
| 621-1180R    | 115 Vac Input 32-Point               | F  | F      | F  |
| 621-1200R    | 230 Vac/Vdc Input 8-Point            | X  | X      | X  |
| 621-1201R    | 230 Vac/Vdc Isolated Input 6-Point   | X  | X      | X  |
| 621-1250R    | 240 Vac Input 16-Point               | F  | F      | X  |
| 621-1251R    | 240 Vac Isolated Input 16-Point      | F  | F      | X  |
| 621-1500R    | 24 Vac/Vdc Input 8-Point             | X  | X      | X  |
| 621-1550R    | 24 Vac/Vdc Input 16-Point            | X  | X      | X  |

*Continued on next page*

## 6.2 Agency Approval List, Continued

Agency approval  
status list, continued

Table 6-1 Agency Approval Status, continued

| Model Number | Description                                | UL  | CSA OL | FM |
|--------------|--------------------------------------------|-----|--------|----|
| 621-2100R    | 115 Vac Output 8-Point                     | X   | S      | X  |
| 621-2101R    | 115 Vac Isolated Output 6-Point            | X   | X      | S  |
| 621-2102R    | 115 Vac Self-Protect Point Output 8-Point  | N/P | F      | F  |
| 621-2150R    | 115 Vac Output 16-Point                    | F   | X      | X  |
| 621-2175R    | 115 Vac Output 32-Point                    | F   | F      | F  |
| 621-2200R    | 230 Vac Output 8-Point                     | F   | X      | X  |
| 621-2201R    | 230 Vac Isolated Output 6-Point            | X   | X      | X  |
| 621-2500R    | 24 Vac Output 8-Point                      | X   | X      | X  |
| 621-2550R    | 24 Vac Output 16-Point                     | F   | X      | X  |
| 621-3300R    | 5 Vdc Sink Input 8-Point                   | X   | X      | X  |
| 621-3450R    | 12 Vdc Sink Input 8-Point                  | X   | X      | X  |
| 621-3500R    | 12-24 Vdc Sink Input 8-Point               | X   | X      | X  |
| 621-3502R    | 12-24 Vdc Sink Fast Response Input 8-Point | X   | X      | X  |
| 621-3552R    | 24 Vdc Sink Fast Response Input 16-Point   | X   | X      | X  |
| 621-3560R    | 12-24 Vdc Sink Input 16-Point              | X   | X      | X  |
| 621-3580R    | 12-24 Vdc Sink Input 32-Point              | F   | F      | F  |
| 621-3600R    | 48 Vdc Sink Input 16-Point                 | X   | X      | X  |
| 621-3660R    | 48 Vdc Sink Input 32-Point                 | X   | X      | S  |
| 621-4300R    | 5 Vdc Source Input 8-Point                 | X   | X      | X  |
| 621-4350R    | 5 Vdc TTL Output 16-Point                  | F   | F      | F  |
| 621-4500R    | 12-24 Vdc Source Input 8-Point             | X   | X      | X  |
| 621-4502R    | 12-24 Vdc Source Fast Resp. Input 8-Point  | X   | X      | X  |
| 621-4550R    | 24 Vdc Source Input 16-Point               | X   | X      | X  |
| 621-6300R    | 5 Vdc Source Output 8-Point                | X   | X      | X  |
| 621-6350R    | 5 Vdc TTL Source Output 16-Point           | X   | X      | X  |
| 621-6450R    | 12 Vdc Source Output 16-Point              | X   | X      | X  |
| 621-6500R    | 12-24 Vdc Source Output 8-Point            | X   | X      | X  |

*Continued on next page*

## 6.2 Agency Approval List, Continued

Agency approval  
status list, continued

Table 6-1 Agency Approval Status, continued

| Model Number | Description                                 | UL  | CSA OL | FM  |
|--------------|---------------------------------------------|-----|--------|-----|
| 621-6503R    | 12-24 Vdc Source Self Protec Output 8-Point | X   | X      | X   |
| 621-6550R    | 24 Vdc Source Output 16-Point               | X   | X      | X   |
| 621-6551R    | 24 Vdc Low Power Source Output 16-Point     | X   | X      | X   |
| 621-6575R    | 24 Vdc Source Output 32-Point               | N/P | N/P    | N/P |
| 621-6600R    | 48 Vdc Source Output 8-Point                | X   | X      | X   |
| 621-6650R    | 48 Vdc Source Output 16-Point               | X   | X      | X   |
| 621-6700R    | 120 Vdc 0.5 Source Output 8-Point           | X   | X      | X   |
| 621-6701R    | 120 Vdc 2 A Source Output 8-Point           | X   | X      | X   |
| 621-9928R    | Serial Link Selector Module                 | X   | X      | X   |
| 621-9929     | Parallel Bus Selector Module                | X   | X      | X   |
| 621-9932     | 24 Vdc 8 A I/O Card File Power Supply       | X   | X      | X   |
| 621-9933     | 115/230 Vac 15 A I/O Card File Power Supply | X   | X      | X   |
| 621-9934     | 115/230 Vac 8 A I/O Card File Power Supply  | X   | X      | X   |
| 621-9937     | Parallel I/O Module                         | X   | X      | X   |
| 621-9938R    | Serial I/O Module (Dual Port) Module        | X   | X      | X   |
| 621-9939     | Serial Link Module                          | X   | X      | X   |
| 621-9940     | Serial I/O (Single Port) Module             | X   | X      | X   |
| 621-9949     | I/O Terminal block 8-Point                  | X   | X      | N/A |
| 621-9950     | I/O Terminal block 16-Point                 | X   | X      | N/A |
| 621-9990     | Full-Size I/O Card File                     | X   | X      | X   |
| 621-9991     | Half-Size I/O Card File                     | X   | X      | X   |
| 627-1002R    | 3-Port MiniCOP Module                       | X   | X      | S   |
| 627-1003R    | 4-Port MiniCOP Module                       | X   | X      | S   |



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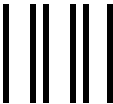
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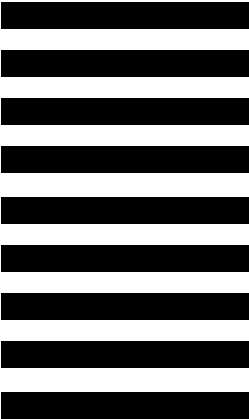
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